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Seal support assembly for a turbine engine

Abstract

A turbine engine is provided. The gas turbine engine defines a radial direction and includes: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment, the seal segment having a seal face configured to form a fluid bearing with the rotor; and a seal support assembly, the seal support assembly including a magnet assembly having a magnet coupled to the carrier or the seal segment for biasing the first seal segment along the radial direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2970808	12/1960	Coppa	N/A	N/A
3146992	12/1963	Farrell	N/A	N/A
3511511	12/1969	Voitik	N/A	N/A
4251185	12/1980	Karstensen	N/A	N/A
4307993	12/1980	Hartel	N/A	N/A
4334822	12/1981	Rossmann	N/A	N/A
4632635	12/1985	Thoman et al.	N/A	N/A
4747603	12/1987	Sugino et al.	N/A	N/A
4856963	12/1988	Klapproth et al.	N/A	N/A
4974821	12/1989	Balsells	N/A	N/A
4983051	12/1990	Hibner et al.	N/A	N/A
5100158	12/1991	Gardner	N/A	N/A
5143384	12/1991	Lipschitz	N/A	N/A
5301957	12/1993	Hwang et al.	N/A	N/A
5388843	12/1994	Sedy	N/A	N/A
5431533	12/1994	Hobbs	N/A	N/A
5509664	12/1995	Borkiewicz	N/A	N/A
5639210	12/1996	Carpenter et al.	N/A	N/A
5975537	12/1998	Turnquist et al.	N/A	N/A
6145843	12/1999	Hwang	N/A	N/A
6202302	12/2000	Descoteaux	N/A	N/A
6210103	12/2000	Ramsay	N/A	N/A
6273671	12/2000	Ress, Jr.	N/A	N/A
6368054	12/2001	Lucas	N/A	N/A
6505837	12/2002	Heshmat	N/A	N/A
6514041	12/2002	Matheny et al.	N/A	N/A
6543992	12/2002	Webster	N/A	N/A
6655696	12/2002	Fang et al.	N/A	N/A
6692006	12/2003	Holder	N/A	N/A
6811154	12/2003	Proctor et al.	N/A	N/A
6877952	12/2004	Wilson	N/A	N/A
6895757	12/2004	Mitchell et al.	N/A	N/A

6896038	12/2004	Arilla et al.	N/A	N/A
7066470	12/2005	Turnquist et al.	N/A	N/A
7079957	12/2005	Finnigan et al.	N/A	N/A
7086649	12/2005	Plona	N/A	N/A
7125223	12/2005	Turnquist et al.	N/A	N/A
7334980	12/2007	Trinks et al.	N/A	N/A
7367776	12/2007	Albers et al.	N/A	N/A
7435049	12/2007	Ghasripoor et al.	N/A	N/A
7438526	12/2007	Enderby	N/A	N/A
7448849	12/2007	Webster et al.	N/A	N/A
7459081	12/2007	Koenig et al.	N/A	N/A
7596954	12/2008	Penda et al.	N/A	N/A
7614792	12/2008	Wade et al.	N/A	N/A
7726660	12/2009	Datta	N/A	N/A
7752849	12/2009	Webster et al.	N/A	N/A
7901186	12/2010	Cornett et al.	N/A	N/A
8002285	12/2010	Justak	N/A	N/A
8047765	12/2010	Wilson et al.	N/A	N/A
8052380	12/2010	Willett, Jr.	N/A	N/A
8056902	12/2010	Roddis et al.	N/A	N/A
8113771	12/2011	Turnquist et al.	N/A	N/A
8142141	12/2011	Tesh et al.	N/A	N/A
8177476	12/2011	Andrew et al.	N/A	N/A
8186945	12/2011	Bhatnagar	415/173.1	F01D 11/22
8210799	12/2011	Rawlings	N/A	N/A
8240986	12/2011	Ebert	N/A	N/A
8434766	12/2012	Zeng et al.	N/A	N/A
8556578	12/2012	Memmen et al.	N/A	N/A
8608427	12/2012	Bock	N/A	N/A
8641045	12/2013	Justak	N/A	N/A
8678742	12/2013	Klingels	N/A	N/A
8790067	12/2013	McCaffrey et al.	N/A	N/A
8864443	12/2013	Narita et al.	N/A	N/A
8985938	12/2014	Petty	N/A	N/A
9045994	12/2014	Bidkar et al.	N/A	N/A
9068471	12/2014	Klingels	N/A	N/A
9103223	12/2014	Uehara et al.	N/A	N/A
9115810	12/2014	Bidkar et al.	N/A	N/A
9145785	12/2014	Bidkar et al.	N/A	N/A
9169741	12/2014	Szwedowicz et al.	N/A	N/A
9200530	12/2014	McCaffrey	N/A	N/A
9255489	12/2015	DiTomasso et al.	N/A	N/A
9255642	12/2015	Bidkar et al.	N/A	N/A
9359908	12/2015	Bidkar et al.	N/A	N/A
9394801	12/2015	Willett, Jr.	N/A	N/A
9435218	12/2015	Casavant et al.	N/A	N/A
9528554	12/2015	Moratz	N/A	N/A
9587746	12/2016	Bidkar et al.	N/A	N/A
9598971	12/2016	Hasnedl et al.	N/A	N/A

9683451	12/2016	Sonokawa et al.	N/A	N/A
9869196	12/2017	Day et al.	N/A	N/A
9869205	12/2017	Ganiger et al.	N/A	N/A
9890650	12/2017	Von Berg et al.	N/A	N/A
9963988	12/2017	Swedowicz et al.	N/A	N/A
9976435	12/2017	Borja et al.	N/A	N/A
10041534	12/2017	Ganiger et al.	N/A	N/A
10060280	12/2017	Crawley, Jr. et al.	N/A	N/A
10077782	12/2017	Zhang et al.	N/A	N/A
10100660	12/2017	Sippel et al.	N/A	N/A
10161259	12/2017	Gibson et al.	N/A	N/A
10184347	12/2018	D'Ambruoso	N/A	N/A
10190431	12/2018	Bidkar et al.	N/A	N/A
10196980	12/2018	Ganiger et al.	N/A	N/A
10323541	12/2018	Ganiger et al.	N/A	N/A
10344612	12/2018	Hudson et al.	N/A	N/A
10352455	12/2018	Berard et al.	N/A	N/A
10385715	12/2018	Wong et al.	N/A	N/A
10415418	12/2018	McCaffrey et al.	N/A	N/A
10415419	12/2018	Sun et al.	N/A	N/A
10422431	12/2018	Chuong et al.	N/A	N/A
10436070	12/2018	McCaffrey	N/A	N/A
10443424	12/2018	McCaffrey	N/A	N/A
10533446	12/2019	Barak et al.	N/A	N/A
10920593	12/2020	Millier et al.	N/A	N/A
10962024	12/2020	Nesteroff et al.	N/A	N/A
10962118	12/2020	Duffy et al.	N/A	N/A
10995861	12/2020	Hilbert et al.	N/A	N/A
11047481	12/2020	Bidkar et al.	N/A	N/A
11193590	12/2020	Black	N/A	N/A
2001/0007632	12/2000	Pesek et al.	N/A	N/A
2007/0053772	12/2006	Couture, Jr. et al.	N/A	N/A
2007/0248452	12/2006	Brisson	415/10	F01D 11/06
2008/0056895	12/2007	Senoo	N/A	N/A
2008/0265513	12/2007	Justak	N/A	N/A
2010/0078893	12/2009	Turnquist et al.	N/A	N/A
2010/0303612	12/2009	Bhatnagar	415/127	F01D 11/22
2012/0177484	12/2011	Lusted et al.	N/A	N/A
2012/0211944	12/2011	Nishimoto et al.	N/A	N/A
2012/0223483	12/2011	Bidkar et al.	N/A	N/A
2012/0248704	12/2011	Fennell et al.	N/A	N/A
2013/0034423	12/2012	Adaickalasamy et al.	N/A	N/A
2014/0008871	12/2013	Bidkar et al.	N/A	N/A
2014/0062024	12/2013	Bidkar et al.	N/A	N/A
2014/0117624	12/2013	Bidkar et al.	N/A	N/A
2014/0119912	12/2013	Bidkar et al.	N/A	N/A

2015/0159498	12/2014	Mukhopadhyay et al.	N/A	N/A
2016/0010480	12/2015	Bidkar et al.	N/A	N/A
2016/0097291	12/2015	Hayford et al.	N/A	N/A
2016/0130963	12/2015	Wilson et al.	N/A	N/A
2016/0138412	12/2015	Rioux	N/A	N/A
2016/0376907	12/2015	O'Leary et al.	N/A	N/A
2017/0051621	12/2016	Ackermann et al.	N/A	N/A
2017/0051834	12/2016	Webster et al.	N/A	N/A
2017/0211402	12/2016	Peters et al.	N/A	N/A
2018/0045066	12/2017	Chuong	N/A	N/A
2018/0058240	12/2017	Chuong et al.	N/A	N/A
2018/0179908	12/2017	D'Alessandro	N/A	F01D 11/001
2018/0372229	12/2017	Bidkar et al.	N/A	N/A
2019/0072186	12/2018	Bidkar et al.	N/A	N/A
2019/0085712	12/2018	Wesling et al.	N/A	N/A
2019/0203842	12/2018	Bidkar et al.	N/A	N/A
2019/0218926	12/2018	DiFrancesco et al.	N/A	N/A
2019/0276851	12/2018	Lin	N/A	N/A
2020/0040735	12/2019	Millier et al.	N/A	N/A
2020/0063588	12/2019	Morliere et al.	N/A	N/A
2020/0102845	12/2019	DiFrancesco et al.	N/A	N/A
2020/0191162	12/2019	Weihard et al.	N/A	N/A
2020/0318489	12/2019	Webb	N/A	N/A
2020/0362716	12/2019	Glahn et al.	N/A	N/A
2021/0207487	12/2020	George et al.	N/A	N/A
2022/0154580	12/2021	Gainger et al.	N/A	N/A
2022/0235667	12/2021	Mizumi et al.	N/A	N/A
2022/0349475	12/2021	Nguyen et al.	N/A	N/A
2022/0389825	12/2021	Johnson et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
108374694	12/2017	CN	N/A
109973658	12/2018	CN	N/A
113446069	12/2020	CN	N/A
4011710	12/1990	DE	N/A
2239423	12/2009	EP	N/A
3042555	12/2016	FR	N/A
3059041	12/2019	FR	N/A
S57195803	12/1981	JP	N/A
S58206807	12/1982	JP	N/A
S60111004	12/1984	JP	N/A
WO2010/112421	12/2009	WO	N/A

OTHER PUBLICATIONS

Arnold Magnetic Technologies, Recoma HT Rare Earth Magnets for High Temperatures, 2020 (Year: 2020). cited by examiner

Bruce et al., Advanced Seal Technology Role in Meeting Next Generation Turbine Engine Goals, National Aeronautics and Space Administration Lewis Research Center, France, May 11-15, 1998, pp. 1-14. cited by applicant
Chupp et al., Sealing in Turbomachinery, NASA/TM-2006-214341, National Aeronautics and Space Administration, Cleveland, OH, 2006, 62 Pages. cited by applicant
Delgado et al., A Review of Engine Seal Performance and Requirements for Current and Future Army Engine Platforms, NASA/TM-2008-215161, 43.SUP.rd .Joint Propulsion Conference Co-Sponsored by AIAA, ASME, SAE, and ASEE, Cincinnati, OH, Jul. 8-11, 2007, 22 Pages. cited by applicant
Grondahl et al., Film Riding Leaf Seals for Improved Shaft Sealing, Proceeding of ASME Turbo Expo 2010: Power for Land, Sea and Air, GT2010-23629, Glasgow, UK, Jun. 14-18, 2010, 8 Pages. cited by applicant
Hamidizadeh, Study of Magnetic Properties and Demagnetization Models of Permanent Magnet for Electric Vehicles Application, Thesis McGill University, 2016, 84 Pages. https://escholarship.mcgill.ca/downloads/qb98mj16r.pdf. cited by applicant
Moore, Lip Seal, Materials Science, Fluoroelastomers Handbook 2006, 16 Pages. Retrieved Dec. 9, 2022 from Weblink https://www.sciencedirect.com/topics/materials-science/lip-seal. cited by applicant
Munson et al., Development of Film Riding Face Seals for a Gast Turbine Engine, Tribology Transactions, vol. 35, Issue 1, 1992, pp. 65-70. cited by applicant
Steinetz et al., Advanced Seal Technology Role in Meeting Next Generation Turbine Engine Goals, RTO AVT Symposium on Design Principles and Methods for Aircraft Gas Turbine Engines, Toulouse, France, May 11-15, 1998, 14 Pages. cited by applicant

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Background/Summary

PRIORITY INFORMATION
(1) The present application claims priority to Polish Patent Application Number P.444189 filed on Mar. 24, 2023.
FIELD
(2) The present disclosure relates to seal support assemblies for a turbine engine.
BACKGROUND
(3) Gas turbine engines, such as turbofan engines, may be used for aircraft propulsion. A turbofan engine generally includes a bypass fan section and a turbomachine such as a gas turbine engine to drive the bypass fan. The turbomachine generally includes a compressor section, a combustion section, and a turbine section in a serial flow arrangement. Both the compressor section and the turbine section are driven by one or more rotor shafts and generally include multiple rows or stages of rotor blades coupled to the rotor shaft. Each individual row of rotor blades is axially spaced from a successive row of rotor blades by a respective row of stator or stationary vanes. A radial gap is formed between an inner surface of the stator vanes and an outer surface of the rotor shaft.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a cross-sectional view of a gas turbine engine in accordance with an exemplary aspect of the present disclosure.

(2) FIG. **2** is a cross sectional, schematic view of a portion of the turbomachine of FIG. **1**.

(3) FIG. **3** is a close-up, schematic, cross-sectional view of a portion of the turbomachine of FIG. **2**, taken along Line **3-3** and FIG. **2**.

(4) FIG. **4** is a close-up, schematic, cross-sectional view of the rotor, carrier, first seal segment, and seal support assembly of FIG. **3**.

(5) FIG. **5** is a partial, perspective view of the assembly of FIG. **4**.

(6) FIG. **6** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(7) FIG. **7** is a perspective view of the assembly of FIG. **6**, with the carrier removed for clarity.

(8) FIG. **8** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with yet another embodiment of the present disclosure.

(9) FIG. **9** is a perspective view of the assembly of FIG. **8**, with the carrier removed for clarity.

(10) FIG. **10** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with still another embodiment of the present disclosure.

(11) FIG. **11** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with yet another embodiment of the present disclosure.

(12) FIG. **12** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with still another embodiment of the present disclosure.

(13) FIG. **13** is a top-looking-down view of the assembly of FIG. **12**.

(14) FIGS. **14**A through **14**H provide various exemplary embodiments of a spring extension in accordance with aspects of the present disclosure.

(15) FIG. **15** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(16) FIG. **16** is a forward-looking-aft, perspective view of the assembly of FIG. **15** with a portion of the carrier cut-away for clarity.

(17) FIG. **17** is a perspective view of the assembly of FIGS. **15** and **16** with the carrier removed for clarity.

(18) FIG. **18** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(19) FIG. **19** is a forward-looking-aft, perspective view of the assembly of FIG. **18** without the rotor or stator for clarity.

(20) FIG. **20** is cross-sectional view of the assembly of FIGS. **18** and **19**.

(21) FIG. **21** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(22) FIG. **22** provides a cross-sectional view of a first seal segment of the seal assembly of the assembly of FIG. **21** along Line **22-22** in FIG. **21**.

(23) FIG. **23** is a close-up, schematic view of a plurality of seal segments in accordance with an aspect of the present disclosure.

(24) FIG. **24** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(25) FIG. **25** is an aft-looking-forward, perspective view of the assembly in FIG. **24**.
(26) FIG. **26** is a perspective view of a spring arrangement of the seal support assembly depicted in FIGS. **24** and **25**.
(27) FIG. **27** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(28) FIG. **28** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(29) FIG. **29** is a perspective, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(30) FIG. **30** is a cross-sectional view of the assembly of FIG. **29** in a reference plane defined by a radial direction and an axial direction.
(31) FIG. **31** is a schematic, close-up, cross-sectional view of an alternative embodiment of a spring arrangement of the present disclosure.
(32) FIG. **32** is a plan view of a plate spring as may be incorporated into one or more of the embodiments of FIGS. **29** through **31**.
(33) FIGS. **33**A through **33**C are schematic views of various plate spring in accordance with the present disclosure as may be incorporated into one or more of embodiments of the present disclosure.
(34) FIG. **34** is a plate spring in accordance with an exemplary embodiment of the present disclosure.
(35) FIG. **35** is a cross-sectional, schematic view of the plate spring of FIG. **34**.
(36) FIG. **36** is a schematic view of a plate spring in accordance with an exemplary embodiment of the present disclosure in a first position.
(37) FIG. **37** is a schematic view of the exemplary plate spring of FIG. **36** in a second position.
(38) FIG. **38** is a schematic view of a plate spring in accordance with an exemplary embodiment of the present disclosure.
(39) FIG. **39** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(40) FIG. **40** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(41) FIG. **41** is a side, close-up view of a spring arrangement of the seal support assembly of FIG. **40**.
(42) FIG. **42** depicts the assembly of FIG. **41** during a nonoperating condition of a turbine engine.
(43) FIG. **43** depicts the assembly of FIG. **41** during a high-power operating condition of the turbine engine.
(44) FIG. **44** depicts the assembly of FIG. **41** during a low-power operating condition of the turbine engine.
(45) FIG. **45** is a spring arrangement in accordance with an exemplary embodiment.
(46) FIG. **46** is a schematic view of a bimetallic material in accordance with an exemplary embodiment.
(47) FIG. **47** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present

disclosure in a first position.

(48) FIG. **48** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **47** in a second position.

(49) FIGS. **49** and **50** provide a schematic view of a seal support assembly of the present disclosure in a first position and in a second position, respectively.

(50) FIGS. **51** and **52** provide a schematic view of another seal support assembly of the present disclosure in a first position and in a second position, respectively.

(51) FIG. **53** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(52) FIG. **54** is a close-up view of a spring arrangement of the seal support assembly of FIG. **53**.

(53) FIG. **55** is a schematic, cross-sectional view of a section of a turbine engine including the assembly of FIG. **53**.

(54) FIG. **56**A is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(55) FIG. **56**B is a close-up, schematic, cross-sectional view of the embodiment of FIG. **56**A in a second position.

(56) FIG. **57** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.

(57) FIG. **58** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(58) FIG. **59** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **58** in a second position.

(59) FIG. **60** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(60) FIG. **61** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **60** in a second position.

(61) FIGS. **62** and **63** provide schematic views of a bimetallic member in accordance with an exemplary aspect of the present disclosure in a first position and in a second position, respectively.

(62) FIG. **64** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(63) FIG. **65** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **64** in a second position.

(64) FIG. **66** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(65) FIG. **67** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **66** in a second position.

(66) FIG. **68** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.

(67) FIG. **69** is a perspective view of a torsional spring and cam of the exemplary seal support assembly of FIG. **68**.

(68) FIG. **70** is an aft-looking-forward, schematic view of a portion of the seal support assembly of FIG. **68**, as viewed along Line **70-70** in FIG. **68**, in the first position.

(69) FIG. **71** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **68** in a second position.
(70) FIG. **72** is a close-up, schematic, cross-sectional view of the exemplary seal support assembly of FIG. **70** in the second position.
(71) FIG. **73** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **68** in a third position.
(72) FIG. **74** is a seal support assembly in accordance with another exemplary embodiment of the present disclosure.
(73) FIG. **75** is a seal support assembly in accordance with another exemplary embodiment of the present disclosure.
(74) FIG. **76** is a schematic view of a cam in accordance with the present disclosure in a first position.
(75) FIG. **77** is a schematic view of a cam in accordance with the present disclosure in a second position.
(76) FIG. **78** is a schematic view of a cam in accordance with the present disclosure in a third position.
(77) FIG. **79** depicts a conjugate cam in accordance with an aspect of the present disclosure in a first position.
(78) FIG. **80** depicts the exemplary conjugate cam of FIG. **78** in a second position.
(79) FIG. **81** depicts the exemplary conjugate cam of FIG. **78** in a third position.
(80) FIG. **82** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(81) FIG. **83** provides a forward-looking-aft, perspective view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(82) FIG. **84** provides a close-up view of a portion of the seal support assembly of FIG. **83**.
(83) FIG. **85** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(84) FIG. **86** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(85) FIG. **87** a tangential spring extension in accordance with an exemplary embodiment of the present disclosure is provided.
(86) FIG. **88** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(87) FIG. **89** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(88) FIG. **90** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(89) FIG. **91** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(90) FIG. **92** is a schematic, forward-looking-aft view of an assembly in accordance with the present disclosure
(91) FIG. **93** is a close-up view of a portion of the assembly of FIG. **92**.

(92) FIG. **94** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(93) FIG. **95** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **94** in a second position.
(94) FIGS. **96** and **97** provide schematic views of a magnet assembly of the seal support assembly of FIGS. **94** and **95** in a first position and in a second position, respectively.
(95) FIGS. **98** and **99** provide schematic views of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure in a first position and in a second position, respectively.
(96) FIG. **100** is a schematic, forward-looking-aft view of an assembly in accordance with the present disclosure
(97) FIG. **101** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(98) FIG. **102** is a schematic view of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure.
(99) FIG. **103** is a schematic view of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure.
(100) FIG. **104** is a schematic view of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure.
(101) FIG. **105** is a schematic view of a magnet of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure.
(102) FIG. **106** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(103) FIG. **107** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **106** in a second position.
(104) FIG. **108** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(105) FIG. **109** is a schematic view of a magnet of a magnet assembly of a seal support assembly in accordance with another exemplary aspect of the present disclosure.
(106) FIG. **110** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(107) FIG. **111** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(108) FIG. **112** is an aft-looking-forward view of the exemplary assembly of FIG. **111**.
(109) FIG. **113** provides a schematic view of a prestressed spring assembly of a seal support assembly of the present disclosure installed with a carrier of a stator in a reference plane defined by a radial direction and a circumferential direction of a turbine engine.
(110) FIG. **114** provides an exploded view of the prestressed spring assembly of FIG. **113**.
(111) FIG. **115** provides a view of the assembly of FIGS. **113** and **114** during an assembly phase.
(112) FIG. **116** provides a view of the assembly of FIGS. **113** and **114** during an operating condition of the turbine engine.
(113) FIG. **117**A provides a schematic view of a spring extension in accordance with an exemplary aspect of the present disclosure.
(114) FIG. **117**B provides a schematic view of a spring extension in accordance with an exemplary

aspect of the present disclosure.
(115) FIG. **118**A provides a schematic view of a spring extension in accordance with an exemplary aspect of the present disclosure.
(116) FIG. **118**B provides a schematic view of a spring extension in accordance with an exemplary aspect of the present disclosure.
(117) FIG. **119** is a close-up, schematic, cross-sectional view of a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(118) FIG. **120** is a perspective view of the seal assembly and carrier of FIG. **120**.
(119) FIG. **121** is a view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(120) FIG. **122** is a close-up view of a section of the exemplary assembly of FIG. **121**.
(121) FIG. **123** is a close-up, cross-sectional view of a section of the assembly in FIG. **122**, as viewed along line **123-123** in FIG. **122**.
(122) FIG. **124** is a perspective view of portion of a prestressed spring assembly of FIGS. **121** through **123**.
(123) FIG. **125** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(124) FIG. **126** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **125** in a second position.
(125) FIG. **127** is a close-up view of a section of a pneumatic engagement assembly of FIGS. **125** and **126** (in the position depicted in FIG. **125**).
(126) FIG. **128** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(127) FIG. **129** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(128) FIG. **130** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(129) FIG. **131** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(130) FIG. **132** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **131** in a second position.
(131) FIG. **133** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(132) FIG. **134** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **133** in a second position.
(133) FIG. **135** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(134) FIG. **136** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(135) FIG. **137** is a cross-sectional view of the exemplary assembly of FIG. **136** along Line **137-137** in FIG. **136**.

(136) FIG. **138** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(137) FIG. **139** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **138** in a second position.
(138) FIG. **140** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure in a first position.
(139) FIG. **141** is a close-up, schematic, cross-sectional view of the embodiment of FIG. **140** in a second position.
(140) FIG. **142** is a close-up, schematic, cross-sectional view of a rotor, a stator having a carrier, a seal assembly, and a seal support assembly in accordance with another embodiment of the present disclosure.
(141) FIG. **143** is a schematic, cross-sectional view of the exemplary assembly of FIG. **142** along Line **143-143** in FIG. **142**.

DETAILED DESCRIPTION

(142) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.
(143) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.
(144) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.
(145) The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.
(146) The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.
(147) The term “gas turbine engine” or “turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.
(148) The term “combustion section” refers to any heat addition system for a turbomachine. For example, the term combustion section may refer to a section including one or more of a deflagrative combustion assembly, a rotating detonation combustion assembly, a pulse detonation combustion assembly, or other appropriate heat addition assembly. In certain example embodiments, the combustion section may include an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or other appropriate combustion system, or combinations thereof.
(149) The terms “low” and “high”, or their respective comparative degrees (e.g., -er, where applicable), when used with a compressor, a turbine, a shaft, or spool components, etc. each refer to relative speeds within an engine unless otherwise specified. For example, a “low turbine” or “low speed turbine” defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a “high turbine” or “high speed turbine” of the engine.
(150) The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For

example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

(151) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

(152) The term “spring extension” refers to an object that is configured to deform elastically and store mechanical energy as a result of such deformation. A spring extension may be configured to deform linearly through extension or compression, which is referred to herein as a “linear spring”; may be configured to deform in a twisting manner through rotation about its axis, which is referred to herein as a “torsional spring extension”; or in any other suitable manner.

(153) The term “proximate” refers to being closer to one end than an opposite end. For example, when used in conjunction with first and second ends; high-pressure and low-pressure sides; or the like, the phrase “proximate the first end,” or “proximate the high-pressure side,” refers to a location closer to the first end than the second end, or closer to the high-pressure side than the low-pressure side, respectively.

(154) The term “adjacent” with respect to a relative position of two like components refers to there being no other like components positioned therebetween. The term “adjacent” with respect to a relative position of two different components refers to there being no intervening structure separating the two components.

(155) The term “shape memory alloy material” and “shape memory alloy (SMA)” generally refer to a metal alloy that experiences a temperature-related or strain-related, solid-state, micro-structural phase change. An SMA material may change from one physical shape to another physical shape. The temperature at which a phase change occurs generally is called the critical or transition temperature of the SMA. The SMA material may be constructed of a single SMA or of various SMA materials. In an embodiment, high temperature SMA may define transition temperatures ranging between about 20 degrees Celsius and about 1400 degrees Celsius. The transition temperature of the SMA may be tunable to specific applications.

(156) In some embodiments, a component said to be formed of a SMA may include the SMA material as a major constituent, e.g., in an amount greater than 50 weight percent (“wt. %”) of the component. In certain embodiments, the component may be essentially composed of the SMA material (e.g., at least 90 wt. %, such as at least 95 wt. %, such as 100 wt. %).

(157) A SMA material is generally an alloy capable of returning to its original shape after being deformed. For instance, SMA materials may define a hysteresis effect where the loading path on a stress-strain graph is distinct from the unloading path on the stress-strain graph. Thus, SMA materials may provide improved hysteresis damping as compared to traditional elastic materials.

(158) A SMA material may also provide varying stiffness, in a pre-determined manner, in response to certain ranges of temperatures. The change in stiffness of the shape memory alloy may be due to a temperature related, solid state micro-structural phase change that enables the alloy to change from one physical shape to another physical shape. The changes in stiffness of the SMA material may be developed by working and annealing a preform of the alloy at or above a temperature at which the solid state micro-structural phase change of the shape memory alloy occurs. Such may allow a component formed of a SMA to act as a spring extension having a desired stiffness profile.

(159) In the manufacture of a component comprising SMA (also referred to as an SMA component) intended to change stiffness during operation of a gas turbine engine, the component may be formed to have one operative stiffness (e.g., a first stiffness) below a transition temperature and have another stiffness (e.g., a second stiffness) at or above the transition temperature.

(160) The term “temperature-dependent shape memory alloy material” refers to a SMA characterized by a temperature-dependent phase change. These phases include a martensite phase and an austenite phase. The martensite phase generally refers to a relatively lower temperature phase. Whereas the austenite phase generally refers to a relatively higher temperature phase. The

martensite phase is generally more deformable, while the austenite phase is generally less deformable. When the shape memory alloy is in the martensite phase and is heated to above a certain temperature, the shape memory alloy begins to change into the austenite phase. The temperature at which this phenomenon starts is referred to as the austenite start temperature (As). The temperature at which this phenomenon is completed is called the austenite finish temperature (Af). When the shape memory alloy, which is in the austenite phase, is cooled, it begins to transform into the martensite phase. The temperature at which this transformation starts is referred to as the martensite start temperature (Ms). The temperature at which the transformation to martensite phase is completed is called the martensite finish temperature (Mf). As used herein, the term “transition temperature” without any further qualifiers may refer to any of the martensite transition temperature and austenite transition temperature. Further, “below transition temperature” without the qualifier of “start temperature” or “finish temperature” generally refers to the temperature that is lower than the martensite finish temperature, and the “above transition temperature” without the qualifier of “start temperature” or “finish temperature” generally refers to the temperature that is greater than the austenite finish temperature.
(161) In some embodiments, a SMA component (such as a spring extension formed of an SMA material) may define a first stiffness at a first temperature and define a second stiffness at a second temperature, wherein the second temperature is different from the first temperature. Further, in some embodiments, one of the first temperature or the second temperature is below the transition temperature and the other one may be at or above the transition temperature. Thus, in some embodiments, the first temperature may be below the transition temperature and the second temperature may be at or above the transition temperature. While in some other embodiments, the first temperature may be at or above the transition temperature and the second temperature may be below the transition temperature. Further, various embodiments of SMA components described herein may be configured to have different first stiffnesses and different second stiffnesses at the same first and second temperatures.
(162) The term “strain dependent shape memory alloy material” refers to a SMA characterized by a strain-dependent phase change. These phases similarly include a martensite phase and an austenite phase, which function in a similar manner as with the temperature dependent shape memory alloy materials, but instead of defining a transition temperature, the strain dependent SMAs define a transition strain.
(163) Non-limiting examples of SMAs that may be suitable for forming various embodiments of the SMA components described herein may include nickel-titanium (NiTi) and other nickel-titanium based alloys such as nickel-titanium hydrogen fluoride (NiTiHf) and nickel-titanium palladium (NiTiPd). However, it should be appreciated that other SMA materials may be equally applicable to the current disclosure. For instance, in certain embodiments, the SMA material may include a nickel-aluminum based alloys, copper-aluminum-nickel alloy, or alloys containing zinc, zirconium, copper, gold, platinum, and/or iron. The alloy composition may be selected to provide the desired stiffness effect for the application such as, but not limited to, damping ability, transformation temperature and strain, the strain hysteresis, yield strength (of martensite and austenite phases), resistance to oxidation and hot corrosion, ability to change shape through repeated cycles, capability to exhibit one-way or two-way shape memory effect, and/or a number of other engineering design criteria. Suitable shape memory alloy compositions that may be employed with the embodiments of present disclosure may include, but are not limited to NiTi, NiTiHf, NiTiPt, NiTiPd, NiTiCu, NiTiNb, NiTiVd, TiNb, CuAlBe, CuZnAl and some ferrous based alloys. In some embodiments, NiTi alloys having transition temperatures between 5 degrees C. and 150 degrees C. are used. NiTi alloys may change from austenite to martensite upon cooling.
(164) Moreover, SMA materials may also display superelasticity characteristics. Superelasticity may generally be characterized by recovery of large strains, potentially with some dissipation. For instance, martensite and austenite phases of the SMA material may respond to mechanical stress as

well as temperature induced phase transformations. For example, SMAs may be loaded in an austenite phase (e.g. above a certain temperature). As such, the material may begin to transform into the (twinned) martensite phase when a critical stress is reached. Upon continued loading and assuming isothermal conditions, the (twinned) martensite may begin to detwin, allowing the material to undergo plastic deformation. If the unloading happens before plasticity, the martensite may generally transform back to austenite, and the material may recover its original shape by developing a hysteresis.

(165) The term “bimetallic material” refers to a material having a first layer formed of a first material and a second layer formed of a second material, with the first and second materials configured to expand differently in response to temperature, strain, or a combination thereof. For example, the first material may define a first coefficient of thermal expansion and the second material may define a second coefficient of thermal expansion different than the first coefficient of thermal expansion. Additionally or alternatively one of the first material or the second material may be a SMA material configured to expand differently than the other of the first material or the second material in response to operating conditions to which the bimetallic material is expected to be exposed.

(166) The present disclosure is generally related to a seal member support system for a turbomachine of a gas turbine engine. A turbomachine generally includes a compressor section including a low-pressure compressor and a high-pressure compressor, a combustion section, and a turbine section including a high-pressure turbine and a low-pressure turbine arranged in serial-flow order. Each of the low-pressure compressor, the high-pressure compressor, the high-pressure turbine and the low-pressure turbine include sequential rows of stationary or stator vanes axially spaced by sequential rows of rotor blades. The rotor blades are generally coupled to a rotor shaft and the stator vanes are mounted circumferentially in a ring configuration about an outer surface of the rotor shaft. Radial gaps are formed between the outer surface of the rotor shaft and an inner portion of each ring or row of stator vanes.

(167) During operation, it is desirable to control (reduce or prevent) compressed air flow or combustion gas flow leakage through these radial gaps. Ring seals are used to form a film bearing seal to seal these radial gaps. Ring seals generally include a plurality of seal shoe or seal member segments. As pressure builds in the compressor section and/or the turbine section, the seal members are forced radially outwardly and form a bearing seal between the outer surface of the rotor shaft and the respective seal members. To reduce wear on the rotor shaft and/or the seal members, it is desirable to maintain a positive radial clearance between the seal members and the outer surface of the rotor shaft under all operating conditions of the turbomachine. However, at low delta pressure operating conditions and transients like during start-up, stall, rotor vibration events, or during sudden pressure surges within the turbomachine, the film bearing stiffness may be low or suddenly change thus leading to seal member/rotor rubs.

(168) Disclosed herein is a seal member support system to hold the seal members in a retracted position radially away from the rotor shaft during these low delta pressure operating conditions. Various embodiments presented work on a tangential spring-based retraction mechanism. In an assembly or low-pressure condition, the seal members are held in the retracted position, radially outward from the outer surface of the rotor shaft. As a pressure delta across a backside surface of the seal members increases, the seal members/segments move radially inwardly to a desired radial position to seal the respective radial gap (e.g., the seal rides on an air film). As the pressure delta across the seal members decreases, the seal members return to the retracted condition, thus reducing the potential for rotor scrub and damage or excessive wear to the seal members.

(169) Referring now to the drawings, wherein identical numerals indicate the same elements throughout the figures, FIG. **1** is a schematic cross-sectional view of a gas turbine engine in accordance with an exemplary embodiment of the present disclosure. More particularly, for the embodiment of FIG. **1**, the gas turbine engine is a high-bypass turbofan jet engine, sometimes also

referred to as a “turbofan engine.” As shown in FIG. **1**, the gas turbine engine **10** defines an axial direction A (extending parallel to a longitudinal centerline **12** provided for reference), a radial direction R. and a circumferential direction C extending about the longitudinal centerline **12**. In general, the gas turbine engine **10** includes a fan section **14** and a turbomachine **16** disposed downstream from the fan section **14**.

(170) The exemplary turbomachine **16** depicted generally includes a substantially tubular outer casing **18** that defines an annular inlet **20**. The outer casing **18** encases, in serial flow relationship, a compressor section including a booster or low-pressure (LP) compressor **22** and a high-pressure (HP) compressor **24**; a combustion section **26**; a turbine section including a high-pressure (HP) turbine **28** and a low-pressure (LP) turbine **30**; and a jet exhaust nozzle section **32**. A high-pressure (HP) shaft **34** (which may additionally or alternatively be a spool) drivingly connects the HP turbine **28** to the HP compressor **24**. A low-pressure (LP) shaft **36** (which may additionally or alternatively be a spool) drivingly connects the LP turbine **30** to the LP compressor **22**. The compressor section, combustion section **26**, turbine section, and jet exhaust nozzle section **32** together define a working gas flowpath **37**.

(171) For the embodiment depicted, the fan section **14** includes a fan **38** having a plurality of fan blades **40** coupled to a disk **42** in a spaced apart manner. As depicted, the fan blades **40** extend outwardly from disk **42** generally along the radial direction R R. Each fan blade **40** is rotatable relative to the disk **42** about a pitch axis P by virtue of the fan blades **40** being operatively coupled to a suitable pitch change mechanism **44** configured to collectively vary the pitch of the fan blades **40**, e.g., in unison. The gas turbine engine **10** further includes a power gear box **46**, and the fan blades **40**, disk **42**, and pitch change mechanism **44** are together rotatable about the longitudinal centerline **12** by LP shaft **36** across the power gear box **46**. The power gear box **46** includes a plurality of gears for adjusting a rotational speed of the fan **38** relative to a rotational speed of the LP shaft **36**, such that the fan **38** may rotate at a more efficient fan speed.

(172) Referring still to the exemplary embodiment of FIG. **1**, the disk **42** is covered by rotatable front hub **48** of the fan section **14** (sometimes also referred to as a “spinner”). The front hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**.

(173) Additionally, the exemplary fan section **14** includes an annular fan casing or outer nacelle **50** that circumferentially surrounds the fan **38** and/or at least a portion of the turbomachine **16**. It should be appreciated that the outer nacelle **50** is supported relative to the turbomachine **16** by a plurality of circumferentially-spaced outlet guide vanes **52** in the embodiment depicted. Moreover, a downstream section **54** of the outer nacelle **50** extends over an outer portion of the turbomachine **16** so as to define a bypass airflow passage **56** therebetween.

(174) During operation of the gas turbine engine **10**, a volume of air **58** enters the gas turbine engine **10** through an associated inlet **60** of the outer nacelle **50** and fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of air **62** is directed or routed into the bypass airflow passage **56** and a second portion of air **64** as indicated by arrow **64** is directed or routed into the working gas flowpath **37**, or more specifically into the LP compressor **22**. The ratio between the first portion of air **62** and the second portion of air **64** is commonly known as a bypass ratio. A pressure of the second portion of air **64** is then increased as it is routed through the HP compressor **24** and into the combustion section **26**, where it is mixed with fuel and burned to provide combustion gases **66**.

(175) The combustion gases **66** are routed through the HP turbine **28** where a portion of thermal and/or kinetic energy from the combustion gases **66** is extracted via sequential stages of HP turbine stator vanes **68** that are coupled to the outer casing **18** and HP turbine rotor blades **70** that are coupled to the HP shaft **34**, thus causing the HP shaft **34** to rotate, thereby supporting operation of the HP compressor **24**. The combustion gases **66** are then routed through the LP turbine **30** where a second portion of thermal and kinetic energy is extracted from the combustion gases **66** via sequential stages of LP turbine stator vanes **72** that are coupled to the outer casing **18** and LP

turbine rotor blades **74** that are coupled to the LP shaft **36**, thus causing the LP shaft **36** to rotate, thereby supporting operation of the LP compressor **22** and/or rotation of the fan **38**.
(176) The combustion gases **66** are subsequently routed through the jet exhaust nozzle section **32** of the turbomachine **16** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **62** is substantially increased as the first portion of air **62** is routed through the bypass airflow passage **56** before it is exhausted from a fan nozzle exhaust section **76** of the gas turbine engine **10**, also providing propulsive thrust. The HP turbine **28**, the LP turbine **30**, and the jet exhaust nozzle section **32** at least partially define a hot gas path **78** for routing the combustion gases **66** through the turbomachine **16**.
(177) It should be appreciated, however, that the exemplary gas turbine engine **10** depicted in FIG. **1** is by way of example only, and that in other exemplary embodiments, the gas turbine engine **10** may have any other suitable configuration. For example, although the gas turbine engine **10** depicted is configured as a ducted gas turbine engine (e.g., including the outer nacelle **50**), in other embodiments, the gas turbine engine **10** may be an unducted gas turbine engine (such that the fan **38** is an unducted fan, and the outlet guide vanes **52** are cantilevered from, e.g., the outer casing **18**). Additionally, or alternatively, although the gas turbine engine **10** depicted is configured as a geared gas turbine engine (e.g., including the power gear box **46**) and a variable pitch gas turbine engine (e.g., including a fan **38** configured as a variable pitch fan), in other embodiments, the gas turbine engine **10** may additionally or alternatively be configured as a direct drive gas turbine engine (such that the LP shaft **36** rotates at the same speed as the fan **38**), as a fixed pitch gas turbine engine (such that the fan **38** includes fan blades **40** that are not rotatable about a pitch axis P), or both. It should also be appreciated, that in still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable gas turbine engine. For example, in other exemplary embodiments, aspects of the present disclosure may (as appropriate) be incorporated into, e.g., a turboprop gas turbine engine, a turboshaft gas turbine engine, or a turbojet gas turbine engine.
(178) Referring now to FIG. **2**, a cross sectional, schematic view of a portion of the turbomachine **16** of FIG. **1** is provided. As will be appreciated, the exemplary turbomachine **16** generally includes a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** disposed between the rotor **100** and the stator **102**, and a seal support assembly **108**. The rotor **100** may be any rotor of the turbomachine **16**, such as the LP shaft **36**, the HP shaft **34**, etc. By way of example, referring briefly back to FIG. **1**, Circles SA have been added to FIG. **1** to provide example locations that the seal assembly **106** and seal support assembly **108** of the present disclosure may be incorporated into a turbomachine of the present disclosure.
(179) Referring still to FIG. **2**, and as will be explained in more detail below, the exemplary seal assembly **106** includes a plurality of seal segments **110** arranged along the circumferential direction C. Each seal segment **110** of the plurality of seal segments **110** has a seal face **112** configured to form a fluid bearing with the rotor **100**, and more specifically a radial fluid bearing (e.g., configured to constrain the rotor **100** along the radial direction R).
(180) As will also be explained in more detail below, the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and a first seal segment **110**A of the plurality of seal segments **110** to support the plurality of seal segments **110** of the seal assembly **106**. The seal support assembly **108** may further include similar spring arrangements **114** extending between the carrier **104** and the other seal segments **110** of the plurality of seal segments **110**.
(181) Further, referring now to FIG. **3**, a close-up, schematic, cross-sectional view is depicted, taken along Line **3-3** and FIG. **2**. In particular, FIG. **3** depicts the first seal segment **110**A of the plurality of seal segments **110** positioned between the rotor **100** and the carrier **104** of the stator **102**.
(182) As will be appreciated, the stator **102** further includes a stator vane **116** and the seal assembly **106** is, in the embodiment depicted, positioned at an inner end of a stator vane **116** along the radial

direction R of the turbomachine **16**. The turbomachine **16** further includes a first stage **118** of rotor blades **120** and a second stage **122** of rotor blades **120** spaced along the axial direction A of the gas turbine engine **10**. The seal assembly **106** is positioned between the first stage **118** of rotor blades **120** and the second stage **122** of rotor blades **120** along the axial direction A.

(183) In the embodiment depicted, the seal assembly **106** is positioned within a turbine section of the gas turbine engine **10**, such as within the HP turbine **28** or the LP turbine **30**. In such a manner, it will be appreciated that the rotor **100** may be a rotor coupled to the HP turbine **28**, such as the HP shaft **34**, or a rotor coupled to the LP turbine **30**, such as the LP shaft **36**. More specifically, still, in the embodiment affected, the rotor **100** is a connector extending between a disk **124** of the first stage **118** of rotor blades **120** and a disk **124** of the second stage of rotor blades **120**.

(184) It will further be appreciated that the seal assembly **106** defines a high-pressure side **126** and a low-pressure side **128**. The seal assembly **106** is operable to prevent or minimize an airflow from the high-pressure side **126** to the low-pressure side **128** between the rotor **100** and the seal assembly **106**. In particular, it will be appreciated that the first seal segment **110**A depicted includes the seal face **112** configured to form a fluid bearing with the rotor **100** to support the rotor **100** along the radial direction R and prevent or minimize the airflow from the high-pressure side **126** to the low-pressure side **128** between the rotor **100** and the seal assembly **106**.

(185) As will be appreciated, the first seal segment **110**A may be in fluid communication with a high-pressure air source to provide a high-pressure fluid flow to the seal face **112** to form the fluid bearing with the rotor **100**. In at least certain exemplary aspects, the high-pressure air source may be the working gas flowpath **37** through the gas turbine engine **10** and the seal assembly **106**, and more specifically the first seal segment **110**A, may be in fluid communication with the high-pressure air source, e.g., at the high-pressure side **126** of the seal assembly **106**.

(186) In particular, for the embodiment depicted, referring back briefly also to FIG. **1**, the gas turbine engine **10** further includes a high-pressure air duct **130** extending from the high-pressure air source and in fluid communication with seal assembly **106**. As noted, the high-pressure air source is the working gas flowpath **37**, and more specifically is a portion of the working gas flowpath defined by the HP compressor **24** of the compressor section (see FIG. **1**). The high-pressure air duct **130** extends to and through the stator vane **116** and to a high-pressure cavity **132** defined at the high-pressure side **126** of the seal assembly **106** (e.g., between the stator **102** and the rotor **100**). A high-pressure airflow from the high-pressure air duct **130** may pressurize the high-pressure cavity **132** to prevent gasses from the working gas flowpath **37** (which may be combustion gasses) from entering the high-pressure cavity **132** and damaging one or more components exposed thereto. The high-pressure airflow may also feed the seal assembly **106**. For example, the exemplary first seal segment **110**A defines a plurality of air ducts **134** extending therethrough, extending between one or more inlets in airflow communication with the high-pressure cavity **132** and one or more outlets in airflow communication with the seal face **112** to provide a necessary high-pressure airflow to form the fluid bearing with the rotor **100**.

(187) It will be appreciated, however, that in other exemplary embodiments, the seal assembly **106** may be integrated into, e.g., a compressor section of the gas turbine engine **10**. In such a case, the high-pressure side **126** may be positioned on a downstream side or aft side of seal assembly **106**, and the low-pressure side **128** may be positioned on an upstream side forward side of the seal assembly **106**.

(188) Referring now also to FIG. **4**, a close-up, schematic, cross-sectional view is provided of the rotor **100**, carrier **104**, first seal segment **110**A, and seal support assembly **108** of FIG. **3**. As will be appreciated, the exemplary first seal segment **110**A depicted further includes a lip **136** and a body **138**. The lip **136** extends from the body **138** along the axial direction A of the gas turbine engine **10** on the high-pressure side **126** of the seal assembly **106**. The lip **136** includes an outer pressurization surface **140** along the radial direction R of the gas turbine engine **10**. For the embodiment depicted, the outer pressurization surface **140** is in airflow communication with the working gas flowpath **37**

of the gas turbine engine **10**, and more specifically is exposed to the high-pressure cavity **132** and thus is in fluid communication with the working gas flowpath **37** of the gas turbine engine **10** from the high-pressure side **126** of the seal assembly **106**. The outer pressurization surface **140** is a radially outer surface of the lip **136**, and as will be appreciated, as a pressure within the high-pressure cavity **132** increases, a radially-inward force exerted on the outer pressurization surface **140** (and the first seal segment **110**A) correspondingly increases.

(189) The seal support assembly **108**, more specifically, the spring arrangement **114** of the seal support assembly **108**, extends between the carrier **104** and the first seal segment **110**A to counter a pressure on the outer pressurization surface **140** during operation of the gas turbine engine **10**, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.

(190) For example, the seal support assembly **108** may generally define a resistance along the radial direction R of the gas turbine engine **10**. The gas turbine engine **10** may be operable in a high power operating mode and in a low power operating mode. The high power operating mode may be any operating mode in which a higher amount of thrust is provided from the gas turbine engine **10** as compared to when the gas turbine engine **10** is operated in the low power operating mode. For example, the high power operating mode may be, e.g., a takeoff operating mode, a climb operating mode, a cruise operating mode, or the like. By contrast, the low power operating mode may be, e.g., an idle operating mode, a descent operating mode, or the like.

(191) The gas turbine engine **10** may define a first high-pressure at the high-pressure side **126** of the seal assembly **106** (e.g., within the high-pressure cavity **132**) when the gas turbine engine **10** is operated in the high power operating mode, and may further define the second high-pressure at the high-pressure side **126** of the seal assembly **106** (e.g., within the high-pressure cavity **132**) when the gas turbine engine **10** is operated in the low power operating mode. The seal support assembly **108** is configured to hold the first seal segment **110**A at a radial distance away from the rotor **100** when the gas turbine engine **10** defines the second high-pressure. By contrast, the seal support assembly **108** is configured to move the first seal segment **110**A (or rather, allow the first seal segment **110**A to move) towards the rotor **100** when the gas turbine engine **10** defines the first high-pressure. In such a manner, the seal support assembly **108** may allow for the first seal segment **110**A to be moved closer to the rotor **100** during the high-pressure operating mode as compared to during the low-pressure operating mode.

(192) Such a configuration may allow for a higher radial clearance between the first seal segment **110**A and the rotor **100** during low-pressure operating conditions and transients, which may allow for accommodation of, e.g., rotor vibrations with a reduced amount of rub between the rotor **100** and the first seal segment **110**A. Such a configuration may also allow for a lower radial clearance between the first seal segment **110**A and the rotor **100** during high-pressure operating conditions when, e.g., rotor **100** vibrations may be less severe. As will be appreciated, the first seal segment **110**A may be more effective at preventing or minimizing airflow from the high-pressure side **126** to the low-pressure side **128** with a lower radial clearance.

(193) Referring still to FIG. **4**, and now also to FIG. **5**, providing a partial, perspective view of the assembly of FIG. **4** with the carrier **104** removed for clarity, it will be appreciated that the exemplary spring arrangement **114** of the seal support assembly **108** depicted includes a plurality of spring extensions **142** extending between the carrier **104** and the first seal segment **110**A.

(194) More specifically, as is depicted, the spring arrangement **114** further includes a base **144** extending along the circumferential direction C of the gas turbine engine **10**, with the plurality of spring extensions **142** coupled to or formed integrally with the base **144** and extending between a proximal end **146** at the base **144** and a distal end **148** (see FIG. **4**). The plurality of spring extensions **142** are spaced along the circumferential direction C. In the embodiment shown, the spring arrangement **114** includes at least two spring extensions **142** extending between the base **144** and the first seal segment **110**A and up to, e.g., 20 spring extensions **142** extending between the

base **144** and the first seal segment **110**A. For example, the spring arrangement **114** may include at least three and up to 10 spring extensions **142** extending between the base **144** and the first seal segment **110**A.

(195) The base **144** is coupled to or formed integrally with the carrier **104**, and more specifically, is coupled to the carrier **104** (see FIG. **4**). More specifically, still, it will be appreciated that the carrier **104** includes a radial extension **150** defining a radial face, and the base **144** of the spring arrangement **114** is coupled to the carrier **104** at the radial face of the radial extension **150** (see FIG. **4**).

(196) Further, for the embodiment shown, the first seal segment **110**A defines a slot **152** in the body **138** of the first seal segment **110**A. The slot **152** extends along the axial direction A and the circumferential direction C, and the respective distal ends **148** of the plurality of spring extensions **142** are each positioned in the slot **152** of the first seal segment **110**A. Notably, in the embodiment shown, the first seal segment **110**A and the respective distal ends **148** of the plurality of spring extensions **142** together define a clearance gap **154** along the axial direction A (see FIG. **4**). Such a configuration may allow the first seal segment **110**A to move relative to the carrier **104** along the axial direction A.

(197) Further, the base **144** of the spring arrangement **114** is coupled to or formed integrally with the carrier **104** proximate the high-pressure side **126** of the seal assembly **106**, while the plurality of spring extensions **142** contact the first seal segment **110**A proximate the low-pressure side **128** of the seal assembly **106**. Such a configuration may ensure that the plurality of spring extensions **142** provide a desired amount of flexibility during operation of the gas turbine engine **10**. Notably, however, each of the plurality of spring extensions **142** define an extended contact surface with the first seal segment **110**A within the slot **152** of the first seal segment **110**A. The extended contact surface extends, for the embodiment shown, along the axial direction A to an axial halfway point **156** of the seal face **112** of the first seal segment **110**A. In such a manner, the spring arrangement **114** may prevent or minimize a twist of the first seal segment **110**A relative to the rotor **100** as a result of a force on the outer pressurization surface **140** of the lip **136** of the first seal segment **110**A.

(198) Referring now to FIGS. **6** and **7**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **6** provides a close-up, schematic, cross-sectional view of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108** having a spring arrangement **114**. FIG. **7** provides a perspective view of the assembly of FIG. **6**, with the carrier **104** removed for clarity. The embodiment of FIGS. **6** and **7** may be configured in a similar manner as the exemplary embodiment described above with reference to, e.g., FIGS. **4** and **5**, and the same or similar numbers may refer to the same or similar parts.

(199) For example, for the embodiment of FIGS. **6** and **7**, the seal support assembly **108** includes a spring arrangement **114** having a base **144** and a plurality of spring extensions **142** extending from the base **144** from respective proximal ends **146** to respective distal ends **148**. The plurality of spring extensions **142** are formed integrally with the base **144** at the respective proximal ends **146**.

(200) However, for the embodiment depicted, the plurality of spring extensions **142** are further fixedly coupled to, or formed integrally with, the body **138** of the first seal segment **110**A proximate a low-pressure side **128** of the seal assembly **106**. More specifically, for the embodiment shown, the plurality of spring extensions **142** are formed integrally with the body **138** of the first seal segment **110**A proximate the low-pressure side **128** of the seal assembly **106**.

(201) As such, it will further be appreciated that the base **144** of the spring arrangement **114** does not extend continuously between the plurality of seal segments **110** like the embodiment described above with reference to FIGS. **4** and **5**. Instead, for the embodiment shown, the spring arrangement **114** is a first spring arrangement **114**A for the first seal segment **110**A, and the seal support assembly **108** includes a plurality of spring arrangements **114** spaced along the circumferential

direction C. Each spring arrangement **114** of the plurality of spring arrangements **114** includes a base **144** coupled to or formed integrally with the carrier **104** and a plurality of spring extensions **142** extending from the base **144** to a respective seal segment **110**.
(202) Notably, for the embodiment shown, the seal support assembly **108** further includes an annular flange **158** extending along the circumferential direction C between the base **144** of each one of the plurality of spring arrangements **114** and the carrier **104**. The annular flange **158** may extend at least 170 degrees along the circumferential direction C, such as at least 180 degrees along the circumferential direction C, such as at least 300 degrees along the circumferential direction C, such as 360 degrees along the circumferential direction C. In such a manner, the annular flange **158** may maintain relative circumferential positions of the plurality of spring arrangements **114**.
(203) Referring now to FIGS. **8** and **9**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **8** provides a close-up, schematic, cross-sectional view of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108** having a spring arrangement **114**. FIG. **9** provides a perspective view of the assembly of FIG. **8**, with the carrier **104** removed for clarity. The embodiment of FIGS. **8** and **9** may be configured in a similar manner as the exemplary embodiment described above with reference to, e.g., FIGS. **6** and **7**.
(204) For example, the exemplary embodiment of FIGS. **8** and **9** includes the seal support assembly **108** having a spring arrangement **114** extending between the carrier **104** (see FIG. **8**) and the first seal segment **110**A. However, for the embodiment shown, the spring arrangement **114** includes a base **144** and a single spring extension **142** extending from the base **144** at a proximal end **146** to the first seal segment **110**A at a distal end **148**.
(205) Notably, as with the embodiment of FIGS. **6** and **7**, the embodiment of FIGS. **8** and **9** includes a separate spring arrangement **114** for each seal segment **110** of the seal assembly **106**, including a first spring arrangement **114**A extending between the carrier **104** and the first seal segment **110**A (see, e.g., FIG. **9**).
(206) The exemplary spring extension **142** for the embodiment depicted in FIGS. **8** and **9** includes a first segment **160** and a second segment **162**. The first and second segments **160**, **162** extend parallel to each other, and further extend along the axial direction A. The exemplary spring extension **142** further includes a third segment **164**, also extending parallel to the first segment **160** and second segment **162** along the axial direction A. The spring extension **142** includes a first bend **166** between the first segment **160** and the second segment **162** and a second bend **168** between the second segment **162** and the third segment **164**. Such a configuration may provide for the radial extension **150** of the first seal segment **110**A in response to a pressure on an outer pressurization surface **140** during operation of the gas turbine engine **10**.
(207) Further, for the embodiment shown, the distal end **148** of the spring extension **142** contacts the first seal segment **110**A at a location aligned along the axial direction A of the gas turbine engine **10** with an axial halfway point **156** of the seal face **112**. Such a configuration may prevent or minimize a twisting of the first seal segment **110**A during operation.
(208) Notably, the distal end **148** of the spring extension **142** defines an extended contact line with the first seal segment **110**A. The extended contact line extends along the circumferential direction C (see FIG. **9**) at least half of a circumferential length of a radially outer surface of the first seal segment **110**A. In particular, for the embodiment shown, the extended contact line extends along the circumferential direction C at least 75% of the circumferential length of the radially outer surface of the first seal segment **110**A, and more specifically extends the entirety of the circumferential length of the radially outer surface of the first seal segment **110**A. Such a configuration may prevent or minimize undesired movement of the first seal segment **110**A relative to the rotor **100** during operation.
(209) Referring now to FIG. **10**, an assembly in accordance with yet another exemplary embodiment of the present disclosure is provided. FIG. **10** provides a close-up, schematic, cross-

sectional view of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108** having a spring arrangement **114** in accordance with an exemplary embodiment of the present disclosure. The embodiment of FIG. **10** may be configured in a similar manner as the exemplary embodiment described above with reference to, e.g., FIGS. **8** and **9**.

(210) For example, the exemplary spring arrangement **114** of the seal support assembly **108** includes a base **144** and a spring extension **142**. The spring extension **142** extends between the base **144** and the first seal segment **110**A. The base **144** is coupled to the carrier **104**. For the embodiment depicted, the spring extension **142** includes a first segment **160** and a second segment **162**. The first segment **160** and the second segment **162** each extend from the base **144** to the first seal segment **110**A. Notably, for the embodiment shown, the first segment **160** and the second segment **162** each extend at least partially along the axial direction A of the gas turbine engine **10**. That is, the first segment **160** extends forward from the base **144** and the second segment **162** extends aft from the base **144**.

(211) More specifically, the first segment **160** of the spring extension **142** contacts the first seal segment **110**A proximate a high-pressure side **126** of the seal assembly **106** and the second segment **162** of the spring extension **142** contacts the first seal segment **110**A proximate a low-pressure side **128** of the seal assembly **106**. More specifically, still, the first seal segment **110**A defines a first slot **152**A extending along the axial direction A and the circumferential direction C of the gas turbine engine **10** from the high-pressure side **126** of the seal assembly **106**. The first seal segment **110**A further includes a second slot **152**B extending along the axial direction A and the circumferential direction C from the low-pressure side **128** of the seal assembly **106**. The first segment **160** includes a first distal end **148**A positioned in the first slot **152**A and the second segment **162** includes a second distal end **148**B positioned in the second slot **152**B. Such a configuration may assist with maintaining an orientation of the seal assembly **106** during operation.

(212) Referring now to FIG. **11**, an assembly in accordance with yet another exemplary embodiment of the present disclosure is provided. FIG. **11** provides a close-up, schematic, cross-sectional view of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108** having a spring arrangement **114** in accordance with another exemplary embodiment of the present disclosure. The embodiment of FIG. **11** may be configured in a similar manner as one or more of the exemplary embodiments described above.

(213) For example, in the embodiment shown, the spring arrangement **114** of the seal support assembly **108** includes a base **144** and a spring extension **142** extending from the base **144** to the first seal segment **110**A. The base **144** is coupled to the carrier **104** proximate a high-pressure side **126** of the seal assembly **106**. However, for the embodiment shown, the carrier **104** further includes a carrier axial extension **170** position at least partially between the spring extension **142** and the first seal segment **110**A.

(214) Additionally, in the embodiment shown the seal assembly **106** includes a circumferential seal **172** positioned between the first seal segment **110**A and the carrier axial extension **170** of the carrier **104**. In particular, for the embodiment depicted, the circumferential seal **172** is a first circumferential seal **172**A and the seal assembly **106** further includes a second circumferential seal **172**B. The first circumferential seal **172**A is more specifically configured as a piston seal extending along a circumferential direction C positioned between the carrier **104** and each of a plurality of seal segments **110** of the seal assembly **106** (including the first seal segment **110**A; see, e.g., FIG. **2**). The second circumferential seal **172**B is a brush seal positioned between the first seal segment **110**A and the carrier **104**. The second circumferential seal **172**B also extends along the circumferential direction C and more specifically is positioned between a radial extension **150** of the carrier **104** and each of the plurality of seal segments **110** of the seal assembly **106** (including the first seal segment **110**A).

(215) It will be appreciated, however, that in other exemplary embodiments, the seal assembly **106** may only include one of the first circumferential seal **172**A or the second circumferential seal **172**B, and further, that one or both of the first circumferential seal **172**A and second circumferential seal **172**B may be configured in any other suitable manner.
(216) Inclusion of one or both of the first circumferential seal **172**A and the second circumferential seal **172**B may minimize a transfer of an airflow from the high-pressure side **126** of the seal assembly **106** (and more specifically from a high-pressure cavity **132**) to a low-pressure side **128** of the seal assembly **106**.
(217) Referring now to FIGS. **12** and **13**, an assembly in accordance with yet another exemplary embodiment of the present disclosure is provided. FIG. **12** provides a close-up, schematic, cross-sectional view of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108** having a spring arrangement **114** in accordance with an exemplary embodiment of the present disclosure. FIG. **13** provides a top-looking-down view of the assembly of FIG. **12**.
(218) The exemplary embodiment of FIGS. **12** and **13** may be configured in substantially the same manner as the exemplary embodiment described above with reference to FIG. **11**. For example, the exemplary embodiment of FIGS. **12** and **13** includes a first circumferential seal **172**A and a second circumferential seal **172**B positioned between the carrier **104** and the plurality of seal segments **110** (see, e.g., FIG. **13**).
(219) However, for the embodiment shown, a base **144** of the spring arrangement **114** of the seal support assembly **108** is coupled to the carrier **104** proximate a low-pressure side **128** of the seal assembly **106**, as opposed to being coupled to the carrier **104** proximate a high-pressure side **126** of the seal assembly **106**.
(220) Referring now briefly to FIGS. **14**A through **14**H, it will be appreciated that a spring extension **142** of a spring arrangement **114** of the present disclosure may have any suitable shape or geometry to provide a desired amount of resistance along a radial direction R for one or more seal segments **110** of a seal assembly **106**.
(221) For example, in certain exemplary embodiments, the spring extensions **142** may extend parallel to one another and parallel to an axis of the engine (see, e.g., FIG. **14**A); may extend parallel to one another at an angle relative to an axis of the engine (see, e.g., FIG. **14**B); may extend in a nonparallel direction to another (see, e.g., FIG. **14**C); may include cutouts or other openings (see, e.g., FIG. **14**D); may extend in a nonlinear direction relative to an axis of the engine (see, e.g., FIG. **14**E); may define a rectangular shape (see, e.g., FIG. **14**F); may define a tapered shape between a proximal end **146** and a distal end **148** (see, e.g., FIG. **14**G); may define a converging-diverging shape between a proximal end **146** and a distal end **148** (see, e.g., FIG. **14**H); or a combination thereof.
(222) Further, it will be appreciated that any one or more features of the seal support assemblies **108** described hereinabove may be combined with any one or more other features of the seal support assemblies **108** described hereinabove.
(223) Referring now to FIGS. **15** and **16**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **15** provides a schematic, cross-sectional view of a section of a gas turbine engine, and more specifically of a section of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** positioned between the rotor **100** and the stator **102**, and a seal support assembly **108**. The view of FIG. **15** is of a reference plane defined by a radial direction R and an axial direction A of the turbine engine. FIG. **16** provides a forward-looking-aft, perspective view of the assembly of FIG. **15** with a portion of the carrier **104** cut-away for clarity. The assembly of FIGS. **15** and **16** may be configured in a similar manner as one or more of the other exemplary assemblies described above.
(224) For example, the seal support assembly **108** includes a spring arrangement **114** having a spring extension **142** extending between the carrier **104** and a first seal segment **110**A of the seal

assembly **106** to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between a seal face **112** and the rotor **100** during operation of the turbine engine. For the embodiment depicted, and as may be most clearly seen in FIG. **16**, the spring extension **142** defines at least two points of contacts with the carrier **104**, at least two points of contact with the first seal segment **110**A, or both.

(225) More specifically, for the embodiment shown the spring extension **142** includes at least two spring segments defining the at least two points of contact with the carrier **104** and further includes at least two spring segments defining the at least two points of contacts with the first seal segment **110**A. More specifically, still, in the embodiment shown, the spring extension **142** includes a first segment **160** and a second segment **162** separately extending to the first seal segment **110**A and defining in part a first point of contact with the first seal segment **110**A and a second point of contact with the first seal segment **110**A. Further, the spring extension **142** includes a third segment **164** and a fourth segment **174** separately extending to the carrier **104** and defining in part a third point of contact with the carrier **104** and a fourth point of contact with the carrier **104**.

(226) In the embodiment shown, the spring extension **142** further includes a central segment **176**, with the first segment **160**, the second segment **162**, the third segment **164**, and the fourth segment **174** each extending from the central segment **176**. The central segment **176** is a closed shape (e.g., circular, square, ovular, rectangular, polygonal, etc.) in a cross-sectional direction (e.g., in a plane defined by a circumferential direction C and the radial direction R for the embodiment shown; see FIG. **16**), and more specifically is an ovular shape for the embodiment depicted.

(227) Notably, one or more of the segments **160**, **162**, **164**, **174** of the spring extension **142** defines a length along the axial direction A at a respective point of contact with the first seal segment **110**A or the carrier **104** which may assist in reducing a twisting of the first seal segment **110**A during operation of the gas turbine engine, e.g., in response to receiving an increasing pressure on the outer pressurization surface **140** of the lip **136** of the first seal segment **110**A. More specifically, referring particularly to FIG. **15**, the first segment **160** and the second segment **162** each define an axial length **178** along the axial direction A between 5% and 100% of an axial length **180** of a seal face **112** of the first seal segment **110**A at the first point of contact and at the second point of contact respectively. Similarly, the third segment **164** and the fourth segment **174** each define an axial length **182** along the axial direction A between 5% and 100% of the axial length **180** of the seal face **112** of the first seal segment **110**A at the third point of contact and at the fourth point of contact respectively.

(228) Referring briefly to FIG. **17**, a perspective view of the assembly of FIGS. **15** and **16** is provided with the carrier **104** removed for clarity. As will be appreciated, the seal assembly **106** includes a plurality of seal segments **110** and the seal support assembly **108** includes a plurality of spring arrangements **114**, with each spring arrangement **114** including a spring extension **142** extending between the respective seal segment **110** and the carrier **104**. In such a manner, each of the plurality of seal segments **110** may be movable relative to one another and independently movable relative to the carrier **104** (not depicted in FIG. **17**; see, e.g., FIG. **16**). The spring extensions **142** of each of the plurality of spring arrangements **114** may be configured in a similar manner as described above with reference to FIGS. **15** and **16**.

(229) Referring now to FIGS. **18** and **19** an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **18** provides a schematic, cross-sectional view of a section of a gas turbine engine, and more specifically of a section of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** positioned between the rotor **100** and the stator **102**, and a seal support assembly **108**. The view of FIG. **18** is of a reference plane defined by a radial direction R and a circumferential direction C of the turbine engine. FIG. **19** provides a forward-looking-aft perspective view of the assembly of FIG. **18** without the rotor **100** or stator **102**, for clarity. The assembly of FIGS. **18** and **19** may be configured in a similar manner as one or more of

the exemplary assemblies described above.
(230) For example, the exemplary seal support assembly **108** depicted includes a spring arrangement **114** extending between the carrier **104** (FIG. **18**) and a first seal segment **110**A of the seal assembly **106** to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine. The spring arrangement **114** includes a spring extension **142** extending between the carrier **104** and the first seal segment **110**A of the seal assembly **106**. The spring extension **142** defines at least two points of contact with the carrier **104**, at least two points of contact with the first seal segment **110**A, or both.
(231) In particular, for the embodiment shown, the spring extension **142** extends between a first end **184** and a second end **186**. The first end **184** and second end **186** together define the at least two points of contact with the first seal segment **110**A. In the embodiment shown, the spring extension **142** is coupled to the first seal segment **110**A using one or more mechanical fasteners **194** at the first end **184** and at the second end **186**.
(232) Alternatively, however, the spring extension **142** may be coupled to the first seal segment **110**A in any other suitable manner (e.g., brazing).
(233) Further, for the embodiment shown, the spring extension **142** additionally defines a point of contact with the carrier **104** at a location between the first end **184** and the second end **186**. The point of contact with the carrier **104** is, for the embodiment depicted, halfway between the first end **184** and the second end **186**. In the embodiment shown, the spring arrangement **114** further includes a bolt assembly having a bolt **188**, a washer **190**, and a nut **192**. The bolt assembly may attach the spring extension **142** to the carrier **104**, forming at least in part the point of contact with the carrier **104**.
(234) As will be appreciated from the view of FIG. **19**, the spring extension **142** defines an axial length at the first end **184** and at the second end **186**, and more specifically, at the points of contact with the first seal segment **110**A. The axial length may be between about 5% and 100% of an axial length of a spring face of the first seal segment **110**A at the respective points of contact (see, e.g., axial length **180** depicted in FIG. **15**, described above).
(235) The spring extension **142** depicted in FIGS. **18** and **19** includes a plurality of bends **196** between the first end **184** and the point of contact with the carrier **104** and between the second end **186** and the point of contact with the carrier **104**. It will further be appreciated that the spring extension **142** is configured as a spring plate having a generally rectangular shape (e.g., a rectangular plate bent to form the shape depicted).
(236) In other embodiments, the spring extension **142** may have any other suitable shape or configuration, such as one or the exemplary configurations depicted in FIGS. **14**A through **14**H.
(237) Referring briefly to FIG. **20**, a zoomed out, cross-sectional view is depicted of the seal assembly **106** and seal support assembly **108** of FIGS. **18** and **19**. As will be appreciated, the seal assembly **106** includes a plurality of seal segments **110** and the seal support assembly **108** includes a plurality of spring arrangements **114**, with each spring arrangement **114** including a spring extension **142** extending between the respective seal segment **110** and the carrier **104**. In such a manner, each of the plurality of seal segments **110** may be movable relative to one another and independently movable relative to the carrier **104**. The spring extensions **142** of each of the plurality of spring arrangements **114** may be configured in a similar manner as described above with reference to FIGS. **18** and **19**.
(238) Referring now to FIGS. **21** and **22**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **21** provides an aft-looking-forward view of a stator **102** having a carrier **104**, a seal assembly **106**, and a seal support assembly **108** in accordance with another exemplary embodiment of the present disclosure, and FIG. **22** provides a cross-sectional view of a first seal segment **110**A of the seal assembly **106** along Line **22-22** in FIG.

21. The assembly of FIGS. **21** and **22** may be configured in a similar manner as one or the exemplary assemblies described hereinabove.
(239) For example, in the embodiment depicted, the seal assembly **106** includes a plurality of seal segments **110** (including the first seal segment **110**A) arranged along a circumferential direction C of a gas turbine engine for supporting rotation of a rotor **100** (not shown; see above embodiments). In addition, the seal support assembly **108** includes a spring arrangement **114** having a spring extension **142** extending between the carrier **104** and the plurality of seal segments **110** to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.
(240) However, for the embodiment of FIGS. **21** and **22**, the spring extension **142** extends continuously across the plurality of seal segments **110** in the circumferential direction C. For example, the spring extension **142** may extend continually across at least two seal segments **110**, and up to all of the plurality of seal segments **110** (e.g., 360 degrees).
(241) As will be appreciated from the embodiment of FIG. **21**, the spring extension **142** may be, e.g., a plate spring extension bent to form the shape depicted (e.g., similar to the configurations discussed above with reference to, for example, FIGS. **18** through **20**).
(242) Additionally or alternatively, the spring extension **142** may have any suitable shape or configuration, such as one or the exemplary configurations depicted in FIGS. **14**A through **14**H.
(243) As noted, the spring extension **142** extends between the carrier **104** and the seal segments **110**. Referring particularly to FIG. **21**, the carrier **104** includes a plurality of carrier tabs **198** spaced along the circumferential direction C, and the spring extension **142** is coupled to the carrier **104** through the plurality of carrier tabs **198**. In the embodiment shown, each one of the carrier tabs **198** is generally positioned at a circumferential location between two adjacent seal segments **110**. The spring extension **142** may be coupled to the plurality of carrier tabs **198** through one or more mechanical fasteners.
(244) Additionally, or alternatively, the carrier tabs **198** may define a geometry configured to constrain the spring extension **142** along a radial direction R during operation (similar to the seal segment tabs **200** described below).
(245) Further, each of the plurality of seal segments **110**, for the embodiment shown, includes one or more seal segment tabs **200** to couple the spring extension **142** to the respective seal segment **110**. In particular, referring particularly to the first seal segment **110**A, the first seal segment **110**A includes a first seal segment tab **200**A with the spring extension **142** coupled to the first seal segment **110**A at least in part using the first seal segment tab **200**A.
(246) More particularly, the first seal segment tab **200**A includes an inner side **202** along the radial direction R and the spring extension **142** is positioned adjacent to the inner side **202** of the first seal segment tab **200**A. In such a manner, the spring extension **142** may bias the first spring extension **142** against movement inwardly along the radial direction R. The inner side **202**, for the embodiment depicted, defines a geometry to constrain the spring extension **142** along the radial direction R, and further along an axial direction A. In particular, the inner side **202** defines a concave shape (e.g., a hook or U-shape geometry) when viewed in the reference plane of FIG. **22** defines by the axial and radial directions A. R.
(247) Further, for the embodiment depicted, the first seal segment **110**A includes a second seal segment tab **204** positioned proximate a first circumferential end **206** of the first seal segment **110**A and a third seal segment tab **208** positioned proximate a second circumferential end **210** of the first seal segment **110**A. The second seal segment tab **204** and third seal segment tab **208** each includes an outer surface **212** along the radial direction R with the spring extension **142** configured to contact the outer surfaces **212** during at least certain operations. In such a manner, the spring extension **142** may bias the first spring extension **142** against movement outwardly along the radial direction R more than a predetermined amount during the at least certain operations.

(248) In such a manner, the spring extension **142** may bias the plurality of seal segments **110** outwardly along the radial direction R when a relatively low-pressure is exerted on an outer pressurization surface **140** of a lip **136** of the respective seal segment **110** (see FIG. **22**), and may allow for the plurality of seal segments **110** to move inwardly along the radial direction R when a relatively high-pressure is exerted on the outer pressurization surface **140** of the lip **136** of the respective seal segment **110**. Such configuration may allow for the seal assembly **106** (and more particularly a seal face **112** of each of the plurality of seal segments **110**) to define desired clearances with the rotor **100** throughout a variety of operating conditions of the gas turbine engine.
(249) Briefly, referring to FIG. **23**, a close-up, schematic view is provided of a plurality of seal segments **110** in accordance with an aspect of the present disclosure. In the embodiment shown, a seal support assembly **108** may include a plurality of tangential spring extensions **214** extending between a channel **216** of one seal segment **110** and a channel **216** of an adjacent seal segment **110** to bias the seal segments **110** away from one another in the circumferential direction C. Such a configuration, when used in conjunction with, e.g., the embodiment of FIGS. **21** and **22**, may assist with maintaining alignment of the plurality of seal segments **110**.
(250) Referring now to FIGS. **24** and **25**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **24** provides a cross-sectional view of a stator **102** having a carrier **104**, a seal assembly **106**, and a seal support assembly **108** in accordance with another exemplary embodiment of the present disclosure. The view of FIG. **24** is in a cross-sectional plane defined by an axial direction A of the turbine engine and a radial direction R of the turbine engine. FIG. **25** is an aft-looking-forward, perspective view of the assembly in FIG. **24**. The assembly of FIGS. **24** and **25** may be configured in a similar manner as one or more of the exemplary embodiments described hereinabove.
(251) For example, the seal assembly **106** includes a plurality of seal segments **110** including a first seal segment **110**A, and the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and the first seal segment **110**A to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.
(252) For the embodiment shown, the carrier **104** further includes an axial extension, or rather a carrier axial extension **170** extending generally along the axial direction A. Similarly, the first seal segment **110**A includes a segment axial extension **218**, located outward of the carrier axial extension **170** along the radial direction R of the turbine engine. The spring arrangement **114** is positioned within a gap **220** defined between the carrier axial extension **170** and the segment axial extension **218** along the radial direction R. In such a manner, it will be appreciated that the spring arrangement **114** is compressed with an inward movement of the first seal segment **110**A along the radial direction R relative to the carrier **104**.
(253) In the embodiment depicted, the carrier axial extension **170** is coupled to, and extends from a radial wall **222** of the carrier **104** positioned proximate a low-pressure side **128** of the seal assembly **106**, and the segment axial extension **218** is coupled to, and extends from, a radial wall **224** of the first seal segment **110**A positioned proximate a high-pressure side **126** of the seal assembly **106**. Notably, the carrier **104** further includes a radial wall, also referred to herein as a radial extension **150**, proximate the high-pressure side **126** of the seal assembly **106**. Although not depicted, the seal assembly **106** may include a seal between the radial wall of the carrier **104** proximate the high-pressure side **126** and the first seal segment **110**A.
(254) Referring now briefly to FIG. **26**, a perspective view of the spring arrangement **114** depicted in FIGS. **24** and **25** is provided. In the embodiment shown, the spring arrangement **114** is configured as a plate having a support **226** and a plurality of spring segments **228** extending from the support **226**. The plurality of spring segments **228** are cantilevered from the support **226** in the embodiment shown. In particular, the plurality of spring segments **228** may be punched, cut, or

otherwise extruded from the support **226** and bent to form the plurality of spring segments **228**. Each of the plurality of spring segments **228** therefore extends between a proximal end **230** coupled to or formed integrally with the support and a distal end **232**. The support **226** may be coupled to the segment axial extension **218** and the distal end **232** may be configured to contact the segment axial extension **218** (see FIGS. **24** and **25**) of the first seal segment **110**A.
(255) The spring arrangement **114** may be configured to extend between a plurality of seal segments **110** of the seal assembly **106**.
(256) Alternatively, the spring arrangement **114** may be a first spring arrangement **114**A of a plurality of spring arrangements **114** dedicated to an individual seal segment **110** of the plurality of seal segments **110**.
(257) Referring now to FIGS. **27** and **28**, two additional embodiments of an assembly in accordance with exemplary embodiments of the present disclosure are provided. FIGS. **27** and **28** each provide schematic views of the assembly in a reference plane defined by a radial direction R and a circumferential direction C of the gas turbine engine **10**.
(258) Referring particularly FIG. **27**, the assembly includes a stator **102** having a carrier **104**, a seal assembly **106** having a first seal segment **110**A, and a seal support assembly **108**. The seal support assembly **108** includes a spring arrangement **114** having a spring extension **142** extending between the carrier **104** and the first seal segment **110**A. The spring extension **142** defines at least two points of contact with the carrier **104**, at least two points of contact with the first seal segment **110**A, or both.
(259) More specifically, for the embodiment shown, the spring extension **142** extends between a first end **184** and a second end **186**, and the first end **184** and the second end **186** defining the at least two points of contact with the carrier **104**. In the embodiment shown the first end **184** and the second end **186** are connected to the carrier **104** through a hinged pin connection (e.g., a connection allowing rotation about an axis), and the spring extension **142** is configured as a leaf spring.
(260) Referring now particular to FIG. **28**, the assembly is configured in substantially the same manner as the embodiment of FIG. **27**, however for the embodiment of FIG. **28**, a first end **184** and a second end **186** of a spring extension **142** are connected to a carrier **104** through a sliding pin connection (e.g., a connection allowing rotation about an axis and movement linearly in a direction perpendicular to the axis).
(261) It will be appreciated that in other example embodiments, the first end **184**, the second end **186**, or both may be coupled through any other suitable connection, such as a combination of a hinged pin connection and a sliding pin connection.
(262) Referring now to FIGS. **29** and **30**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **29** provides a perspective, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a seal assembly **106**, and a seal support assembly **108**. The seal assembly **106** is configured to be positioned between the carrier **104** and a rotor **100** (not shown; see embodiments described above). FIG. **30** provides a cross-sectional view of the assembly of FIG. **29** in a reference plane defined by a radial direction R and an axial direction A of the turbine engine. The assembly of FIGS. **29** and **30** may be configured in a similar manner as one or more of the exemplary assemblies described above.
(263) For example, the seal support assembly **108** includes a spring arrangement **114** and the seal assembly **106** includes a plurality of seal segments **110**. The plurality of seal segments **110** includes a first seal segment **110**A. However, in the embodiment of FIGS. **29** and **30**, the spring arrangement **114** includes a plate spring **234** coupled to the carrier **104** and to the first seal segment **110**A. The plate spring **234** extends within a reference plane perpendicular to the radial direction R of the turbine engine.
(264) For the embodiment shown, the plate spring **234** is coupled to the carrier **104** proximate a high-pressure side **126** of seal assembly **106** and proximate a low-pressure side **128** of the seal

assembly **106**. More specifically, for the embodiment shown, the carrier **104** includes a forward radial wall **236** (also referred to hereinabove as a radial extension **150**) and aft radial wall **238**. The forward radial wall **236** defines a first slot **240** extending in a tangential direction (e.g., a direction perpendicular to the radial direction R) and the aft radial wall **238** defines a second slot **242** also extending in the tangential direction. For the embodiment shown, the plate spring **234** is positioned within the first slot **240** and the second slot **242** to couple the plate spring **234** to the carrier **104**. More particularly, for the embodiment shown the plate spring **234** is press fit to the carrier **104**, and more particularly, still, press fit into the first slot **240** and the second slot **242** of the carrier **104**.

(265) It will be appreciated, however, that in other exemplary embodiments, the seal plate may be coupled to the carrier **104** in any other suitable manner. For example, in other embodiments, the seal plate may be coupled to the carrier **104** using one or more mechanical fasteners (e.g., bolts), or coupled to the carrier **104** through a combination of mechanical fasteners, press fit, etc.

(266) In addition, the plate spring **234** is coupled to the first seal segment **110**A at a location inward of an outer perimeter **244** of the plate spring **234**. In particular, for the embodiment shown, the plate spring **234** defines a center **246** (e.g., a location identified by a middle 25% of a crosswise measure and a middle 25% of a lengthwise measure), and the plate spring **234** is coupled to the first seal segment **110**A at the center **246** of the plate spring **234**.

(267) In order to attach the first seal segment **110**A to the plate spring **234**, the first seal segment **110**A includes an attachment column **248** extending outward along the radial direction R of the turbine engine. The plate spring **234** is coupled to the attachment column **248**. More specifically, the attachment column **248** includes a first section **250**, a ledge **252**, and a post **254**. The ledge **252** extends tangentially from the radial direction R between the first section **250** and the post **254**, and the post **254** extends outwardly along the radial direction R from the ledge **252**. The post **254** extends through the plate spring **234** and the plate spring **234** is pressed against the ledge **252**. In particular, the plate spring **234** includes an attachment ring **256** defining an attachment opening, and the post **254** extends through the attachment opening in the attachment ring **256**. The first seal segment **110**A further includes a retainer **258** coupled to the post **254** to couple the plate spring **234** to the first seal segment **110**A. The retainer **258** may be, e.g., a clip, a nut, a through bolt, or any other suitable mechanical fastener.

(268) In such a manner, the plate spring **234** may provide a biasing force to the first seal segment **110**A along the radial direction R.

(269) Referring briefly to FIG. **31**, a close-up, cross-sectional view of an alternative embodiment of the spring arrangement **114** of FIGS. **29** and **30** is provided. For the embodiment of FIG. **31**, a spring arrangement **114** is provided, further including a deflection limiter **260** positioned outward of the plate spring **234** along the radial direction R of the turbine engine, inward of the plate spring **234** along the radial direction R of the turbine engine, or both. More specifically, for the embodiment depicted, the deflection limiter **260** is positioned both outward and inward of the plate spring **234** along the radial direction R. The deflection limiter **260** is, in the embodiment shown, coupled to the carrier **104**, and more specifically, coupled to the forward radial wall **236** of the carrier **104** and to the aft radial wall **238** of the carrier **104** at common attachment points with the plate spring **234**. The deflection limiter **260** may define a stiffness greater than a stiffness of the plate spring **234**, and may provide additional resistance to this plate spring **234** to prevent the plate spring **234** and first seal segment **110**A from moving more than a desired amount along the radial direction R. Such a configuration may reduce or minimize a chance of rub between the first seal segment **110**A and the rotor **100** of the turbine engine during operation of the turbine engine.

(270) Referring now to FIG. **32**, a plan view of an exemplary plate spring **234** is provided as may be incorporated into one or more of the embodiments of FIGS. **29** through **31**. It will be appreciated that the exemplary plate spring **234** depicted includes a plurality of flex members **262**, an outer perimeter **244**, and an attachment ring **256**. The plurality of flex members **262** extend from the outer perimeter **244** to one or more attachment points defined with a first seal segment **110**A. In

particular, the exemplary plate spring **234** depicted in FIG. **32** is configured to define a single attachment point with the first seal segment **110**A at the attachment ring **256**. For the embodiment shown, the plate spring **234** includes four flex members **262** extending from the outer perimeter **244** to the attachment ring **256**, with each of the plurality of flex member defining a serpentine path between the outer perimeter **244** and the attachment ring **256**.

(271) For the embodiment depicted in FIG. **32**, the plate spring **234** defines a rectangular shape in the reference plane depicted (e.g., a reference plane perpendicular to the radial direction R). It will be appreciated, however, that in other embodiments, the plate spring **234** may additionally or alternatively define any other suitable shape to provide a desired amount of resistance along the radial direction R, resistance to twisting, etc.

(272) For example, referring now to FIGS. **33**A, **33**B and **33**C, in other exemplary embodiments, a seal support assembly **108** may include a spring arrangement **114** having a plate spring **234** configured in any other suitable manner. For example, the plate spring **234** may define a circular cross-sectional shape. In such a manner, the plate spring **234** may again include a plurality of flex members **262** extending from an outer perimeter **244** to an attachment ring **256**, with the plurality of flex members **262** defining a serpentine path from the outer perimeter **244** to the attachment ring **256**. With such a configuration, the plate spring **234** may be coupled, e.g., to a carrier **104** having, e.g., a circular opening.

(273) Referring now to FIG. **34**, a plate spring **234** in accordance with an exemplary embodiment of the present disclosure is provided. The plate spring **234** may be configured to be incorporated into one or more of the exemplary embodiments described herein.

(274) The plate spring **234** is depicted in a free condition. When in the free condition, the plate spring **234** defines a curvature relative to a reference plane extending perpendicular to a radial direction R of the turbine engine. In particular, it will be appreciated that the plate spring **234** defines a first radius of curvature when in the free condition. In such a manner, when the plate spring **234** is coupled to the carrier **104** and the first seal segment **110**A, it may be in a pre-strained condition.

(275) In particular, referring now to FIG. **35**, a cross-sectional view is depicted of the plate spring **234** of FIG. **34**. The view of FIG. **35** may be a forward-looking-aft view along an axial direction A of the turbine engine. The top line **264** shows the plate spring **234** in the free condition. The middle line **266** shows the plate spring **234** installed, e.g., coupled to the carrier **104** and the first seal segment **110**A. The bottom line **268** shows the plate spring **234** in an engaged position (e.g., when a pressure on an outer pressurization surface **140** of a lip **136** of a seal segment **110** increases as the turbine engine is moved from, for example, a low-power operating mode to a high-power operating mode; see, e.g., FIG. **29**).

(276) As will be appreciated, the plate spring **234** moves towards a more planar geometry when moved from the free condition to the installed condition, and further moves towards a more planar geometry when moved from the installed condition to the engaged position. For example, the plate spring **234** may define a second radius of curvature when in the installed position, with the second radius of curvature being greater than the first radius of curvature. The plate spring **234** may further define a third radius of curvature when in the engaged position, with the third radius of curvature being greater than the second radius of curvature.

(277) Installing the plate spring **234** in a pre-strained condition, may allow for a desired resistance profile for a seal support assembly of the present disclosure. For example, the plate spring **234** may not provide a constant resistance, e.g., along the radial direction R, such that by installing the plate spring **234** in a pre-strained condition allows for use of a smaller or lighter spring to achieve the desired resistance, e.g., along the radial direction R.

(278) It will be appreciated, however, that in other exemplary embodiments, the plate spring **234** may be installed in a pre-strained condition in other suitable manners. For example, referring now to FIGS. **36** and **37**, a top-down view along a radial direction R is provided of a plate spring **234** in

accordance with another exemplary embodiment of the present disclosure. FIG. **36** provides a view of the plate spring **234** in a free condition and FIG. **37** provides a view of the plate spring **234** in an installed position and/or an engaged position.

(279) The plate spring **234** of FIGS. **36** and **37** includes an outer perimeter **244** and a plurality of flex members **262** extending from the outer perimeter **244** towards an attachment ring **256** defining an attachment point defined with a seal segment. As shown, the flex members **262** do not extend all the way to the attachment point when in the free position. The flex members **262** are strained to extend to the attachment ring **256**, such that when the plate spring **234** is coupled to the first seal segment **110**A the plate spring **234** is in a pre-strained condition.

(280) It will be appreciated that coupling the plate spring **234** to the carrier **104** and the first seal segment **110**A in such a manner may provide a more desirable stiffness for the spring arrangement **114** and resistance along a radial direction R to the first seal segment **110**A, without requiring thicker and/or heavier plate springs.

(281) Briefly, it will be appreciated that although for the exemplary embodiments described above with terms to FIGS. **29** through **37**, the plate springs **234** depicted include a single attachment ring **256** defining a single attachment point with the first seal segment **110**A, in other exemplary embodiments, a plate spring **234** may be provided with a plurality of attachment rings **256** defining a plurality of attachment points. For example, referring now briefly to FIG. **38**, a plate spring **234** in accordance with another exemplary embodiment of the present disclosure is provided.

(282) The plate spring **234** generally includes an outer perimeter **244** and a plurality of flex members **262** extending from the outer perimeter **244**. The plurality of flex members **262** includes a first flex member **262**A extending from the outer perimeter **244** to a first attachment ring **256**A defining a first attachment point with the first seal segment **110**A (not shown) and a second flex member **262**B extending from the outer perimeter **244** to a second attachment ring **256**B defining a second attachment point with the first seal segment **110**A (not shown). The plate spring **234** of FIG. **38** may be coupled to the first seal segment **110**A in a similar manner as the plate spring **234** described above with reference to FIGS. **29** and **30** (e.g., the first seal segment **110**A may include a separate attachment column **248** for each attachment point).

(283) For the embodiment of FIG. **38**, the first attachment ring **256**A and first attachment point is spaced from the second attachment ring **256**B and second attachment point along an axial direction A of the turbine engine. Such may reduce or minimize a twisting of the first seal segment **110**A during operating conditions of the turbine engine.

(284) Additionally or alternatively, the first attachment point may be spaced from the second attachment point along the circumferential direction C of the gas turbine engine **10**.

(285) Referring now to FIG. **39**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **39** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** positioned between the rotor **100** and the stator **102**, and a seal support assembly **108**. The view of FIG. **15** is of a reference plane defined by an axial direction A and a radial direction R of the turbine engine. The assembly of FIG. **39** may be configured in a similar manner as one or more of the exemplary assemblies described above.

(286) For example, the seal assembly **106** includes a first seal segment **110**A and the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and the first seal segment **110**A, with the spring arrangement **114** having a plate spring **234**. More specifically, for the embodiment shown, the spring arrangement **114** includes a first plate spring **234**A and a second plate spring **234**B, each of the first plate spring **234**A and second plate spring **234**B extending between the carrier **104** and the first seal segment **110**A. The first plate spring **234**A extends between the carrier **104** and the first seal segment **110**A proximate a high-pressure side **126** of the seal assembly **106** and the second plate spring **234**B extends between the carrier **104** and the first seal segment **110**A proximate a low-pressure side **128** of the seal assembly **106**.

(287) The first plate spring **234**A and second plate spring **234**B may be configured in a similar manner as one or the exemplary plate springs **234** described above with reference to FIGS. **29** through **38**. However, for the embodiment of FIG. **39**, the first plate spring **234**A and the second plate spring **234**B extend along the radial direction R. Further, for the embodiment shown, the plate springs **234**A, **234**B are each coupled to the first seal segment **110**A at respective outer perimeters **244** of the respective plate springs **234**A, **234**B.
(288) Inclusion of the first plate spring **234**A and the second plate spring **234**B extending along the radial direction R may provide for desired stiffness for the first seal segment **110**A along the radial direction R, while also combating undesirable twist of the first seal segment **110**A.
(289) Referring now to FIG. **40**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **40** provides a perspective, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. FIG. **40** provides a cross-sectional view of the assembly in a reference plane defined by a radial direction R and an axial direction A of the turbine engine. The assembly of FIG. **40** may be configured in a similar manner as one or more of the exemplary assemblies described above.
(290) For example, in the embodiment of FIG. **40**, the seal assembly **106** includes a plurality of seal segments **110**, with the plurality of seal segments **110** including a first seal segment **110**A. Additionally, the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and the first seal segment **110**A to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between a seal face **112** and the rotor **100** during operation of the turbine engine.
(291) For the embodiment depicted, the spring arrangement **114** includes one or more elements formed of a shape memory alloy material, of a bimetallic material, or both. In particular, for the embodiment of FIG. **40**, the spring arrangement **114** includes a first spring extension **142**A formed of a shape memory alloy material. In at least one exemplary embodiment, the shape memory alloy material may be a temperature-dependent shape memory alloy material, a strain-dependent shape memory alloy material, or both.
(292) In particular, for the embodiment of FIG. **40**, the spring arrangement **114** includes the first spring extension **142**A and further includes a second spring extension **142**B. The first spring extension **142**A and the second spring extension **142**B together extend between the carrier **104** and the first seal segment **110**A in series.
(293) For example, referring briefly to FIG. **41**, a side, close-up view of the spring arrangement **114** of FIG. **40** is depicted. As is shown, the spring arrangement **114** includes the first spring extension **142**A and the second spring extension **142**B collectively extending between the carrier **104** and the first seal segment **110**A in series. For the embodiment shown, the first spring extension **142**A includes a plurality of individual leaf springs stacked along the radial direction R. The second spring extension **142**B similarly includes a plurality of individual leaf springs stacked along the radial direction R.
(294) In one exemplary embodiment, the first spring extension **142**A is formed of a first shape memory alloy material and the second spring extension **142**B is formed of a second shape memory alloy material different than the first shape memory alloy material. For example, the first shape memory alloy material may be a temperature-dependent shape memory alloy material or a strain-dependent shape memory alloy material, and the second shape memory alloy material may be the other of the temperature-dependent shape memory alloy material or the strain-dependent shape memory alloy material.
(295) Notably, in at least certain exemplary embodiments, the spring arrangement **114** may be in thermal communication with a working gas flowpath of the turbine engine (see, e.g., the embodiment described above with reference to, for example, FIGS. **1** through **3**). In particular, for

the embodiment shown, the seal assembly **106** defines a high-pressure side **126** and a low-pressure side **128**, and is exposed to a high-pressure cavity **132** at the high-pressure side **126**. The high-pressure side **126** is in fluid communication with the working gas flowpath through the high-pressure cavity **132** at a location at least partially upstream of the stator **102** (e.g., a portion of the working gas flowpath defined by a compressor, such as a high-pressure compressor, such as a downstream stage of the high-pressure compressor). In the embodiment shown, the spring arrangement **114** is in thermal communication with the high-pressure cavity **132** (which receives airflow from the working gas flowpath), is in airflow communication with the high-pressure cavity **132** (which receives airflow from the working gas flowpath), or both.

(296) Inclusion of a spring arrangement **114** in accordance with the exemplary embodiment of FIG. **40** may allow for the seal support assembly **108** to position the first seal segment **110**A to define a variable radial clearance with the rotor **100** based on an operating condition of the gas turbine engine **10**.

(297) For example, reference will now be made to FIGS. **42** through **44**. FIGS. **42** through **44** provide schematic views of the exemplary assembly of FIG. **40** at three different operating conditions. Notably, for the embodiment depicted, the first spring extension **142**A is formed of a temperature-dependent shape memory alloy material and the second spring extension **142**B is formed of a strain-dependent shape memory alloy material.

(298) FIG. **42** depicts the assembly during a nonoperating condition, such as a pre-startup operating condition of the turbine engine. During the nonoperating condition, there is little or no pressure change from a high-pressure side **126** to a low-pressure side **128** of the seal assembly **106**, and a temperature of the seal support assembly **108** is close to an ambient temperature (e.g., within 50 degrees F. of the ambient temperature). During the nonoperating condition, the first seal segment **110**A defines a first radial clearance with the rotor **100**, as is indicated by reference line **270**.

(299) FIG. **43** depicts the assembly during a high-power operating condition of the turbine engine. During the high-power operating condition, there is a relatively high-pressure change from the high-pressure side **126** to the low-pressure side **128** of the seal assembly **106** and the temperature the seal support assembly **108** is also relatively high (e.g., high relative to a temperature of the seal support assembly **108** during the nonoperating condition and during a low-power operating condition, described below). As noted, the first spring extension **142**A is formed of a temperature-dependent shape memory alloy material, such that the relatively high temperature of the spring support assembly will result in a reduction in a stiffness of the first spring extension **142**A, allowing for the first seal segment **110**A to move inwardly along the radial direction R. Similarly, the second spring extension **142**B is formed of a strain-dependent shape memory alloy material, such that the relatively high-pressure change from the high-pressure side **126** to the low-pressure side **128** of the seal assembly **106** (creating an increased pressure on the outer pressurization surface **140** of the lip **136** of the first seal segment **110**A; see FIG. **40**) may result in a reduction in a stiffness of the second spring extension **142**B, further allowing for the first seal segment **110**A to move inwardly along the radial direction R. During the high power operating condition, the first seal segment **110**A defines a second radial clearance with the rotor **100**, as is indicated by reference line **272**.

(300) FIG. **44** depicts the assembly during a low-power operating condition of the turbine engine. During the low-power operating condition, there is a relatively low-pressure change from the high-pressure side **126** to the low-pressure side **128** of the seal assembly **106** (more than during the nonoperating condition and less than during the high-power operating condition). Further, during the low-power operating condition, the temperature of the seal support assembly **108** is relatively low (e.g., low relative to the temperature of the seal support assembly **108** during the high-power operating condition, but higher than the temperature of the seal support assembly **108** during the nonoperating condition). Such operating conditions may result in an increase in the stiffness of the first spring extension **142**A and further will increase the stiffness of the second spring extension

142B relative to the high power operating condition. Such will cause the first seal segment **110**A to move outwardly along the radial direction R relative to the position and FIG. **43**. During the low-power operating condition, the first seal segment **110**A defines a third radial clearance of the rotor **100**, as is indicated by reference line **274**.

(301) As will be appreciated, the first radial clearance is greater than the second radial clearance and the third radial clearance; the second radial clearance is less than the first and third radial clearances; and the third radial clearance is less than the first radial clearance but greater than the second radial clearance.

(302) With such a configuration, the first seal segment **110**A may define the greatest radial clearance during, e.g., a startup operating mode wherein the rotor **100** is more susceptible to high vibrations. Further, the seal segment **110** may define the least radial clearance during, e.g., a high-power operating mode such as cruise, takeoff, or climb, where the rotor **100** is least susceptible to high vibrations. In such a manner, the seal support assembly **108** may reduce wear on the first seal segment **110**A, while also increasing a sealing efficiency of the first seal segment **110**A.

(303) It will be appreciated that although the embodiments described above with respect to FIGS. **40** through **44** describe the first spring extension **142**A and the second spring extension **142**B as each being formed of a shape memory alloy material, in other exemplary embodiments, one or both of the first spring extension **142**A or the second spring extension **142**B may be formed of a different material, such as a traditional metallic spring material.

(304) Further, as briefly mentioned above, in other embodiments the spring arrangement **114** may include a spring extension formed of a bimetallic material. Referring briefly to FIG. **45**, a spring arrangement **114** in accordance with another exemplary embodiment is provided having a spring extension **142**, with the spring extension **142** formed of a bimetallic material.

(305) As shown, the bimetallic material includes a first layer **276** and a second layer **278**. In at least certain exemplary embodiments, the first layer **276** may be a first shape memory alloy material and the second layer **278** may be a second shape memory alloy material. For example, in one embodiment, the first layer **276** may be a strain-dependent shape memory alloy material and the second layer **278** may be a temperature-dependent shape memory alloy material. Additionally, or alternatively, one or both of the first layer **276** or second layer **278** may be formed of a metallic material having a different coefficient of thermal expansion than the other of the first layer **276** or second layer **278**.

(306) Further, referring to FIG. **46**, the bimetallic material include one or more additional layers. For example, the bimetallic material may include a high temperature substrate **280**, a shape memory alloy material layer **282**, and optionally a high temperature coating **284**. It will be appreciated that the view of FIG. **46** shows the bimetallic material in an assembly position (e.g., the straight orientation) as well as in an operating condition (e.g., the bent orientation).

(307) The bimetallic materials depicted in FIG. **45** and FIG. **46** may be incorporated into one or more of the spring extensions **142** described hereinabove.

(308) Referring now to FIGS. **47** and **48**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **47** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. FIG. **47** provides a view of the assembly in a first position, and FIG. **48** provides a view of the assembly and a second position. The assembly of FIGS. **47** and **48** may be configured in a similar manner as one or more of the exemplary assemblies described above.

(309) For example, the seal assembly **106** includes a first seal segment **110**A and the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and the first seal segment **110**A to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control

of a radial clearance gap defined between a seal face **112** and the rotor **100** during operation of the turbine engine.
(310) In particular, the carrier **104** includes a carrier axial extension **170** and the first seal segment **110**A similarly includes a segment axial extension **218** located outward of the carrier axial extension **170**. The spring arrangement **114** extends between the carrier **104** and the first seal segment **110**A, and more specifically between the carrier axial extension **170** and the segment axial extension **218**. Accordingly, increasing a radial height of the spring arrangement **114** moves the first seal segment **110**A farther away from the rotor **100** along the radial direction R and conversely, decreasing a radial height of the spring arrangement **114** moves the first seal segment **110**A closer to the rotor **100** along the radial direction R. The configuration of FIG. **47** depicts the assembly at a low-power operating condition of the turbine engine and FIG. **48** depicts the assembly at a high-power operating condition of the turbine engine. As will be appreciated, an increased pressure at the outer pressurization surface **140** of the lip **136** of the first seal segment **110**A may move the first seal segment **110**A from the position shown in FIG. **47** to the position shown in FIG. **48**.
(311) Referring now to FIGS. **49** and **50**, a first embodiment of the spring arrangement **114** described with reference to FIGS. **47** and **48** is provided. For the embodiment shown, the spring arrangement **114** includes a first cam **286**A and a second cam **286**B. The first cam **286**A and the second cam **286**B are rotatable about a common rotational axis **288**. The spring arrangement **114** includes a spring extension **142** extending between the first cam **286**A in the second cam **286**B.
(312) The first cam **286**A and the second cam **286**B are rotatably coupled to one of the first seal segment **110**A or the carrier **104**, and the first cam **286**A and the second cam **286**B extend to and engage with a surface of the other of the first seal segment **110**A or the carrier **104**. In particular, for the embodiment depicted the first cam **286**A and the second cam **286**B are rotatably coupled to the carrier axial extension **170** of the carrier **104**, and extend to and engage with a surface of the segment axial extension **218** of the first seal segment **110**A.
(313) Notably, the spring extension **142** is configured to bias the first cam **286**A towards the second cam **286**B, decreasing an angle therebetween. Accordingly, for the embodiment of FIG. **49**, the turbine engine may be in a low power operating condition, and for the embodiment of FIG. **50**, the turbine engine may be in a high-power operating condition.
(314) Referring now to FIGS. **51** and **52**, another embodiment of the spring arrangement **114** described above with reference to FIGS. **47** and **48** is provided. For the embodiment of FIGS. **51** and **52**, the spring arrangement **114** includes a cam **286** extending between a first end **290** and a second end **292**. The cam **286** defines the rotational axis **288** and the spring arrangement **114** includes a first spring extension **142**A and a second spring extension **142**B. The first spring extension **142**A extends from the cam **286** at the first end **290** of the cam **286** or between the first end **290** of the cam **286** and the rotational axis **288** of the cam **286**. The second spring extension **142**B extends from the cam **286** at the second end **292** of the cam **286** or between the second end **292** of the cam **286** and the rotational axis **288**. The first spring extension **142**A and the second spring extension **142**B are further coupled to one of the carrier **104** or the first seal segment **110**A. In the embodiment shown, the first spring extension **142**A and the second spring extension **142**B are each further coupled to the carrier **104**.
(315) Notably, the first spring extension **142**A and the second spring extension **142**B are configured to bias the cam **286** towards a radial orientation, pushing the first seal segment **110**A outward along the radial direction (see, e.g., FIGS. **47** and **48**). Accordingly, for the embodiment of FIG. **51**, the turbine engine may be in a low power operating condition, and for the embodiment of FIG. **52**, the turbine engine may be in a high-power operating condition.
(316) Referring now to FIG. **53**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **53** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a seal assembly **106** configured to be positioned between the carrier **104** and a rotor **100** (not shown, and

a seal support assembly **108**. The assembly of FIG. **53** may be configured in a similar manner as one or more of the exemplary assemblies described above.
(317) For example, the seal assembly **106** includes a first seal segment **110**A and the seal support assembly **108** includes a spring arrangement **114** extending between the carrier **104** and the first seal segment **110**A to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** (not depicted) during operation of the turbine engine.
(318) In particular, for the embodiment depicted, the spring arrangement **114** includes a spring extension **142** arranged in tension, such that a lengthening of the spring extension **142** results in movement of the first seal segment **110**A inwardly along the radial direction R and closer to the rotor **100**.
(319) Referring now to FIG. **54**, a close-up view of the spring arrangement **114** of FIG. **53** is provided. The exemplary spring arrangement **114** of FIG. **54** generally includes a first body extension **294** fixed along the radial direction R relative to the carrier **104** and a second body extension **296** fixed along the radial direction R relative to the first seal segment **110**A. In particular, for the embodiment depicted, the first body extension **294** is coupled to the carrier **104** and the second body extension **296** is coupled to the first seal segment **110**A. The first body extension **294** is coupled to the carrier **104** through one or more mechanical fasteners **297**, which may be, e.g., a bolt.
(320) In the embodiment shown, the first body extension **294** defines a complementary geometry to the second body extension **296**, such that the first body extension **294** is slidable relative to the second body extension **296**. For example, the first body extension **294** may define a cylindrical external surface **298** and the second body extension **296** may have a cylindrical internal surface **300**, with the cylindrical external surface **298** having an outer diameter slightly less than an inner diameter of the cylindrical internal surface **300**.
(321) Further, for the embodiment shown, the spring extension **142** is a helical spring arranged coaxially with the first body extension **294** and the second body extension **296**. Additionally, the spring arrangement **114** includes a first spring retention member **302** at one end and a second spring retention member **304** at an opposite end. The first spring retention member **302** may be coupled to the carrier **104**, the first body extension **294**, or both. The second spring retention member **304** may be coupled to the first seal segment **110**A, the second body extension **296**, or both.
(322) Further, it will be appreciated that for the embodiment depicted, the spring arrangement **114** defines a movable attachment point with the first seal segment **110**A. In particular, the spring arrangement **114** forms a spherical joint **306** with the first seal segment **110**A for connecting the spring arrangement **114** to the first seal segment **110**A.
(323) Although the spring extension **142** is depicted as a uniform helical spring in the embodiment of FIGS. **53** and **54**, in other embodiments, the spring extension **142** may have any other suitable shape. For example, in other embodiments the spring extension **142** may be configured as a bellows, is a compound spring, as a nonlinear spring, etc.
(324) Referring now briefly to FIG. **55**, a cross-sectional view of the seal assembly **106** and seal support assembly **108** of FIGS. **53** and **54** is provided. As will be appreciated, for the embodiment depicted the seal assembly **106** includes a plurality of seal segment **110** spaced along the circumferential direction C and the seal support assembly **108** includes a plurality of spring arrangements **114**, each spring arrangement **114** extending between a respective one of the plurality of seal segments **110** and the carrier **104** for supporting the respective one of the plurality of seal segments **110** along the radial direction R.
(325) Referring now to FIGS. **56**A and **56**B, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **56**A provides a schematic, cross-sectional

view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. FIG. **56**A provides a view of the assembly in a first position, and FIG. **56**B provides a view of the assembly and a second position. The assembly of FIGS. **56**A and **56**B may be configured in a similar manner as one or more of the exemplary assemblies described above.

(326) For example, the seal assembly **106** includes a first seal segment **110**A, the first seal segment **110**A including a seal face **112** configured to form a fluid bearing with the rotor **100**. Additionally, the seal support assembly **108** extends between the carrier **104** and the first seal segment **110**A to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.

(327) For the embodiment of FIGS. **56**A and **56**B, the seal support assembly **108** includes a flexible extension **308** and a driver extension **310**. The driver extension **310** is coupled to and extends from the flexible extension **308**. The driver extension **310** is formed of a first material, the first material being: (a) a material defining a different coefficient of thermal expansion than the material forming the driver extension **310**; (b) a shape memory alloy material; (c) a bimetallic material; or (d) a combination thereof. The driver extension **310** is positioned to move the flexible extension **308** during an operation of the turbine engine. More specifically, as will be explained in more detail below, the driver extension **310** is positioned to move the flexible extension **308** to correspondingly move the first seal segment **110**A during operation of the turbine engine based on, e.g., an operating condition of the turbine engine (e.g., a temperature or a pressure change across the seal assembly **106**).

(328) More specifically, for the embodiment depicted, the flexible extension **308** is a first flexible extension **308**A and the seal support assembly **108** further includes a second flexible extension **308**B. The driver extension **310** extends between the first flexible extension **308**A and the second flexible extension **308**B.

(329) More specifically, still, the seal support assembly **108** further includes an outer support section **312** along the radial direction R of the turbine engine and an inner support section **314** along the radial direction R of the turbine engine. The outer support section **312** is coupled to or formed integrally with the carrier **104** and the inner support section **314** is coupled to or formed integrally with the first seal segment **110**A, as will be explained in more detail below. The driver extension **310** extends between a first connection point **316** with the first flexible extension **308**A and a second connection point **318** with the second flexible extension **308**B. The seal support assembly **108** defines an extension axis **320** along the radial direction R. The extension axis **320** indicates a direction of travel along the radial direction R of the first seal segment **110**A provided by the seal support assembly **108**. The first flexible extension **308**A and the second flexible extension **308**B are coupled to the outer support section **312** at one or more locations closer to the extension axis **320** than the first connection point **316** and the second connection point **318**.

(330) Referring still to the embodiment FIGS. **56**A and **56**B, the first material defines a coefficient of thermal expansion less than a coefficient of thermal expansion of the material forming the first flexible extension **308**A and the second flexible extension **308**B. For example, in certain exemplary embodiments, the first material may be a shape memory alloy material or metal or metal alloy having a coefficient of thermal expansion less than the coefficient of thermal expansion of the material forming the first flexible extension **308**A and the second flexible extension **308**B.

(331) Notably, in at least certain exemplary embodiments, the seal support assembly **108** may be in thermal communication with a working gas flowpath of the turbine engine (see, e.g., the embodiment described above with reference to, e.g., FIGS. **1** through **3**). In particular, for the embodiment shown, the seal assembly **106** defines a high-pressure side **126** and a low-pressure side **128**, and is exposed to a high-pressure cavity **132** at the high-pressure side **126**. The high-pressure

side **126** is in fluid communication with the working gas flowpath through the high-pressure cavity **132** at a location at least partially upstream of the stator **102** (e.g., a portion of the working gas flowpath defined by a compressor, such as a high-pressure compressor, such as a downstream stage of the high-pressure compressor). In the embodiment shown, the seal support assembly **108** is in thermal communication with the high-pressure cavity **132** (which receives airflow from the working gas flowpath), is in airflow communication with the high-pressure cavity **132** (which receives airflow from the working gas flowpath), or both. In such a manner, it will be appreciated that the configuration of FIGS. **56**A and **56**B allows for movement of the first seal segment **110**A relative to the rotor **100** along the radial direction R to be controlled passively based on an operating temperature of the turbine engine.

(332) For example, referring particularly to FIG. **56**A, the assembly is depicted during, e.g., a low-power operating condition of the gas turbine engine **10**. During this operating condition, a temperature of a working gas through the working gas flowpath through which the stator **102** is positioned may be relatively low, allowing for the first seal segment **110**A to be held at a radially outer position relative to the rotor **100**, such that the seal face **112** of the first seal segment **110**A defines a relatively large radial clearance with the rotor **100**.

(333) By contrast, referring particularly to FIG. **56**B, the assembly is depicted during, e.g., a high-power operating condition. During this operating condition, the temperature of the working gas through the working gas flowpath through which the stator **102** is positioned may be relatively high. With the increase in temperature, in combination with the lower coefficient of thermal expansion of the driver extension **310** relative to the first flexible extension **308**A and second flexible extension **308**B in the configuration depicted and described above, the first seal segment **110**A is moved inwardly along the radial direction R to reduce the radial clearance between the seal face **112** of the first seal segment **110**A and the rotor **100**.

(334) In such a manner, the assembly depicted in FIGS. **56**A and **56**B may provide for a passive control of the radial clearance between the seal face **112** of the first seal segment **110**A and the rotor **100** based on a temperature of a working gas through a working gas flowpath of the engine.

(335) Briefly, referring still to FIGS. **56**A and **56**B, as noted above, the seal support assembly **108** is coupled to the first seal segment **110**A. In particular, the seal support assembly **108** includes an inner support section **314** coupled to the first seal segment **110**A through a hanger attachment. In particular, the first seal segment **110**A includes a hanger **324** having an axial ledge **326** spaced from a body **138** of the first seal segment **110**A and defining an opening **328**. The hanger **324** further defines a cavity **330** between the axial ledge **326** in the body **138** of the first seal segment **110**A. The inner support section **314** of the seal support assembly **108** is positioned within the cavity **330** and defines a length in the axial direction A wider than a length of the opening **328**. In such a manner, the inner support section **314** of the seal support assembly **108** may be coupled to the first seal segment **110**A, which may allow for the seal support assembly to maintain a radial position of the first seal segment **110**A.

(336) Notably, for the embodiment depicted, the seal support assembly **108** includes a leaf spring support **332** positioned between the axial ledge **326** and the inner support section **314**. In such a manner, the leaf spring support **332** may couple the inner support section **314** (and, e.g., the first flexible extension **308**A, the second flexible extension **308**B, and the driver extension **310**) to the first seal segment **110**A. The leaf spring support **332** may allow for some movement of the first seal segment **110**A along the radial direction R as a result of, e.g., a pressure change from a high-pressure side **126** of the seal assembly **106** to a low-pressure side **128** of the seal assembly **106**. In such manner, the leaf spring support **332** may allow for a tighter radial clearance during the high-pressure operation of the turbine engine.

(337) It will be appreciated, however, that in other exemplary embodiments, the seal support assembly **108** may be configured in other suitable manners. For example, referring now to FIG. **57**, a seal support assembly **108** in accordance with another exemplary embodiment of the present

disclosure is provided. The seal support assembly **108** of FIG. **57** may be configured in a similar manner as exemplary seal support assembly **108** described above with reference to FIGS. **56**A and **56**B.
(338) However, for the embodiment of FIG. **57**, a first flexible extension **308**A and a second flexible extension **308**B are coupled to an outer support section **312** at one or more locations farther from an extension axis **320** than a first connection point **316** and a second connection point **318** of the driver extension **310**. For example, for the embodiment of FIG. **57**, the first flexible extension **308**A and the second flexible extension **308**B each includes a converging section **334** and a diverging section **336**, each with respect to the extension axis **320**. The driver extension **310** is coupled to the first flexible extension **308**A at a location between the converging section **334** and the diverging section **336** of the first flexible extension **308**A. The driver extension **310** is further coupled to the second flexible extension **308**B at a location between the converging section **334** and the diverging section **336** of the second flexible extension **308**B.
(339) In such a manner, it will be appreciated that a first material forming the driver extension **310** defines a coefficient of thermal expansion greater than a coefficient of thermal expansion of a material forming the first flexible extension **308**A and the second flexible extension **308**B. Accordingly, an increase in a temperature of, e.g., a working gas through a working gas flowpath through which the stator **102** is positioned will create an inward movement of the first seal segment **110**A in a manner similar to as described above with reference to FIGS. **56**A and **56**B.
(340) It will be appreciated, however, that in other exemplary embodiments, the seal support assembly **108** may be configured in still other suitable manners. For example, referring generally to FIGS. **58** through **61** and **64** through **69**, various alternative exemplary embodiments of a seal support assembly **108** in accordance with exemplary aspects of the present disclosure are provided. Each of these exemplary embodiments may be configured in a similar manner as the exemplary seal support assembly **108** described above with terms to, e.g., FIGS. **56**A through **57**.
(341) For example, with reference to FIGS. **58** and **59**, a seal support assembly **108** is depicted in a first, low power operating edition (FIG. **58**) and a second, high-power operating condition (FIG. **59**). For the embodiment of FIGS. **58** and **59**, a driver extension **310** of the seal support assembly **108** is arranged in series with a flexible extension **308**. More specifically, the seal support assembly **108** includes a first driver extension **310**A and a first flexible extension **308**A arranged in series, as well as a second driver extension **310**B and a second flexible extension **308**B also arranged in series.
(342) With reference to FIGS. **60** and **61**, a seal support assembly **108** is again depicted in a first, low power operating condition (FIG. **60**) and a second, high-power operating condition (FIG. **61**). The embodiment of FIGS. **60** and **61** includes a first flexible extension **308**A having a converging section **334** and a diverging section **336**, and similarly, a second flexible extension **308**B includes a converging section **334** and a diverging section **336**. For the exemplary seal support assembly **108** depicted, the seal support assembly **108** further includes a first support member **338**A extending between the converging section **334** and diverging section **336** of the first flexible extension **308**A as well as a second support member **338**B extending between the converging section **334** and diverging section **336** of the second flexible extension **308**B.
(343) In at least certain exemplary embodiments, the first support member **338**A, the second support member **338**B, or both may be configured as a bimetallic member. For example, referring to FIGS. **62** and **63**, a bimetallic member in accordance with an exemplary aspect of the present disclosure is depicted. The bimetallic member includes a first layer **276** and a second layer **278**. The first layer **276** is formed of a first material having a coefficient of thermal expansion greater than a material forming the second layer **278**. In such manner, when a temperature of the bimetallic member is increased (see FIG. **63**) a length of the bimetallic member may be affected. In certain embodiments, the length of the bimetallic member may be increased, and in another exemplary embodiment, the length of the bimetallic member may be decreased, as a result of being exposed to

an increase in temperature.

(344) Referring now to FIGS. **64** and **65**, a seal support assembly **108** is again depicted in a first, low power operating condition (FIG. **64**) and a second, high-power operating condition (FIG. **65**). The embodiment of FIGS. **64** and **65** may be configured in substantially the same manner as the embodiment described above with reference to FIG. **57**. However, for the embodiment of FIGS. **64** and **65**, the driver extension **310** of the seal support assembly **108** extending between a first flexible extension **308**A and a second flexible extension **308**B is formed of a bimetallic member. For the embodiment depicted, the bimetallic number may be configured to expand in length when exposed to elevated temperatures relative to the first flexible extension **308**A and the second flexible extension **308**B (see FIG. **68** relative to FIG. **64**).

(345) Referring now to FIGS. **66** and **67**, yet another exemplary embodiment of a seal support assembly **108** is provided in accordance with an exemplary aspect of the present disclosure. The exemplary embodiment of FIGS. **66** and **67** may be configured in substantially the same manner as the exemplary embodiment described above with reference to FIGS. **60** and **61**. For the embodiment of FIGS. **66** and **67**, the first support member **338**A and second support member **338**B are each configured as bimetallic members. Further, for the embodiment depicted, the first support member **338**A and the second support member **338**B are further configured as a first driver extension **310**A and a second driver extension **310**B.

(346) Referring now to FIG. **68**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **68** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. The assembly of FIG. **68** may be configured in a similar manner as one or more of the exemplary assemblies described above.

(347) For example, the seal assembly **106** of the exemplary embodiment of FIG. **68** includes a first seal segment **110**A, the first seal segment **110**A having a seal face **112** configured to form a fluid bearing with the rotor **100**. In at least certain exemplary embodiments, the first seal segment **110**A may be one of a plurality of seal segments **110** of the seal assembly **106** arranged, e.g., along the circumferential direction C of the turbine engine.

(348) However, the seal support assembly **108** for the embodiment shown includes a torsional spring extension **340** extending from the carrier **104**, from the first seal segment **110**A, or both to bias the first seal segment **110**A along the radial direction R. More specifically, it will be appreciated that the seal assembly **106** defines a high-pressure side **126** and a low-pressure side **128**. Additionally, the first seal segment **110**A includes a body **138** and a lip **136** extending from the body **138** along an axial direction A of the turbine engine on the high-pressure side **126** of the seal assembly **106**. The lip **136** includes an outer pressurization surface **140** along the radial direction R of the turbine engine. The torsional spring extension **340** of the seal support assembly **108** is provided to at least partially counter a pressure on the outer pressurization surface **140** during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.

(349) More specifically, referring particularly to FIG. **69**, a perspective view of a portion of the seal support assembly **108** of FIG. **68** is provided. As noted, the seal support assembly **108** includes the torsional spring extension **340** extending from the carrier **104**, from the first seal segment **110**A, or both. In particular, for the embodiment shown, the torsional spring extension **340** extends between a base **342** and a distal end **344**. The base **342** of the torsional spring extension **340** is coupled to the carrier **104**. For the embodiment shown, the seal support assembly **108** further includes a cam **346** coupled to the distal end **344** of the torsional spring extension **340**.

(350) More specifically, referring back to FIG. **68**, it will be appreciated that the first seal segment **110**A includes a support extension **348** extending at least partially along the axial direction A of the turbine engine. The support extension **348** defines an inner support surface **350** along the radial

direction R, and the cam **346** is positioned to engage the inner support surface **350**.
(351) More specifically, for the embodiment of FIG. **68**, it will be appreciated that the torsional spring extension **340** is a first torsional spring extension **340**A positioned at the high-pressure side **126** of the seal assembly **106**, and that the seal support assembly **108** further includes a second torsional spring extension **340**B positioned at the low-pressure side **128**. Further, the cam **346** is a first cam **346**A coupled to the distal end **344** of the first torsional spring extension **340**A (see, e.g., FIG. **69**), and the seal support assembly **108** further includes a second cam **346**B coupled to a distal end **344** of the second torsional spring extension **340**B (see, e.g., FIG. **69**). The second torsional spring extension **340**B is coupled at a base **342** to the carrier **104** (see, e.g., FIG. **69**).
(352) Additionally, the support extension **348** is a first support extension **348**A positioned at the high-pressure side **126** of the seal assembly **106**, and the first seal segment **110**A further includes a second support extension **348**B positioned at the low-pressure side **128** of the seal assembly **106**. The second support extension **348**B similarly includes an inner support surface **350**. The first cam **346**A is positioned to engage the inner support surface **350** of the first support extension **348**A and the second cam **346**B is positioned to engage the inner support surface **350** of the second support extension **348**B.
(353) Such a configuration may prevent or minimize a twisting of the first seal segment **110**A during operation of the turbine engine.
(354) Referring now to FIG. **70**, an aft-looking-forward, schematic view of a portion of the seal support assembly **108** of FIG. **68** is provided along Line **70-70**.
(355) As will be appreciated from the view of FIG. **70**, the seal support assembly **108** further includes a third torsional spring extension **340**C spaced from the first torsional spring extension **340**A along the circumferential direction C of the turbine engine. The seal support assembly **108** includes a third cam **346**C coupled to the third torsional spring extension **340**C. The third cam **346**C is positioned to engage the inner support surface **350** of the support extension **348** of the first seal segment **110**A.
(356) As will be appreciated from the arrows **352** in FIG. **70**, the first cam **346**A and the third cam **346**C are each configured to rotate in a common circumferential direction C in the aft-looking-forward view of FIG. **70**. More specifically, the first torsional spring extension **340**A defines a first extension axis **354**A and the third torsional spring extension **340**C defines a third extension axis **354**C. The first cam **346**A is configured to rotate about the first extension axis **354**A and the third cam **346**C is configured to rotate about the third extension axis **354**C, each in a clockwise direction in response to an increase in downward force by the first seal segment **110**A in the aft-looking-forward view of FIG. **70**.
(357) Briefly referring back to FIG. **68**, it will be appreciated that the seal support assembly **108** may further include a fourth torsional spring extension and a forth cam (not shown) spaced from the second torsional spring extension **340**B and second cam **346**B along the circumferential direction C, arranged in a similar manner as the first torsional spring extension **340**A and third torsional spring extension **340**C described above with reference to FIG. **70**.
(358) As will be appreciated, the seal support assembly **108** may be configured to passively control a radial gap defined between the seal face **112** of the first seal segment **110**A and the rotor **100** during operation of the turbine engine based on, e.g., an operating condition of the turbine engine. More specifically, for the embodiment of FIGS. **68** through **70**, the seal support assembly **108** is configured to passively control the radial gap based on a pressure on the outer pressurization surface **140** of the lip **136** of the first seal segment **110**A, which may create a higher amount of inward force along the radial direction R on the first seal segment **110**A.
(359) As will be appreciated, during a first operating condition, such as a pre-startup operating condition, a pressure may be relatively low or zero. During a second operating condition, such as a low-power operating condition, the pressure may be higher than during the first operating condition. Further, during a third operating condition, such as a high-power operating condition, the

pressure may be higher than during the first operating condition and the second operating condition.

(360) The view depicted in FIGS. **68** and **70** may be representative of a position of the first seal segment **110**A and seal support assembly **108** during the first operating condition. In this configuration, the radial clearance gap between the rotor **100** and the seal face **112** may be relatively large to, e.g., accommodate anticipated vibrations, rotor bow, etc.

(361) Referring to FIGS. **71** and **72**, the above embodiment is depicted in a position representative of the first seal segment **110**A and the seal support assembly **108** during the second operating condition. In this configuration, the radial clearance gap may be less than shown in FIG. **68**, but still large enough to accommodate anticipated vibrations at the low-power operating condition without creating a rub between the first seal segment **110**A and the rotor **100**.

(362) Referring to FIG. **73**, the above embodiment is depicted in a position representative of the first seal segment **110**A and the seal support assembly **108** during the third operating condition. In this configuration, the radial clearance gap may be less than the radial clearance gap shown in FIG. **71**, such that the turbine engine may operate at an increased efficiency during this operating condition where a relatively low amount of vibration is expected.

(363) It will be appreciated, however, that the embodiment described above with reference to FIGS. **68** through **73** is provided by way of example only. In other exemplary embodiments, the assembly may have any other suitable configuration.

(364) For example, referring to FIGS. **74** and **75**, two additional embodiments of a seal support assembly **108** of the present disclosure are provided. Each of the seal support assemblies **108** in FIGS. **74** and **75** may generally include a first torsional spring extension **340**A and a first cam **346**A coupled to the first torsional spring extension **340**A, as well as a third torsional spring extension **340**C and a third cam **346**C coupled to the third torsional spring extension **340**C. However, for the embodiments shown, and as is indicated by the directional arrows **352**, the first cam **346**A and third cam **346**C are configured to rotate in opposite circumferential directions in the aft-looking-forward views depicted.

(365) In particular, with reference to FIG. **74**, the first cam **346**A and third cam **346**C are configured to rotate in circumferential directions toward one another, and referring to FIG. **75**, the first cam **346**A and third cam **346**C are configured to rotate in circumferential directions away from one another.

(366) Moreover, it will be appreciated that in the embodiments described above, each cam **346**, such as the first cam **346**A, generally defines an outer engagement surface **358** (see, e.g., FIG. **70**). In the embodiments described above, the outer engagement surface **358** has a pear shape.

(367) However, in other example embodiments, the seal support assembly **108** may be configured in any other suitable manner. For example, referring to FIGS. **76** through **78**, a seal support assembly **108** in accordance with another exemplary embodiment is depicted. The seal support assembly **108** generally includes a torsional spring extension **340** extending from the carrier **104** and a cam **346** coupled to the torsional spring extension **340**. The cam **346** is positioned to engage in inner support surface **350** of a support extension **348** of a first seal segment **110**A.

(368) In the embodiment of FIGS. **76** through **78**, the cam **346** defines an outer engagement surface **358**, with the outer engagement surface **358** having a non-pear shape. In particular, the engagement surface includes a first bump **360**, a second bump **362**, and a flat **364**. FIG. **76** depicts the first bump **360** of the engagement surface contacting the inner support surface **350** of the support extension **348**, FIG. **77** depicts the second bump **362** of the engagement surface contacting the inner support surface **350** of the support extension **348**, and FIG. **78** depicts the flat **364** of the engagement surface contacting the inner support surface **350** of the support extension **348**.

(369) Inclusion of the outer engagement surface **358** having a nonuniform transition across a length of the outer engagement surface **358** may allow for the seal support assembly **108** to provide a various amount of resistance along the radial direction R based on a relative radial position of the

first seal segment **110**A to the carrier **104**. In such a manner, the seal support assembly **108** may bias the first seal segment **110**A towards one or more predetermined positions along the radial direction during operation of the turbine engine (e.g., corresponding to three separate operating conditions of the turbine engine).

(370) Further, in still other exemplary embodiments, the seal support assembly **108** may include in the other suitable configuration to provide a desired resistance along the radial direction R. For example, referring now to FIGS. **79** through **81**, a seal support assembly **108** in accordance with another exemplary embodiment of the present disclosure is provided having a torsional spring extension **340** and a cam **346** coupled to a distal end **344** of the torsional spring extension **340**. However, for the embodiment of FIGS. **79** through **80**, the cam **346** is a conjugate cam. The conjugate cam includes a first cam member **366** and a second cam member **368**, with the first cam member **366** and second cam member **368** defining different shapes, different orientations, or both. FIG. **79** depicts the conjugate cam in a first position, FIG. **86** the conjugate cam in a second position, and FIG. **81** depicts the conjugate cam in a third position. The conjugate cam may similarly allow for the seal support assembly **108** to provide a various amount of resistance along the radial direction R based on a radial position of the first seal segment **110**A relative to the carrier **104**. In such a manner, the seal support assembly **108** may bias the first seal segment **110**A towards one or more predetermined radial positions during operation of the turbine engine.

(371) Moreover, in still other exemplary embodiments, the seal support assembly **108** may include other suitable configurations to provide a desired resistance along the radial direction R. For example, referring briefly to FIG. **82**, a seal support assembly **108** in accordance with another exemplary embodiment of the present disclosure is provided.

(372) For the embodiment of FIG. **82**, the seal support assembly **108** includes a first torsional spring extension **340**A and a first cam **346**A coupled to a distal end of the first torsional spring extension **340**A, and a second torsional spring extension **340**B and a second cam **346**B coupled to a distal end **344** of the second torsional spring extension **340**B. Moreover, for the embodiment depicted, a first seal segment **110**A of the seal assembly **106** includes a first pad **370** defining an inner support surface **350** along the radial direction R and a second pad **372** also defining an inner support surface **350** along the radial direction R. The first cam **346**A is positioned to engage with the inner support surface **350** of the first pad **370** and the second cam **346**B is positioned to engage with the inner support surface **350** of the second pad **372**. Notably, the inner support surface **350** of the first pad **370** and the inner support surface **350** of the second pad **372** are nonlinear, and instead defined a curve. Such a configuration may further provide for nonlinear force based on a radial displacement of the first seal segment **110**A.

(373) Moreover, for the embodiment of FIG. **82**, it will be appreciated that the seal support assembly **108** is mounted in a prestressed condition (e.g., a spring extension is mounted such that at a neutral state of the seal support assembly **108**, the spring extension is in a stressed state, such as compressed, tensioned, torqued, etc.). More specifically, the seal support assembly **108** includes a compression plate **374** configured to pre-compress the first torsional spring extension **340**A and the second torsional spring extension **340**B. A plurality of mounting bolts **376** and fasteners **378** are provided to maintain the seal support assembly **108** in the prestressed condition. The compression plate **374** may be a similar manner as the compression plate described below.

(374) Referring now to FIGS. **83** and **84**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **83** provides a forward-looking-aft, perspective view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. FIG. **84** provides a close-up of a portion of the assembly of FIG. **83**. The assembly of FIGS. **83** and **84** may be configured in a similar manner as one or more of the exemplary assemblies described above.

(375) For example, the seal assembly **106** includes a plurality of seal segments **110** supported at

least in part by the carrier **104**. The plurality of seal segments **110** includes a first seal segment **110**A and a second seal segment **110**B, each of the first seal segment **110**A and the second seal segment **110**B having a seal face **112** (not visible in the views of FIGS. **83** and **84**) configured to form a fluid bearing with the rotor **100**.

(376) The seal support assembly **108** includes, for the embodiment shown, a tangential spring extension **380** extending between the first seal segment **110**A and the second seal segment **110**B for biasing the first seal segment **110**A away from the second seal segment **110**B in a circumferential direction C of the turbine engine.

(377) In particular, referring particularly to FIG. **83**, it will be appreciated that the first seal segment **110**A extends between a first circumferential end **382** and a second circumferential end **384**, and the second seal segment **110**B similarly extends between a first circumferential end **386** and a second circumferential end **388**. The tangential spring extension **380** is coupled to the first seal segment **110**A proximate a second circumferential end **384** and to the second seal segment **110**B proximate the first circumferential end **386**.

(378) More particularly, referring particularly to FIG. **84**, it will be appreciated that the tangential spring extension **380** is configured as a bent plate spring. The bent plate spring extends between a first end **390** and a second end **392**, and includes a middle section **394**, a first segment **396** extending between the middle section **394** in the first end **390** and a second segment **398** extending between the middle section **394** and the second end **392**. The tangential spring extension **380** is fixedly coupled to the first seal segment **110**A at the first end **390** and to the second seal segment **110**B at the second end **392** using mechanical fasteners **400**. Each mechanical fastener **400** may be one or more of a pin, a screw, a bolt, a rivet, or the like.

(379) Further still, it will be appreciated that for the embodiment depicted, the tangential spring extension **380** defines in part a first point of contact with the first seal segment **110**A and a second point of contact with the second seal segment **110**B. The tangential spring extension **380** defines an axial length **402** along an axial direction A at the first point of contact and at the second point of contact to prevent or minimize a twisting of the first seal segment **110**A in the second seal segment **110**B during operation of the turbine engine. For example, as will be appreciated from the description herein above, the seal face **112** of the first seal segment **110**A further defines an axial length (see, e.g., axial length **180** depicted in FIG. **15**). The axial length **402** of the tangential spring extension **380** at the first point of contact and at the second point of contact is between 5% and 100% of the axial length of the seal face **112** of the first seal segment **110**A.

(380) Referring now also to FIG. **85**, a schematic, forward-looking-aft view of the assembly in FIG. **84** is provided. As will further be appreciated from the view of FIG. **85**, the tangential spring extension **380** is further engaged with the carrier **104** to support the seal assembly **106**, and more specifically to support the first seal segment **110**A and the second seal segment **110**B along the radial direction R. In the embodiment shown, the tangential spring extension **380** further includes a pin **404** fixedly or slidably coupled to the carrier **104** and the tangential spring extension **380** is engaged with the pin **404**. In particular, for the embodiment of FIG. **85**, the seal support assembly **108** includes a radial spring **406** and the tangential spring extension **380** is engaged with the pin **404** through the radial spring **406**. For example, for the embodiment depicted, the pin **404** is fixedly coupled to the carrier **104** and includes a pinhead **408** spaced from the carrier **104**. The middle section **394** of the tangential spring extension **380** is positioned outward of the pinhead **408** along the radial direction R, and the radial spring **406** extends between the middle section **394** of the tangential spring extension **380** and the pinhead **408** to bias the tangential spring extension **380** outwardly along the radial direction R. In such a manner, it will be appreciated that the radial spring **406** is positioned inward of the carrier **104** along the radial direction R for the embodiment shown.

(381) Further, for the embodiment shown, the middle section **394** includes a ledge **410** positioned inward of the pinhead **408** of the pin **404** along the radial direction R, to act as a deflection limiter

to limit a radially outward movement of the first segment **396** and the second segment **398** during operation of the turbine engine.
(382) It will be appreciated that the seal support assembly **108** may be configured to maintain a circumferential spacing of the plurality of seal segments **110** of the seal assembly **106** during operation of the turbine engine. Further, the seal support assembly **108** may counter a pressure on an outer pressurization surface **140** of the seal segments **110** during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine (see, e.g., FIG. **4**).
(383) It will be appreciated, however, that in other exemplary embodiments, the seal support assembly **108** may have any other suitable configuration.
(384) For example, referring briefly to FIG. **86**, a schematic, forward-looking-aft view of an assembly in accordance with another exemplary embodiment of the present disclosure is provided. The assembly of FIG. **86** may be configured in substantially the same manner as the assembly of FIG. **85**, described above. However, for the embodiment of FIG. **86**, a first end **390** of a tangential spring extension **380** provided may not be fixedly coupled to a second circumferential end **384** of a first seal segment **110**A and similarly a second end **392** of the tangential spring extension **380** may not be fixedly coupled to a first circumferential end **386** of a second seal segment **110**B. Instead, for the embodiment of FIG. **86**, the first seal segment **110**A defines a tangential slot **412** at the second circumferential end **384** adjacent to the second seal segment **110**B, and the second seal segment **110**B defines a tangential slot **414** at the first circumferential end **386** adjacent to the first seal segment **110**A. The first end **390** of the exemplary tangential spring extension **380** depicted is positioned in the tangential slot **412** of the first seal segment **110**A, and the second end **392** of the tangential spring extension **380** is positioned in the tangential slot **414** of the second seal segment **110**B. In such a manner, the tangential spring extension **380** may be slidably coupled to the first seal segment **110**A and to the second seal segment **110**B.
(385) Furthermore, referring briefly to FIG. **87**, a tangential spring extension **380** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary tangential spring extension **380** of FIG. **87** may be configured in substantially the same manner as the exemplary tangential spring extension **380** described above. However, for the embodiment depicted, a first segment **396**, a second segment **398** or both of the tangential spring extension **380** defines one or more stiffness modifiers **416**. The stiffness modifier **416** refers to any feature operable to modify a stiffness of the first segment **396**, the second segment **398**, or both relative to a constant-thickness, solid plate spring design. For example, in the embodiment of FIG. **87**, the one or more of stiffness modifiers **416** includes a plurality of openings defined in the first segment **396** and a plurality of openings defined in the second segment **398**. However, in other example embodiments, the one or more stiffness modifiers **416** may be, e.g., a single opening, one or more ridges formed of the same or different material, one or more sections of increased thickness, etc.
(386) As will further be appreciated from the view of FIG. **87**, the middle section **394** may define a slot **418** to receive a pin configured to mount the tangential spring extension **380** to a carrier **104**.
(387) Moreover, in other exemplary embodiments, the tangential spring extension **380** may have still other suitable configurations. For example, referring to FIGS. **88** and **89**, two additional exemplary tangential spring extensions **142** are depicted in accordance with one or more aspects of the present disclosure.
(388) Referring particularly to FIG. **88**, a first segment **396** of the tangential spring extension **380** depicted and a second segment **398** of the tangential spring extension **380** depicted each include a plurality of bends **420** to provide additional stiffness and/or flexibility in the circumferential direction C.
(389) Referring particularly FIG. **89**, a first segment **396** of the tangential spring extension **380** depicted and a second segment **398** of the tangential spring extension **380** depicted include a first projection **422** in between the middle section **394** and a first end **390** and a second projection **424**

in between the middle section **394** and a second end **392**. For the embodiment of FIG. **89**, a first seal segment **110**A includes a first slot **426** at a second circumferential end **384** with the first projection **422** of the first seal segment **110**A positioned therein, and a second seal segment **110**B includes a second slot **428** at a first circumferential end **386** with the second projection **424** of the second seal segment **110**B positioned therein.

(390) In addition, the first seal segment **110**A includes a contact section **430** between the first end **390** and the first projection **422** configured to contact the second circumferential end **384** of the first seal segment **110**A to provide a circumferential and a radial biasing force to the first seal segment **110**A. Similarly, the second segment **398** includes a contact section **432** between the second end **392** and the second projection **424** to provide a circumferential and a radial biasing force to the second seal segment **110**B. The contact sections **430**, **432** are generally linear surfaces of the respective first and second seal segments **110**A, **110**B defining an angle with a radial direction R such that they slope towards the adjacent seal segment **110** as it moves inwardly along the radial direction R.

(391) Accordingly, it will be appreciated that for the exemplary embodiment of FIG. **89**, the tangential spring extension **380** is engaged with a circumferential end of the first seal segment **110**A and a circumferential end of the second seal segment **110**B.

(392) Further, still, in other exemplary embodiments, the tangential spring extension **380** may be supported relative to the carrier **104** in any other suitable manner. For example, referring now to FIG. **90**, a radial spring **406** of the seal support assembly **108** is positioned at least partially outward of a carrier **104** along a radial direction R of the turbine engine. In particular, for the embodiment of FIG. **90**, the tangential spring extension **380** is coupled to a pin **404** extending through an opening **434** in the carrier **104**. The pin **404** includes a pinhead **411**, and the radial spring **406** extends between the carrier **104** and the pinhead **411** to bias the tangential spring extension **380** outwardly along the radial direction R.

(393) Further, referring now to FIG. **91**, another exemplary embodiment of the present disclosure is provided. For the embodiment shown, a seal support assembly **108** is provided having a wedge body **436** position at least partially between a first seal segment **110**A and a second seal segment **110**B of a seal assembly **106**. The seal support assembly **108** further includes a tangential spring extension **380**, and more specifically, includes a tangential spring extension **380**A and a second tangential spring extension **380**B. The first tangential spring extension **380**A is positioned on a first side **438** of the wedge body **436** and is engaged with the first seal segment **110**A. The second tangential spring extension **380**B is positioned on a second side **440** of the wedge body **436** and is engaged with the second seal segment **110**B.

(394) In particular, for the embodiment shown, a second circumferential end **384** of the first seal segment **110**A defines a recess **442**, and the first side **438** of the wedge body **436** similarly defines a recess **444**. The first tangential spring extension **380**A is positioned within the recess **442** of the first seal segment **110**A and the recess **444** at the first side **438** of wedge body **436**. Further, a first circumferential end **386** of the second seal segment **110**B defines a recess **446** and the second side **440** of the wedge body **436** also defines a recess **448**. The second tangential spring extension **380**B is positioned within the recess **446** of the second seal segment **110**B and the recess **448** at the second side **440** of the wedge body **436**.

(395) Referring now to FIGS. **92** and **93**, another exemplary embodiment of the present disclosure is provided. FIG. **92** provides a schematic, forward-looking-aft view of an assembly in accordance with the present disclosure, and FIG. **93** provides a close-up view of a portion of the assembly of FIG. **92**. The exemplary embodiment of FIGS. **92** and **93** may be configured in a similar manner as one or the exemplary embodiments described herein.

(396) For example, for the embodiment shown, the assembly includes a rotor **100**, a stator **102** having a carrier **104**, a seal assembly **106** positioned between the rotor **100** and the stator **102**, and a seal support assembly **108**. The seal assembly **106** includes a plurality of seal segments **110**

supported at least in part by the carrier **104**, with the plurality of seal segments **110** including a first seal segment **110**A and a second seal segment **110**B. Each of the first seal segment **110**A and the second seal segment **110**B includes a seal face **112** configured to form a fluid bearing with the rotor **100**.

(397) For the embodiment depicted, the seal support assembly **108** includes a first engagement assembly **450**A extending between the first seal segment **110**A and the carrier **104** operable to bias the first seal segment **110**A along the radial direction R. The seal support assembly **108** further includes a second engagement assembly **450**B extending between the first seal segment **110**A and the carrier **104** operable to bias the second seal segment **110**B along the radial direction R. Moreover, the seal support assembly **108** includes a tangential spring extension **380** extending between the first engagement assembly **450**A and the second engagement assembly **450**B for biasing the first seal segment **110**A relative to the second seal segment **110**B in a circumferential direction C of the turbine engine.

(398) More specifically, referring particularly to FIG. **93**, the first engagement assembly **450**A includes a first radial extension **452**A and the second engagement assembly **450**B includes a second radial extension **452**B. More specifically, the first engagement assembly **450**A includes a first spring extension **454**A configured to bias the first radial extension **452**A along the radial direction R, and thus configured to bias the first seal segment **110**A along the radial direction R. Similarly, the second engagement assembly **450**B includes a second spring extension **454**B configured to bias the second radial extension **452**B along the radial direction R, and thus configured to bias the second seal segment **110**B along the radial direction R. The first spring extension **454**A, the second spring extension **454**B, or both may be configured in a similar manner as one or more of the exemplary spring extensions described hereinabove.

(399) The tangential spring extension **380** in the embodiment of FIG. **93** extends between the first radial extension **452**A and the second radial extension **452**B.

(400) More specifically, still, for the embodiment depicted, the seal support assembly **108** further includes a first ring member **456**A coupled to the first engagement assembly **450**A and a second ring member **456**B coupled to the second engagement assembly **450**B. In particular, for the embodiment shown, the first ring member **456**A is coupled to the first radial extension **452**A and the second ring member **456**B is coupled to the second radial extension **452**B. The tangential spring extension **380** extends between the first ring member **456**A and the second ring member **456**B.

(401) In such a manner, the seal support assembly **108** may maintain a circumferential alignment of the plurality of seal segments **110**. Further, the seal support assembly **108** may allow for the seal assembly **106** to operate in the event of a failure of one of the seal segments **110**, in the event of a failure of one of the spring extensions **142**, etc.

(402) Referring now to FIGS. **94** and **95**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **94** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. FIG. **94** provides a view of the assembly in a first position, and FIG. **95** provides a view of the assembly and a second position. The view of FIGS. **94** and **95** is of a reference plane defined by an axial direction A and a radial direction R of the turbine engine. The assembly of FIGS. **94** and **95** may be configured in a similar manner as one or more of the exemplary assemblies described above.

(403) For example, the seal assembly **106** includes a first seal segment **110**A, with the first seal segment **110**A having a seal face **112** configured to form a fluid bearing with the rotor **100**.

(404) However, for the embodiment shown, the seal support assembly **108** includes a magnet assembly **460** having a magnet coupled to the carrier **104** or the first seal segment **110**A for biasing the first seal segment **110**A along the radial direction R relative to the carrier **104**. More specifically, for the embodiment shown, the magnet is a first magnet **462** coupled to the carrier **104**

and the magnet assembly **460** further includes a second magnet **464** coupled to the first seal segment **110**A. The first magnet **462** defines a magnetic field and the second magnet **464** is positioned within the magnetic field of the first magnet **462**. In particular, for the configuration depicted in FIGS. **94** and **95**, the first magnet **462** and second magnet **464** together form a magnetic attraction force to bias the first seal segment **110**A outwardly along the radial direction R towards the carrier **104**.

(405) Referring still to FIGS. **94** and **95**, the seal support assembly **108** further includes a first magnet base **466** and a second magnet base **468**. The first magnet **462** is coupled to and positioned at least partially within the first magnet base **466**, and similarly the second magnet **464** is coupled to and positioned at least partially within the second magnet base **468**. The first magnet base **466** and second magnet base **468** are each formed of a nonferromagnetic material. For example, the first magnet base **466** and second magnet base **468** may be formed of a ceramic material.

(406) More specifically, still, the first magnet base **466** is coupled to the carrier **104** and the second magnet base **468** is coupled to the first seal segment **110**A. The magnetic attraction force formed by the first magnet **462** and the second magnet **464** may be transferred to the carrier **104** and to the first seal segment **110**A through the first magnet base **466** and the second magnet base **468**.

(407) The seal support assembly **108** further includes a particle shield **470** surrounding at least in part the magnet assembly **460**. Specifically, for the embodiment shown, the particle shield **470** is a bellows assembly extending along the radial direction R and coupled to the carrier **104** and the first seal segment **110**A. The bellows assembly extends completely around the magnet assembly **460**. In such a manner, the particle shield **470** may prevent or minimize any metallic/ferromagnetic particles within an airflow surrounding the magnet assembly **460** from attaching to the first magnet **462**, the second magnet **464**, or both.

(408) For the embodiment shown, the first magnet **462** and the second magnet **464** are each configured as permanent magnets. Depending on the location of the seal support assembly **108**, the permanent magnet may define a sufficiently high Curie temperature such that the permanent magnet may withstand anticipated operating conditions. For example, in certain exemplary embodiments, the permanent magnet may define a Curie temperature of at least 1200 degrees Celsius (C), such as at least 1300 degrees C., such as at least 1400 degrees C., such as up to 2000 degrees C. For example, in one or more exemplary embodiments, the permanent magnet may be formed of Alnico alloys, RECOMA HT520™, a rare earth material, or any other suitable material.

(409) Notably, in at least certain exemplary aspects, the Curie temperature defined by the permanent magnet may be at least 15% higher than anticipated maximum operating temperature of the magnet assembly **460**. For example, in certain exemplary embodiments, the Curie temperature defined by the permanent magnet may be at least 25% higher than the anticipated maximum operating temperature of the magnet assembly **460**.

(410) The first seal segment **110**A further includes a body **138** and a lip **136** extending from the body **138** along the axial direction A. The lip **136** includes an outer pressurization surface **140**, and a pressure on the outer pressurization surface **140** may generally increase at higher power operating conditions of the turbine engine. FIG. **94** depicts the first seal segment **110**A at a position indicative of the turbine engine being at a relatively low power operating condition, and FIG. **95** depicts the first seal segment **110**A at a position indicative of the turbine engine being at a relatively high power operating condition. The magnetic attraction force generated between the first magnet **462** and the second magnet **464** may allow for the first seal segment **110**A to define a relatively high radial clearance with the rotor **100** (e.g., a distance between the seal face **112** and the rotor **100**) during the relatively low power operating condition, and may further allow for the first seal segment **110**A to define a relatively low radial clearance with the rotor **100** during the relatively high power operating condition.

(411) Notably, to ensure the first seal segment **110**A may be moved along the radial direction inward as a power of the turbine engine is increased, the magnet assembly **460** is configured such

that the first magnet **462** and second magnet **464** do not contact one another. More specifically, the first magnet base **466** and the second magnet base **468** together form a bumper **472** to prevent the first magnet **462** from contacting the second magnet **464**.

(412) In particular, referring to FIGS. **96** and **97**, schematic views are provided of a portion of the seal support assembly **108** of FIGS. **94** and **95**. The position depicted in FIG. **96** corresponds to the position depicted in FIG. **94** and the positioning FIG. **97** corresponds to the position FIG. **95**.

(413) As is depicted, the first magnet base **466** includes one or more radial walls **474** extending further inwardly along the radial direction R than the first magnet **462**. Similarly, the second magnet base **468** includes one or more radial walls **476** extending farther outwardly along the radial direction R than the second magnet **464**. In such a manner, when the turbine engine is in the relatively low power operating condition, the magnetic attraction force may be prevented from moving the first magnet **462** into contact with the second magnet **464** as the one or more radial walls of the first magnet base **466** may contact the one or more radial walls of the second magnet base **468** before the first magnet **462** and the second magnet **464** make contact (see, specifically, FIG. **96**). Notably, for the embodiment depicted, the one or more radial walls of the first magnet base **466** and the second magnet base **468** each includes a bumper **478** to minimize any damage resulting from such contact.

(414) It will be appreciated, however, that in other exemplary embodiments, the seal support assembly **108** may have any other suitable configuration. For example, referring to FIGS. **98** and **99**, an assembly in accordance with another example embodiment of the present disclosure is provided. The assembly of FIGS. **98** and **99** may be configured in substantially the same manner as exemplary embodiment described above with reference to FIGS. **96** and **97**. However, for the embodiment of FIGS. **98** and **99**, a first magnet **462** is prevented from contacting a second magnet **464** through use of a nonferromagnetic plate **480**. More specifically, a magnet assembly **460** of the seal support assembly **108** depicted includes a nonferromagnetic plate **480** positioned between the first magnet **462** and the second magnet **464**. More specifically, still, for the embodiment depicted, the nonferromagnetic plate **480** is a first nonferromagnetic plate **480**A coupled to a first magnet base **466** over a surface of the first magnet **462** facing the second magnet **464**. The magnet assembly **460** further includes a second nonferromagnetic plate **480**B coupled to a second magnet base **468** over a surface of the second magnet **464** facing the first magnet **462**. In such a manner the first nonferromagnetic plate **480**A and the second nonferromagnetic plate **480**B may prevent contact between the first magnet **462** and the second magnet **464** during, e.g., a low power operating condition. In such a manner, the plates **480**A, **480**B may make it easier for the magnets **462**, **464** to be moved away from one another.

(415) Referring briefly to FIG. **100**, it will be appreciated that in at least certain exemplary embodiments, the seal support assembly **108** may include a plurality of magnet assemblies **460**, with each magnet assembly **460** dedicated to a respective seal segment **110** of the plurality of seal segments **110** of the seal assembly **106**.

(416) Alternatively, although not depicted, a magnet assembly **460** of the seal support assembly **108** of the present disclosure may support two or more seal segments **110**.

(417) Referring now to FIG. **101**, an assembly in accordance with another example embodiment of the presence closure is provided. The exemplary embodiment of FIG. **101** may be configured in a similar manner as one or the exemplary embodiments described above. For example, a seal support assembly **108** is provided having a magnet assembly **460**, with the magnet assembly **460** including a first magnet **462** and a second magnet **464**. In the embodiment shown, the first magnet **462** includes a first surface **482** facing the second magnet **464** and the second magnet **464** includes a second surface **484** facing the first magnet **462**. However, for the embodiment depicted, the first surface **482** has a nonplanar geometry and the second surface **484** similarly has a nonplanar geometry. Further, in the embodiment shown, the first surface **482** is complementary to the second surface **484**. In such a manner, nonplanar features of the first surface **482** fit into correspondingly-

shaped nonplanar features of the second surface **484**.

(418) Further, referring briefly to FIGS. **102** through **104**, the first surface **482**, the second surface **484**, or both may have any other suitable configuration. For example, the first surface **482**, the second surface **484**, or both may define corresponding shaped rectangles, corresponding state triangles, corresponding the shaped semicircles, etc.

(419) Further, referring to FIG. **105**, a magnet in accordance with one or the exemplary embodiments described herein is provided. In the embodiment shown, the magnet defines a circular shape having concentric circular grooves **486** defined in a surface. The surface depicted in FIG. **105** may be a first surface **482** of a first magnet **462** or a second surface **484** of a second magnet **464**. The magnet of FIG. **105** may be incorporated in a magnet assembly **460** with another magnet having a correspondingly-shaped and/or a complementary-shaped surface.

(420) It will be appreciated that in other exemplary embodiments, a surface of a magnet of the present disclosure may have any other suitable shape or arrangement.

(421) Further, in still other exemplary embodiments, still other suitable configurations may be provided. For example, referring now to FIGS. **106** and **107**, an embodiment is provided having a seal assembly **106** with a first seal segment **110**A, a carrier **104** at least partially outward of the first seal segment **110**A in a radial direction R, and a rotor **100** positioned at least partially inward of the first seal segment **110**A along the radial direction R. A seal support assembly **108** is further provided, the seal support assembly **108** including a magnet assembly **460** with a first magnet **462** and a second magnet **464**.

(422) However, for the embodiment of FIGS. **106** and **107**, the first magnet **462** and the second magnet **464** form a magnet repelling force. In particular, for the embodiment depicted, the carrier **104** includes a carrier axial extension **488** and the first seal segment **110**A includes a segment axial extension **490** located outward of the carrier axial extension **488** along the radial direction R. The first magnet **462** is coupled to the carrier **104** through the carrier axial extension **488**, and the second magnet **464** is coupled to the first seal segment **110**A through the segment axial extension **490**.

(423) With such a configuration, the magnetic repelling force formed by the first magnet **462** and the second magnet **464** may maintain the first seal segment **110**A at a radially outer position during, e.g., a low power operating mode of the turbine engine (see, e.g., FIG. **106**), and may allow the first seal segment **110**A to move to a radially inner position during, e.g., a high power operating mode of the turbine engine (see, e.g., FIG. **107**).

(424) Further, it will be appreciated that in the least certain exemplary aspects of the present disclosure a magnet assembly **460** may be coupled to a carrier **104** and/or a first seal segment **110**A and the suitable manner. For example, referring briefly back to FIGS. **94** and **95**, the seal support assembly **108** may include an outer plate **492** coupled to the first magnet base **466**, with the outer plate **492** coupled to the carrier **104** through one or more mechanical fasteners **494**, such as one or more bolts, screws, rivets, etc. Similarly, the seal support assembly **108** may include an inner plate **496** coupled to second magnet base **468**. The inner plate **496** may, in turn, be coupled to the first seal segment **110**A through a first seal segment hanger **498** (e.g., hooks or ledges that extend over an edge of the inner plate **496**).

(425) Additionally, or alternatively, in other exemplary embodiments, the seal support assembly **108** may be coupled to the carrier **104** and the first seal segment **110**A in any other suitable manner. For example, referring briefly to FIG. **108**, in another example embodiment, an outer plate **492** may be coupled to the carrier **104** through a carrier hanger **500** and an inner plate may be coupled to the first seal segment **110**A through one or more mechanical fasteners **502**.

(426) In still other example embodiments, any other suitable combination of the above mechanisms, or any other suitable mechanisms may be provided.

(427) Moreover, one or more of the exemplary magnets above may be formed of a plurality of individual magnets, or magnet sections, arranged in a suitable manner to form a desired magnetic

field. For example, referring briefly to FIG. **109**, a magnet **504** that may be incorporated into one or the exemplary embodiments described above, e.g., as a first magnet **462** or a second magnet **464**, is depicted. In the embodiment of FIG. **109**, the magnet **504** includes a plurality of sections **506** arranged linearly. For example, the plurality of sections **506** may be arranged in a tangential direction (e.g., a direction perpendicular to a radial direction R of a turbine engine). Each of the individual sections **506** of the magnet **504** defines a north pole, or rather, a north pole direction indicated by the arrows **508** in FIG. **109**. Each section **506** defines the north pole direction facing in a unique direction relative to one both adjacent sections **506** of the magnet **504**.
(428) Inclusion of a magnet in accordance with the embodiment of FIG. **109**, may allow for concentration of a magnetic field and more specifically, of the magnetic forces (e.g., attraction or repelling). This may allow for the magnet assembly **460** incorporating such a magnet to function in a desired manner, while minimizing or reducing magnetic particles from being attached to the magnets from outside directions.
(429) Referring briefly to FIG. **110**, an embodiment in accordance with another example embodiment is provided. The embodiment of FIG. **110** is configured in a similar manner as the exemplary embodiments discussed above. However, for the embodiment of FIG. **110**, a magnet assembly **460** of the seal support assembly **108** utilizes an electro-magnet **510**. The magnet assembly **460** additionally includes a sensor **512** configured to sense data indicative of a position of a first seal segment **110**A relative to a carrier **104**, a controller **514** electrically coupled to the electro-magnet **510** and the sensor **512**, a permanent magnet **516**, and a particle shield **470**.
(430) The controller **514** may provide power to the electro-magnet **510** to alter a magnetic attraction force with the permanent magnet **516** (or a magnetic repelling force depending on the configuration), e.g., in response to receiving data from the sensor **512** indicative of the position of the first seal segment **110**A relative to the carrier **104**. In such a manner, the seal support assembly of FIG. **110** may actively control a radial position of the first seal segment **110**A, and thus of a radial clearance defined between a first seal segment **110**A and a rotor **100**.
(431) It will be appreciated, that in other exemplary embodiments, the magnet assembly **460** may have still other suitable configurations.
(432) Referring now to FIGS. **111** and **112**, another exemplary embodiment of the present disclosure is provided. FIG. **111** provides a cross-sectional view of an assembly in accordance with the present disclosure in a plane defined by a radial direction R and an axial direction A of the turbine engine, and FIG. **112** provides an aft-looking-forward view of the assembly of FIG. **111**. The exemplary embodiment of FIGS. **111** and **112** may be configured in a similar manner as one or the exemplary embodiments described herein.
(433) For example, the embodiment depicted includes a stator **102** having a carrier **104**, a seal assembly **106**, and a seal support assembly **108**. The seal assembly **106** is configured to be positioned at least partially inward of the stator **102** and outward of a rotor **100** (not shown). The seal assembly **106** includes a first seal segment **110**A having a seal face **112** configured form of fluid bearing with the rotor **100**.
(434) The seal support assembly **108** includes a prestressed spring assembly **520** extending from the first seal segment **110**A for biasing the first seal segment **110**A along the radial direction R. In particular, for the embodiment shown, the prestressed spring assembly **520** extends between the first seal segment **110**A and the carrier **104** to counter a pressure on an outer pressurization surface **140** of a lip **136** of the first seal segment **110**A during operation of the turbine engine, while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.
(435) Referring particularly to FIG. **111**, for the embodiment depicted, the first seal segment **110**A includes a support extension **522**, with the support extension **522** including a seal segment spring seat **524**. In the embodiment depicted, the first seal segment **110**A further includes a body **138** and a radial extension **150** extending from the body **138** to the support extension **522**. It will therefore

be appreciated that for the embodiment depicted, the support extension **522** is spaced outwardly from the body **138** along the radial direction R.

(436) Further, the prestressed spring assembly **520** generally includes a spring extension **142** and a compression plate **526**. The compression plate **526** includes a compression spring seat **528**, and the spring extension **142** extends between the compression spring seat **528** and the seal segment spring seat **524**. The prestressed spring assembly **520** further includes one or more housings **530** extending outwardly from the compression plate **526** along the radial direction R to the carrier **104**, and a corresponding one or more attachment mechanisms **532** for attaching the one or more housings **530** to the carrier **104**. Accordingly, the compression plate **526** is fixed to the carrier **104**.

(437) In particular, for the embodiment depicted the one or more housings **530** includes two housings **530** spaced along the circumferential direction C (see, e.g., FIG. **112**). The two housings **530** are each hollow and the one or more attachment mechanisms **532** includes two attachment mechanisms **532**, each extending through a respective one of the two housings **530**. The attachment mechanisms **532** maintain the compression plate **526** pressed against the carrier **104** to maintain the spring extension **142** in a pre-stressed state. In such a manner, it will be appreciated that the spring extension **142** may be referred to as a prestressed spring extension.

(438) Such a configuration may provide a desired resistance along the radial direction R for the seal segment **110**. In particular, installing the spring extension **142** in a prestressed condition (also referred to herein as a pre-strained condition), may allow for a desired resistance profile for a seal support assembly **108** of the present disclosure. For example, the spring extension **142** may not provide a constant resistance, e.g., along the radial direction R, such that by installing the spring extension **142** in a prestressed condition allows for use of a smaller or lighter spring to achieve the desired resistance, e.g., along the radial direction R.

(439) Notably, the carrier **104** further includes a bump stop **534** (FIG. **111**) to maintain the prestressed spring assembly **520** in the prestressed condition and prevent movement of the first seal segment **110**A outwardly along the radial direction R more than a predetermined amount. In particular, for the embodiment depicted the bump stop **534** is configured to contact the support extension **522**, the radial extension **150**, or both.

(440) Referring now to FIGS. **113** and **114**, a prestressed spring assembly **520** in accordance with another exemplary embodiment of the present disclosure is provided. FIG. **113** provides a schematic view of the prestressed spring assembly **520** installed with a carrier **104** of a stator **102** in a reference plane defined by a radial direction R and a circumferential direction C of the turbine engine. FIG. **114** provides an exploded view of the prestressed spring assembly **520** of FIG. **113**. Notably, a majority of the first seal segment **110**A is removed and the view of FIG. **113** for clarity, and a size of the first seal segment **110**A is reduced in FIG. **114** for clarity.

(441) The exemplary embodiment of FIGS. **113** and **114** may be configured in a similar manner as exemplary embodiment of FIGS. **111** and **112**. For example, the prestressed spring assembly **520** generally includes a compression plate **526** and one or more housings **530** extending from the compression plate **526** along the radial direction R to a carrier **104**, along with a respective one more attachment mechanisms **532** extending through the one more housings **530** and coupling the compression plate **526** to the carrier **104**. In the embodiment depicted, the one or more attachment mechanisms **532** are configured as attachment studs attached to or formed integrally with the carrier **104** and extending along the radial direction R.

(442) In particular, the one or more housings **530** of the prestressed spring assembly **520** are positioned over the respective one or more attachment mechanisms **532**, and the prestressed spring assembly **520** includes a respective one more mechanical fasteners **536** engaged with the one or more attachment mechanisms **532** to fix the compression plate **526** and housings **530** to the one or more attachment mechanisms **532** at a certain position along the radial direction R relative to the carrier **104**.

(443) Notably, for the embodiment depicted, the spring extension **142** is a first spring extension

142A, and the prestressed spring assembly **520** further includes a second spring extension **142**B, each extending between the compression plate **526** and the first seal segment **110**A. The mechanical fasteners **536** coupled to the attachment mechanisms **532** maintain the first spring extension **142**A and the second spring extension **142**B in the prestressed condition.

(444) For example, referring also to FIG. **115**, providing a view of the assembly of FIGS. **113** and **114** during an assembly phase, it will be appreciated that the prestressed spring assembly **520** includes a compression screw **538** operable with the carrier **104** and the compression plate **526** to move the compression plate **526** towards the carrier **104** along the radial direction R during assembly. In particular, the compression plate **526** defines an assembly opening **540** and the carrier **104** defines a threaded opening **542** aligned with the assembly opening **540** to facilitate assembly of the prestressed spring assembly **520**. The compression screw **538** may be inserted through the assembly opening **540** and into the threaded opening **542** of the carrier **104**. The compression screw **538** includes a lip **544** wider than the assembly opening **540**. Accordingly, rotation of the compression screw **538** may move the lip **544** and therefore the compression plate **526** towards the carrier **104**. In such a manner, the compression screw **538** may be utilized to move the compression plate **526** towards the carrier **104**, such that one or more housings **530** contact the carrier **104**, with the attachment mechanisms **532** extending through the housings **530**. In particular, with the compression screw **538** in place, the one or more mechanical fasteners **536** may be affixed to the attachment mechanisms **532** (e.g., mounting studs) to fix the compression plate **526** in position relative to the carrier **104**. Once the mechanical fasteners **536** are in position, the compression screw **538** may be removed and the mechanical fasteners **536** and attachment mechanisms **532** may maintain the compression plate **526** in position and first spring extension **142**A and second spring extension **142**B in a prestressed condition.

(445) It will be appreciated that using the above structures may allow for incorporation of a spring extension **142** that defines an unloaded length along the radial direction R. The term "unloaded length" refers to a length of the spring extension **142** with no compressive or tension forces acting thereon (e.g., the length of the spring extension **142** prior to the spring extension **142** being installed in the prestressed spring assembly **520**; see FIG. **115**). The spring extension **142** may further define an assembly length. The term "assembly length" refers to a length of the spring extension **142** once installed in the prestressed spring assembly **520** with the turbine engine in a pre-operation mode (e.g., an operating condition which substantially no pressure is being exerted on the first seal segment **110**A along the radial direction R by a pressurized air flow; see FIG. **113**).

(446) Referring now to FIG. **16**, a view of the assembly of FIGS. **113** through **115** is provided with the turbine engine in a high-power operating condition. In the high-power operating condition the first seal segment **110**A, through the support extension **522**, may press down along the radial direction R against the first spring extension **142**A and the second spring extension **142**B to move the first seal segment **110**A closer to the rotor **100** along the radial direction R. The compression plate **526** may be maintained at a fixed position relative to the carrier **104** through this operation through the attachment mechanisms **532**, the housings **530**, and the mechanical fasteners **536**.

(447) As briefly mentioned above, the views of FIGS. **113**, **115**, and **116** include a majority of the first seal segment **110**A removed for clarity. However, these views include the support extension **522** having a seal segment spring seat **524**. The structure coupling to the support extension **522** to the first seal segment **110**A may be similar to the structure described above with reference to FIGS., e.g., FIGS. **111** and **112**.

(448) It will be appreciated that for the embodiments described above, the spring extension **142** is depicted as a helical spring. FIG. **117**A provides a close-up view of a helical spring as may be incorporated as a spring extension **142** in one or more of the above embodiments.

(449) It will be appreciated, however, that in other exemplary embodiments, the spring extension **142**, which may be a prestressed spring extension as noted above, may have any other suitable configuration. For example, in other exemplary embodiments, one or more of the above assemblies

may include a spring extension **142** configured as a progressive helical spring (see, e.g., FIG. **117**B), a bellows assembly (see, e.g., FIG. **118**B), a conical spring, and optimized super elastic spring block (see, e.g., FIG. **118**A), or an optimized metal foam structure.

(450) Further, it will be appreciated that in other exemplary embodiments, the prestressed spring assembly **520** may have still other suitable configurations. For example, referring now to FIGS. **119** and **120**, views of an assembly in accordance with another exemplary embodiment of the present disclosure is provided. The assembly of FIGS. **119** and **120** may be configured in a similar manner as one or more of the exemplary embodiments described above. For example, the embodiment of FIGS. **119** and **120** includes a stator **102** having a carrier **104** and a seal assembly **106** having a plurality of seal segments **110** spaced along the circumferential direction C of the turbine engine. The plurality of seal segment **110** includes a first seal segment **110**A.

(451) Referring particularly to FIG. **119**, the embodiment depicted includes a seal support assembly **108**, with the seal support assembly **108** including a prestressed spring assembly **520** extending from the first seal segment **110**A for biasing the first seal segment **110**A along the radial direction R. More specifically, for the embodiment depicted, the first seal segment **110**A includes a support extension **522**, with the support extension **522** including a seal segment spring seat **524**. In the embodiment depicted, the first seal segment **110**A further includes a body **138** and a radial extension **150** extending from the body **138** to the support extension **522**. It will therefore be appreciated that for the embodiment depicted, the support extension **522** is spaced outwardly from the body **138** along the radial direction R.

(452) Further, the carrier **104** includes an axial extension **527** positioned inward of the support extension **522** of the first seal segment **110**A along the radial direction R. Additionally, as noted above, the prestressed spring assembly **520** generally includes the spring extension **142**. The spring extension **142** extends between the axial extension **527** of the carrier and the support extension **522** of the first seal segment **110**A. In particular, the support extension **522** includes a seal segment spring seat **524** and the axial extension **527** includes an extension spring seat **529**.

(453) Notably, for the embodiment depicted, the body **138** of the first seal segment **110**A may act as a bump stop to prevent outward movement of the first seal segment **110**A past a preset amount, and to maintain the spring extension **142** in the pre-stressed state.

(454) Such a configuration may provide a desired resistance along the radial direction R for the seal segment **110**. In particular, installing the spring extension **142** in a prestressed condition (also referred to herein as a pre-strained condition), may allow for a desired resistance profile for a seal support assembly **108** of the present disclosure.

(455) Referring briefly also to FIG. **120**, it will be appreciated that each seal segment may include a dedicated prestressed spring assembly **520**. Further, it will be appreciated that the carrier **104** includes an outer cover **105** (see FIG. **119**; removed in FIG. **120** for clarity). The outer cover **105** may not be present during assembly, to allow for installation of the individual seal segments **110** and prestressed spring assemblies **520** (see, e.g., FIG. **120**; depicting the assembly during a mid-assembly stage).

(456) Further, it will be appreciated that in still other exemplary embodiments, the prestressed spring assembly **520** may have still other suitable configurations. For example, referring now to FIG. **121**, a view of an assembly in accordance with another exemplary embodiment of the present disclosure is provided. The assembly of FIG. **121** may be configured in a similar manner as one or more of the exemplary embodiments described above. For example, the embodiment of FIG. **121** includes a seal assembly **106** having a plurality of seal segments **110** spaced along the circumferential direction C of the turbine engine. The plurality of seal segment **110** includes a first seal segment **110**A.

(457) Further, still, the embodiment of FIG. **121** includes a seal support assembly **108**, with the seal support assembly **108** including a prestressed spring assembly **520** extending from the first seal segment **110**A for biasing the first seal segment **110**A along the radial direction R. More

specifically, as will be explained in more detail below, the prestressed spring assembly **520** includes a spring extension **142** extending from the first seal segment **110**A along the radial direction R.
(458) Referring now to FIGS. **122** and **123**, FIG. **122** provides a close-up view of the first seal segment **110**A and the prestressed spring assembly **520** of FIG. **121**, and FIG. **123** provides a cross-sectional view of the prestressed spring assembly **520** along Line **123-123** and FIG. **122**. As will be appreciated, the first seal segment **110**A includes a support extension **522**. The support extension **522** extends generally along the axial direction A (see FIG. **123**). The prestressed spring assembly **520** includes a support ring **546** positioned inward of the support extension **522** along the radial direction R and the prestressed spring assembly **520** includes the spring extension **142** extending between the support extension **522** of the first seal segment **110**A and the support ring **546**.
(459) In particular, referring also to FIG. **124**, it will be appreciated that the prestressed spring assembly **520** includes an attachment block **548** configured to couple to the support extension **522**. The attachment block **548** may have a substantially U-shaped, defining an opening **550**. The attachment block **548** may be configured to fit over the support extension **522** of the first seal segment **110**A, such that the support extension **522** is positioned within the opening **550** of the attachment block **548**.
(460) The prestressed spring assembly **520** further includes a pin **552** extending from the attachment block **548** to the support ring **546**. The pin **552** may be slidably coupled to the support ring **546** such that the attachment block **548** and first seal segment **110**A may be moved inwardly along the radial direction R relative to the support ring **546**.
(461) When the prestressed spring assembly **520** is installed with the plurality of seal segments **110** of the seal assembly **106**, the attachment block **548** may be moved towards the support ring **546** to prestressed the spring extension **142**. During operation, as a pressure or force on the first seal segment **110**A inwardly along the radial direction R increases, the spring extension **142** may compress (and the pin **552** may slide inwardly through the support ring **546**) to allow the seal segment **110**A to move inwardly along the radial direction R to reduce a radial clearance gap defined between the seal face **112** (see FIG. **123**) of the first seal segment **110**A and a rotor **100** (not depicted).
(462) Briefly, referring back to FIG. **121**, it will be appreciated that the spring extension **142** is a first spring extension **142**A and the prestressed spring assembly **520** further includes a plurality of spring extensions **142**. Each of the plurality of spring extensions **142** extends from a seal segment **110** of the plurality of seal segments **110** to the support ring **546**, and more specifically, extends from a support extension **522** of each of the respective seal segments **110** to the support ring **546**. In such a manner, it will be appreciated that the support ring **546** extends 360 degrees in the circumferential direction C at a location inward of the support extensions **522** of the plurality of seal segments **110**. The continuous 360 degree structure of the support ring **546** may allow for the support ring **546** to support the seal assembly **106** along the radial direction R without a separate coupling between the support ring **546** and the carrier **104**.
(463) It will be appreciated that in at least certain exemplary embodiments, the present disclosure may provide for a method for assembling a seal segment for a turbine engine of the present disclosure. The method may include: (1) moving a compression plate of a seal support assembly over an attachment mechanism of a carrier, the attachment mechanism of the carrier extending along a radial direction of the turbine engine, the seal support assembly including a spring extension extending between the compression plate and the carrier; and (2) inserting a tightening rod through a tightening rod opening defined in the compression plate and into an engagement opening in the carrier.
(464) The method may further include (3) moving the tightening rod through the engagement opening in the carrier, wherein moving the tightening rod through the engagement opening in the carrier comprises contacting the compression plate with a ledge of the tightening rod to move the compression plate closer to the carrier. In at least certain exemplary aspects, moving the tightening

rod through the engagement opening in the carrier at (3) may include (4) compressing the spring extension to an assembly length from an unloaded length. The assembly length may be less than or equal to 90% of the unloaded length of the spring extension.
(465) The method may further include (5) affixing a mechanical fastener to the attachment mechanism to maintain a maximum distance between the compression plate and the carrier. In at least certain exemplary aspects, the mechanical fastener may define a threaded opening. With such an exemplary aspect, affixing the mechanical fastener to the attachment mechanism at (5) may include (6) rotating the mechanical fastener about the attachment mechanism to engage the threaded opening with threads of the attachment mechanism.
(466) Further, in certain exemplary aspects, the method may further include (7) removing the tightening rod from the engagement opening and from the tightening rod opening after affixing the mechanical fastener to the attachment mechanism
(467) Such an exemplary aspect may provide for a seal assembly having a prestressed seal support, which may provide for various benefits as discussed hereinabove.
(468) Referring now to FIGS. **125** and **126**, another exemplary embodiment of the present disclosure is provided. FIG. **125** provides a schematic, cross-sectional view of a section of a turbine engine, and more specifically of a section of a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. The view of FIG. **125** is in a plane defined by a radial direction R and an axial direction A of a turbine engine. FIG. **125** provides a view of the assembly in a first position, and FIG. **126** provides a view of the assembly in a second position. The exemplary embodiment of FIGS. **125** and **126** may be configured in a similar manner as one or the exemplary embodiments described hereinabove.
(469) For example, the seal assembly **106** includes a plurality of seal segments **110**, the plurality of seal segments **110** having a first seal segment **110**A having a seal face **112** configured form of fluid bearing with the rotor **100**. The first seal segment **110**A includes a lip **136** and a body **138**, with the lip **136** extending away from the body **138** along the axial direction A of the turbine engine and defining an outer pressurization surface **140**. The outer pressurization surface **140** is exposed to a high-pressure cavity **132** on a high-pressure side **126** of the seal assembly **106**. During operation of the turbine engine, the high-pressure cavity **132** may exert a force on the outer pressurization surface **140** inwardly along the radial direction R.
(470) The seal support assembly **108** includes a pneumatic engagement assembly **560** operable to bias the first seal segment **110**A along the radial direction R during operation of the turbine engine. In particular, as will be explained in more detail below, the pneumatic engagement assembly **560** extends between the carrier **104** and the first seal segment **110**A to counter the pressure on the outer pressurization surface **140** during operation of the turbine engine. The pneumatic engagement assembly **560** may additionally provide such functionality while allowing for passive control of a radial clearance gap defined between the seal face **112** and the rotor **100** during operation of the turbine engine.
(471) As is depicted, the pneumatic engagement assembly **560** is, for the embodiment shown, in fluid communication with a high-pressure air supply. More specifically, for the embodiment shown, the high-pressure air supply is the high-pressure cavity **132** positioned at the high-pressure side **126** of the seal assembly **106**. The high-pressure cavity **132** is in fluid communication with a working gas flowpath defined by the turbine engine, at a high-pressure location of the turbine engine (e.g., a high-pressure compressor or a high-pressure turbine; see, e.g., FIG. **3**). More specifically, in the embodiment shown, the pneumatic engagement assembly **560** includes a high-pressure air duct **562** extending to the high-pressure side **126** of the seal assembly **106** and in fluid communication with the high-pressure cavity **132**. During operation, a high-pressure air flow may be provided through the high-pressure air duct **562** of the pneumatic engagement assembly **560**.
(472) It will be appreciated, however, that in other exemplary embodiments, the pneumatic

engagement assembly **560** may instead be in fluid communication with any other suitable high-pressure air source.
(473) More specifically, it will be appreciated that the pneumatic engagement assembly **560** defines a pressure chamber **564** and includes an extension member **566**. The extension member **566** is coupled to the first seal segment **110**A and is movable along the radial direction R. Moreover, the extension member **566** is positioned at least partially within the pressure chamber **564**.
(474) Referring briefly also to FIG. **127**, providing a close-up view of the pneumatic engagement assembly **560** of FIGS. **125** and **126** (in the position depicted in FIG. **125**), the pressure chamber **564** and extension member **566** generally form a piston arrangement. In particular, the extension member **566** includes a pressure head **568** having a pressure surface positioned within the pressure chamber **564**. The pressure head **568** defines a head crosswise measure **570** and the pressure chamber **564** defines a chamber crosswise measure **572**. The head crosswise measure **570** is within 5% of the chamber crosswise measure **572** (e.g., the head crosswise measure **570** equals chamber crosswise measure **572** plus or minus chamber crosswise measure **572** times 0.05). In such a manner, it will be appreciated that the pressure head **568** may generally form a seal with a wall of the pressure chamber **564**.
(475) The pressure chamber **564** generally includes a high-pressure side **574** and a low-pressure side **576**, separated by the pressure head **568**. The high-pressure side **574** is in fluid communication with the high-pressure air duct **562** for receiving the high-pressure airflow during operation the turbine engine. The pneumatic engagement assembly **560** further defines a low-pressure air duct **578**. The low-pressure air duct **578** is in fluid communication with the low-pressure side **576** of the pressure chamber **564**. In the embodiment shown, the low-pressure air duct **578** is in fluid communication with a low-pressure cavity **580** located outward of the first seal segment **110**A and inward of the carrier **104**. As explained in more detail above, the first seal segment **110**A forms a seal with a radial extension **150** of the carrier **104** to eliminate or reduce a high-pressure airflow from the high-pressure cavity **132** to the low-pressure cavity **580** (see the circumferential seal **172** in FIGS. **125** and **126**).
(476) Further, the extension member **566** generally includes rod **582** extending from the pressure head **568** towards the first seal segment **110**A. The pneumatic engagement assembly **560** includes a spring extension **142** positioned within the pressure chamber **564** to bias the pressure head **568** outward along the radial direction R. During operation of the turbine engine a pressure difference between the high-pressure side **574** of the pressure chamber **564** and the low-pressure side **576** of the pressure chamber **564** may generate a downward force along the radial direction R that may overcome the outward force exerted on the pressure head **568** by the spring extension **142** (see, e.g., FIG. **126**).
(477) Referring still to FIG. **127**, the extension member **566** is coupled to the first seal segment **110**A, such that movement of the pressure head **568** and extension member **566** along the radial direction R within the pressure chamber **564** may correspondingly move the first seal segment **110**A along the radial direction R.
(478) In particular, for the embodiment depicted, the extension member **566** includes an engagement guide **584** positioned at an end of the rod **582**, more specifically, positioned at an inner end of the rod **582** along the radial direction R. The first seal segment **110**A includes a hanger **586**, with the hanger **586** defining a cavity **588**. More specifically, the hanger **586** includes a tangential extension **590** spaced from the body **138** of the first seal segment **110**A along the radial direction R. The separation of the tangential extension **590** from the body **138** forms the cavity **588**, such that the cavity **588** is defined between the tangential extension **590** and the body **138**. The engagement guide **584** of the extension member **566** is positioned within the cavity **588** of the hanger **586**.
(479) As will be appreciated, for the embodiment shown, the cavity **588** defines a height **592** along the radial direction R. Similarly, the engagement guide **584** defines the thickness **594** along the radial direction R. The height **592** of the cavity **588** is greater than the thickness **594** of the

engagement guide **584** along the radial direction R. Such a configuration may allow for the first seal segment **110**A to form a hydrodynamic fluid bearing with the rotor **100** during operation of the turbine engine, e.g., at a high-power operating condition (see, e.g., FIG. **126**). For example, the first seal segment **110**A may move slightly along the radial direction R relative to the extension member **566** during such an operating condition (see, e.g., FIG. **126**).

(480) In at least certain exemplary embodiments, such as the embodiment of FIGS. **125** through **127**, the engagement guide **584** may be a circular member (e.g., as viewed along the radial direction R) and the cavity **588** may similarly be a circular cavity (e.g., as viewed along the radial direction R).

(481) It will be appreciated, however, that another exemplary embodiments, the pneumatic engagement assembly **560** may have any other suitable configuration.

(482) For example, referring now briefly to FIG. **128**, pneumatic engagement assembly **560** in accordance with another example embodiment of the present disclosure is provided. The pneumatic engagement simile of FIG. **128** may be configured in a similar manner as exemplary pneumatic engagement assembly **560** of FIGS. **125** and **126**. However, for the embodiment of FIG. **128**, the pneumatic engagement assembly **560** does not include a spring extension **142** extending from a pressure head **568**. Instead, pneumatic engagement assembly **560** includes a bellows **596** extending from the pressure head **568** and positioned around a rod **582** of the extension member **566** within the pressure chamber **564**. With such a configuration, the pressure head **568** may not form a seal with a wall of the pressure chamber **564**. Further, with such configuration, a high-pressure side **574** of the pressure chamber **564** may be a portion of the pressure chamber **564** located outward of the pressure head **568** along the radial direction R and outside of the bellows **596**. The low-pressure side **576** of the pressure chamber **564** may be located inward of the bellows **596**, between the bellows **596** and the rod **582**. Notably, the pneumatic engagement assembly **560** includes one or more openings to the pressure chamber **564** to provide a low-pressure airflow to the low-pressure side **576** of the pressure chamber **564**. More specifically, for the embodiment depicted, the one or more openings are defined around the rod **582** of the extension member **566**, and open up into the low-pressure side **576** of the pressure chamber **564**. In such a manner, the openings may be considered a low-pressure air duct **578** of the pneumatic engagement assembly **560**.

(483) Further, referring now to FIG. **129**, a pneumatic engagement assembly **560** in accordance with another exemplary embodiment of the present disclosure is provided. The pneumatic engagement assembly **560** of FIG. **129** may be configured in a similar manner as exemplary pneumatic engagement assembly **560** of FIG. **128**. However, for the embodiment depicted, an engagement guide **584** of the extension member **566** defines a thickness **594** along the radial direction R at least equal to a height **592** of a cavity **588** of a hanger **586** of a first seal segment **110**A. Further, for the embodiment of FIG. **129**, an inner surface **598** along the radial direction R of the engagement guide **584** defines a rounded profile. Such a configuration may allow for first seal segment **110**A to twist slightly relative to the extension member **566** during operation of the turbine engine.

(484) Further, still, referring now to FIG. **130**, a pneumatic engagement assembly **560** in accordance with yet another exemplary embodiment of the present disclosure is provided. The pneumatic engagement assembly **560** of FIG. **130** may also be configured in a similar manner as exemplary pneumatic engagement assembly **560** of FIG. **128**. However, for the embodiment of FIG. **130**, the engagement guide **584** is a spherical member, and a cavity **588** of a hanger **586** of a first seal segment **110**A defines a complementary spherical shape. Such a configuration may allow for the first seal segment **110**A to twist relative to the extension member **566** and/or rotate relative to the extension member **566**.

(485) Referring now to FIGS. **131** and **132**, a pneumatic engagement assembly **560** in accordance with yet another exemplary embodiment of the present disclosure is provided. The exemplary pneumatic engagement assembly **560** of FIGS. **131** and **132** may be configured in a similar manner

as exemplary pneumatic engagement assembly **560** of FIGS. **125** and **126**.
(486) In particular, FIG. **131** depicts an assembly in accordance with an exemplary embodiment of the present disclosure in a first position and FIG. **132** depicts the assembly of FIG. **131** and a second position. The first position may correspond to a low power operating condition of a turbine engine, and the second position may correspond to a high power operating condition of the turbine engine.
(487) The embodiment of FIGS. **131** and **132** includes a seal support assembly **108** having a pneumatic engagement assembly **560**. The exemplary pneumatic engagement assembly **560** includes a pressure chamber **564** similar to the embodiment described above. However, for the embodiment of FIGS. **131** and **132**, the pressure chamber **564** is a first pressure chamber **564**A and the pneumatic engagement assembly **560** further includes a second pressure chamber **564**B. The second pressure chamber **564**B is located downstream of the first pressure chamber **564**A.
(488) Additionally, the pneumatic engagement assembly **560** includes a retraction member **600** positioned within the first pressure chamber **564**A movable between a first position (see FIG. **131**) and a second position (see FIG. **132**) along a length of the first pressure chamber **564**A. For the embodiment depicted, the retraction member **600** includes a first pressure head **602** and a spring extension **604**. The first pressure chamber **564**A defines a high-pressure side **574** and low-pressure side **576** on opposing sides of the pressure head **568**. In the embodiment depicted, the spring extension **604** is positioned on the low-pressure side **576**. The pressure head **568** may form a seal with a wall of the first pressure chamber **564**A, creating a fluid seal between the low-pressure side **576** and the high-pressure side **574**. The high-pressure side **574** is in fluid communication with a high-pressure air source, such as a high-pressure cavity **132** through a high-pressure air duct **562** of the pneumatic engagement assembly **560**.
(489) The first pressure chamber **564**A further defines an air outlet **606** in fluid communication with the second pressure chamber **564**B. The air outlet **606** is located at a location between the first position and the second position of the retraction member **600**. In such a manner, when the first pressure chamber **564**A receives a high-pressure airflow at a sufficiently high-pressure to overcome a spring force of the spring extension **604**, moving the retraction member **600** from the first position (see FIG. **131**) to the second position (see FIG. **132**), the air outlet of the first pressure chamber **564**A may become in fluid communication with the high-pressure air source through the first pressure chamber **564**A and high-pressure air duct **562**.
(490) Referring particularly to FIG. **132**, when the retraction member **600** is moved to the second position, the high-pressure air may flow through the air outlet **606** of the first pressure chamber **564**A to the second pressure chamber **564**B. The second pressure chamber **564**B may be configured in a similar manner as the pressure chamber **564** described above with reference to FIGS. **125** and **126**. Accordingly, in at least the embodiment depicted, the pneumatic engagement assembly **560** includes a second extension member **608** positioned at least partially within the second pressure chamber **564**B, with the second extension member **608** including a second pressure head **610** in a rod **582** extending along the radial direction R from the second pressure head **610** to the first seal segment **110**A. The second pressure head **610** may form a seal with a wall of the second pressure chamber **564**B. A spring extension **612** is positioned within the second pressure chamber **564**B on a low-pressure side **576** of the second pressure chamber **564**B to bias the second extension member **608** outward along the radial direction R. A spring resistance of the spring extension **612** position within the second pressure chamber **564**B may be less than a spring resistance of the spring extension **604** positioned within the first pressure chamber **564**A.
(491) Accordingly, in the embodiment depicted, once a pressure of the airflow provided to the first pressure chamber **564**A is sufficient to move the retraction member **600** to the second position (see FIG. **132**), the airflow may further be sufficient to move the second extension member **608** downward along the radial direction R to close a radial gap defined between a seal face **112** of the first seal segment **110**A and the rotor **100**. In such a manner, the pneumatic engagement assembly

560 may provide a substantially stepwise movement of the first seal segment **110**A.
(492) In still other exemplary embodiments, the pneumatic engagement assembly **560** may have still other suitable configurations. For example, referring now to FIGS. **133** and **134** an assembly in accordance with another example embodiment of the present disclosure is provided. The exemplary assembly of FIGS. **133** **134** may be configured in a similar manner as one or the exemplary embodiments described above.
(493) For example, the exemplary embodiment of FIGS. **133** and **134** includes a seal support assembly **108** having a pneumatic engagement assembly **560**. The pneumatic engagement assembly **560** includes a pressure chamber **564** and an extension member **566** positioned at least partially within the pressure chamber **564**. More specifically, the extension member **566** is movable along the radial direction R, and includes a pressure head **568** positioned within the pressure chamber **564**. Further, the pressure chamber **564** defines a high-pressure side **574** and a low-pressure side **576** on opposing sides of the pressure head **568**. However, for the embodiment of FIGS. **133** and **134**, the high-pressure side **574** of the pressure chamber **564** is positioned inward along the radial direction R from the low-pressure side **576**. Accordingly, as higher pressure airflow is provided to the high-pressure side **574** of the pressure chamber **564**, the extension member **566** is configured to move outwardly along the radial direction R. In order to have such movement translate to inward movement of the first seal segment **110**A, the pneumatic engagement assembly **560** includes a lever assembly **614**, with the extension member **566** coupled to the first seal segment **110**A through the lever assembly **614**. For the embodiment shown, the lever assembly **614** generally includes a lever extension **616** and a radial extension **618** extending from the carrier **104** and defining a pivot point **620** for the lever extension **616**. The lever extension **616** extends between the radial extension **618** and an attachment point **622** to the first seal segment **110**A.
(494) As will be appreciated, the view of FIG. **133** may represent a position of the first seal segment **110**A during a low-power operating condition of the turbine engine, and the view of FIG. **134** may represent a position of the first seal segment **110**A during a high-power operating condition of the turbine engine.
(495) Referring now to FIG. **135**, a pneumatic engagement assembly **560** in accordance with another exemplary embodiment of the present disclosure is provided. The exemplary embodiment of FIG. **135** may be configured in a similar manner as one or the exemplary embodiments described above.
(496) For example, the embodiment of FIG. **135** includes a seal assembly **106**, a carrier **104**, and a seal support assembly **108** configured to support the seal assembly **106** relative to the carrier **104**. In particular, the seal assembly **106** includes a first seal segment **110**A. Further, the seal support assembly **108** includes the pneumatic engagement assembly **560**.
(497) However, for the embodiment of FIG. **135**, the pneumatic engagement assembly **560** includes a diaphragm **624** positioned between a high-pressure cavity **132** and a low-pressure cavity **580**. In addition, the pneumatic engagement assembly **560** includes a first engagement rod **626**, a second engagement rod **628**, a third engagement rod **630**, and a spring extension **142**. The first engagement rod **626** is coupled to the diaphragm **624** and configured to move in a direction perpendicular to a radial direction R of the turbine engine. The second engagement rod **628** and third engagement rod **630** are pivotably coupled to the first engagement rod **626**. In particular, for the embodiment shown, the first engagement rod **626** extends between a first end **632** coupled to the diaphragm **624** and a second end **634** defining a pivot connection **636** with the second engagement rod **628** and the third engagement rod **630**. The second engagement rod **628** is further coupled to the carrier **104** and the third engagement rod **630** is further coupled to the first seal segment **110**A.
(498) The spring extension **142** extends between the carrier **104** and the first seal segment **110**A to bias the first seal segment **110**A outwardly along the radial direction R.
(499) During operation, an increase in pressure in the high-pressure cavity **132** relative to the low-pressure cavity **580** may cause the diaphragm **624** to deflect towards the low-pressure cavity **580**,

moving the rod **582** further into low-pressure cavity **580**, and pressing the second engagement rod **628** and third engagement rod **630** away from one another to move the first seal segment **110**A inwardly along the radial direction R relative to the carrier **104**.

(500) Notably, in other embodiments, the spring extension **142** may additionally or alternatively be coupled to the second engagement rod **628** and/or the third engagement rod **630**.

(501) Further, referring now to FIGS. **136** and **137**, an assembly in accordance with another exemplary embodiment of the present disclosure is provided. The assembly of FIGS. **136** and **137** may be configured in a similar manner as one or more of the embodiments described hereinabove. The view of FIG. **136** is in a reference plane defined by an axial direction A of the turbine engine and a radial direction R of the turbine engine. The view of FIG. **137** is in a reference plane defined by the axial direction A and a circumferential direction C of the turbine engine.

(502) For example, the embodiment of FIGS. **136** and **137** includes a seal assembly **106**, a carrier **104**, and a seal support assembly **108** configured to support the seal assembly **106** relative to the carrier **104**. In particular, the seal assembly **106** includes a first seal segment **110**A. Further, the seal support assembly **108** includes a pneumatic engagement assembly **560**. However, for the embodiment of FIG. **135**, the pneumatic engagement assembly **560** includes a pressurized sliding chamber **640**. The pressurized sliding chamber **640** includes a slider **642** positioned at least partially within the pressurized sliding chamber **640**. The pressurized sliding chamber **640** defines a high-pressure side **644** and a low-pressure side **646** on opposing sides of the slider **642**. The high-pressure side **644** may be in fluid communication with a high-pressure cavity **132** and the low-pressure side **646** may be in fluid communication with a low-pressure cavity **580**. The high-pressure cavity **132** may be in fluid communication with a working gas flowpath of the turbine engine, such that as the turbine engine moves from a low-power operating condition to a high-power operating condition, a pressure within the high-pressure cavity **132** (and within the high-pressure side **644** of the pressurized sliding chamber **640**) increases.

(503) The slider **642** includes a radial extension **650** extending inwardly along the radial direction R to the first seal segment **110**A. The first seal segment **110**A defines an engagement surface **652** having a slope relative to a direction in which the pressurized sliding chamber **640** extends. More particularly, for the embodiment depicted the pressurized sliding chamber **640** extends in the axial direction A and the first seal segment **110**A defines the engagement surface **652** having the slope relative to the axial direction A.

(504) A spring extension **142** extends between the first seal segment **110**A and the carrier **104** to bias the first seal segment **110**A outwardly along the radial direction R.

(505) In such a manner, it will be appreciated that as a pressure within the high-pressure side **644** of the pressurized sliding chamber **640** increases, the slider **642** may move along a length of the pressurized sliding chamber **640**, and the radial extension **650** may engage with the engagement surface **652** of the first seal segment **110**A, moving the first seal segment **110**A inwardly along the radial direction R.

(506) Moreover, in still other exemplary embodiments a pneumatic engagement assembly **560** may have still other suitable configurations. For example, referring now to FIGS. **138** and **139**, another example embodiment of the present disclosure is provided, with a pneumatic engagement assembly **560** moved between a first position (see FIG. **138**) and a second position see FIG. **139**). The embodiment of FIGS. **138** and **139** may be configured in substantially the same manner as the example embodiment described above with reference to FIGS. **136** and **137**. For example, the exemplary embodiment of FIGS. **138** and **139** includes the pneumatic engagement assembly **560** defining a pressurized sliding chamber **640** coupled to the carrier **104** and including a slider **642** position at least partially within the pressurized sliding chamber **640**.

(507) Additionally, the pneumatic engagement assembly **560** includes a radial member **654** extending to the first seal segment **110**A. The slider **642**, however, in the embodiment depicted, includes an engagement surface **656** having a slope relative to a direction in which the pressurized

sliding chamber **640** extends. In particular, for the embodiment depicted, the slider **642** includes the engagement surface **656** having the slope relative to the axial direction A. The engagement surface **656** is engaged with the radial member **654** to move the first seal segment **110**A along the radial direction R relative to the carrier **104**. A spring extension **660** is provided to bias the radial member **654** outward along the radial direction R.

(508) In particular, as with the embodiment above, the pressurized sliding chamber **640** defines a high-pressure side **644** and a low-pressure side **646** on opposing sides of the slider **642**, with a pressure differential from the high-pressure side **644** to the low-pressure side **646** driving the slider **642**, and in turn moving the first seal segment **110**A along the radial direction R relative to the carrier **104**. A spring extension **658** is provided within the pressurized sliding chamber **640** to bias the slider **642** towards the high-pressure side **644** of the pressurized sliding chamber **640**.

(509) As will be appreciated, the view of FIG. **138** may correspond to a low power operating condition of the turbine engine, and the view of FIG. **139** may correspond to a high-power operating condition of the turbine engine.

(510) Further, referring briefly to FIGS. **140** and **141**, yet another example embodiment of the present disclosure is provided. The exemplary embodiment of FIGS. **140** and **141** may be configured in substantially the same manner as the exemplary embodiment of FIGS. **138** and **139**. However, for the embodiment of FIGS. **140** and **141**, a radial member is configured to move within an engagement slot **662** defined within the slider **642**.

(511) Referring now to FIG. **142**, an assembly in accordance with another embodiment of the present disclosure is provided. The embodiment of FIG. **142** may be configured in a similar manner as one or the embodiments described herein. For example, the embodiment of FIG. **142** includes a stator **102** having a carrier **104**, a rotor **100**, a seal assembly **106** including a first seal segment **110**A and positioned between the carrier **104** and the rotor **100**, and a seal support assembly **108**. The seal support assembly **108** includes a pneumatic engagement assembly **560** (see FIG. **143**, below).

(512) Referring also to FIG. **143**, providing a schematic, cross-sectional view of a portion of the assembly of FIG. **142** along Line **143-143** in FIG. **142**, it will be appreciated that the pneumatic engagement assembly **560** further includes a unison ring **664**. More specifically, the seal assembly **106** includes a plurality of seal segments **110**, the plurality of seal segments **110** including the first seal segment **110**A and a second seal segment **110**B (the remaining seal segments **110** are omitted in FIG. **143** for clarity). Each of the plurality of seal segments **110** is slidably coupled to the unison ring **664**, as will be described in more detail below.

(513) Further, the pneumatic engagement assembly **560** includes a pneumatic piston assembly **686** having a pressure chamber **564** and an extension member **566** positionally at least partially within the pressure chamber **564**. The extension member **566** includes a pressure head **568** and a rod **582** extending from the pressure head **568**. The pressure chamber **564** defines a high-pressure side **644** and a low-pressure side **646** on opposing sides of the pressure head **568**. The unison ring **664** is coupled to the rod **582** at a pivotable attachment point **668**. Currently, an increase in a pressure change across the high-pressure side **644** and low-pressure side **646** within the pressure chamber **564** may move the rod **582**, and therefore may rotate the unison ring **664**.

(514) The unison ring **664** defines a plurality of angled slots **670** relative to the radial direction R. The unison ring **664** is slidably coupled to the plurality of seal segments **110** through the plurality of angled slots **670** and the pneumatic engagement assembly **560** is operable to rotate the unison ring **664** along a circumferential direction C of the turbine engine to move the plurality of seal segments **110** along the radial direction R. In particular, as will be appreciated from the description above, the pneumatic piston assembly **666** of the pneumatic engagement assembly **560** is configured for rotating the unison ring **664** along the circumferential direction C based on a pressure differential within the pneumatic piston assembly **666**.

(515) Briefly, referring back to FIG. **142**, it will be appreciated that in at least certain exemplary

embodiments, the pneumatic engagement assembly **560** may include a plurality of unison rings **664**. For example, the unison ring **664** may be a first unison ring **664**A and the pneumatic engagement assembly **560** may further include a second unison ring **664**B slidably coupled to each of the plurality of seal segments **110** in a similar manner as the first unison ring **664**A. The first unison ring **664**A and second unison ring **664**B are spaced from one another along an axial direction A of the turbine engine for the embodiment depicted. As will be appreciated, the embodiment depicted, the unison ring **664** extends 360 degrees in the circumferential direction C.
(516) Inclusion of a unison ring **664** in accordance with such an exemplary embodiment may allow for a single pneumatic actuator to move a plurality of seal segments **110** along the radial direction R, potentially reducing a complexity and weight of the turbine engine.
(517) Notably, although the unison rings of FIGS. **142** and **143** are depicted as part of a pneumatic engagement assembly, in other exemplary embodiments, the unison ring may be utilized with one or more spring actuators/spring assemblies, magnet assemblies, or the like.
(518) Further aspects are provided by the subject matter of the following clauses:
(519) A turbine engine defining an axial direction and a radial direction, the turbine engine comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly defining a high-pressure side and a low-pressure side and comprising a plurality of seal segments, the plurality of seal segments including a seal segment having a seal face forming a fluid bearing with the rotor, a lip, and a body, the lip extending from the body along the axial direction of the turbine engine on the high-pressure side and including an outer pressurization surface along the radial direction of the turbine engine; and a seal support assembly comprising a spring arrangement extending between the carrier and the seal segment to counter a pressure on the outer pressurization surface during operation of the turbine engine.
(520) The turbine engine of any preceding clause, wherein the spring arrangement comprises plurality of spring extensions extending between the carrier and the seal segment.
(521) The turbine engine of any preceding clause, wherein the seal segment defines a slot extending along the axial direction and a circumferential direction of the turbine engine, and wherein the plurality of spring extensions each includes a distal end positioned in the slot of the seal segment.
(522) The turbine engine of any preceding clause, wherein the seal segment and the distal ends of the plurality of spring extensions define a clearance gap along the axial direction.
(523) The turbine engine of any preceding clause, wherein the spring arrangement further includes a base extending along a circumferential direction, and wherein the plurality of spring extensions each extend from the base and are spaced along the circumferential direction.
(524) The turbine engine of any preceding clause, wherein the base is coupled to the carrier.
(525) The turbine engine of any preceding clause, wherein the seal support assembly comprises a plurality of spring arrangements spaced along a circumferential direction and an annular flange extending along the circumferential direction, wherein the annular flange is positioned between the plurality of spring arrangements and the carrier.
(526) The turbine engine of any preceding clause, wherein the annular flange extends at least 170 degrees along the circumferential direction.
(527) The turbine engine of any preceding clause, wherein a base of the spring arrangement is coupled to or formed integrally with the carrier proximate the high-pressure side of the seal assembly, and wherein the plurality of spring extensions contact the seal segment proximate the low-pressure side.
(528) The turbine engine of any preceding clause, wherein the plurality of spring extensions are fixedly coupled to, or formed integrally with, the seal segment.
(529) The turbine engine of any preceding clause, wherein the spring arrangement comprises a spring extension and a base, wherein the base is coupled to the carrier, wherein the spring extension has a distal end contacting the seal segment.

(530) The turbine engine of any preceding clause, wherein the spring extension includes a first segment and a second segment extending at least partially along the axial direction of the turbine engine.
(531) The turbine engine of any preceding clause, wherein the distal end of the spring extension contacts the seal segment at a location aligned along the axial direction of the turbine engine with an axial half-way point of the seal face.
(532) The turbine engine of any preceding clause, wherein the carrier defines a radial face, and wherein the base is coupled to the carrier at the radial face.
(533) The turbine engine of any preceding clause, wherein the spring arrangement comprises a spring extension and a base, wherein the base is coupled to the carrier, wherein the spring extension has a first segment and a second segment, wherein the first and second segments extend in parallel to the seal segment.
(534) The turbine engine of any preceding clause, wherein the first segment contacts the seal segment proximate the high-pressure side of the seal assembly, and wherein the second segment contacts the seal segment proximate the low-pressure side of the seal assembly.
(535) The turbine engine of any preceding clause, wherein the seal segment defines a first slot extending along the axial direction and a circumferential direction of the turbine engine from the high-pressure side of the seal assembly, wherein the seal segment further defines a second slot extending along the axial direction and the circumferential direction from the low-pressure side of the seal assembly, wherein the first segment includes a first distal end positioned in the first slot, and wherein the second segment includes a second distal end positioned in the second slot.
(536) The turbine engine of any preceding clause, wherein the seal segment is a first seal segment, wherein the seal assembly further includes a second seal segment positioned adjacent the seal segment along a circumferential direction of the turbine engine and a piston seal extending along the circumferential direction and positioned between the carrier and the first and second seal segments.
(537) The turbine engine of any preceding clause, wherein the spring arrangement comprises a spring extension extending between the carrier and the seal segment, the spring extension defining at least two points of contact with the carrier, at least two points of contact with the seal segment, or both.
(538) The turbine engine of any preceding clause, wherein the spring extension includes at least two seal segments defining the at least two points of contact with the carrier.
(539) The turbine engine of any preceding clause, wherein the spring extension includes at least two seal segments defining the at least two points of contact with the seal segment.
(540) The turbine engine of any preceding clause, wherein the spring extension comprises a first segment and a second segment separately extending to the seal segment and defining in part a first point of contact and a second point of contact with the seal segment.
(541) The turbine engine of any preceding clause, wherein the first segment and the second segment each define a length along the axial direction between 5% and 100% of an axial length of a seal face of the seal segment at the first point of contact and at the second point of contact respectively.
(542) The turbine engine of any preceding clause, wherein the spring extension further comprises a third segment and a fourth segment separately extending to the carrier and defining in part a third point of contact and a fourth point of contact with the carrier.
(543) The turbine engine of any preceding clause, wherein the spring extension further comprises a central segment, and wherein the first segment, the second segment, the third segment, and the fourth segment each extend from the central segment.
(544) The turbine engine of any preceding clause, wherein the spring extension extends between a first end and a second end, and wherein the first and second ends define the at least two points of contact with the seal segment.

(545) The turbine engine of any preceding clause, wherein the spring extension defines a point of contact with the carrier at a location between the first and second ends.
(546) The turbine engine of any preceding clause, wherein the spring extension extends between a first end and a second end, and wherein the first and second ends define the at least two points of contact with the carrier.
(547) The turbine engine of any preceding clause, wherein the first and second ends are connected to the carrier through a hinged pin connection, a sliding pin connection, or a combination thereof.
(548) The turbine engine of any preceding clause, wherein the seal support assembly comprises a spring arrangement having a spring extension extending between the carrier and the plurality of seal segments, wherein the spring extension extends continuously across the plurality of seal segments in a circumferential direction.
(549) The turbine engine of any preceding clause, wherein each seal segment of the plurality of seal segments includes a first tab defining a radially inward surface, and wherein the spring extension contacts the radially inward surface.
(550) The turbine engine of any preceding clause, wherein each seal segment further includes a second tab defining a radially outer surface, and wherein the spring extension further contacts the radially outward surface.
(551) The turbine engine of any preceding clause, wherein the carrier comprises a carrier axial extension, and wherein the seal segment comprises a segment axial extension located outward of the carrier axial extension, and wherein the spring arrangement is positioned within a gap defined between the carrier axial extension and the segment axial extension.
(552) The turbine engine of any preceding clause, wherein the spring arrangement is a plate having a support and plurality of spring segments extending from the support.
(553) The turbine engine of any preceding clause, wherein the plurality of spring segments are cantilevered from the support.
(554) The turbine engine of any preceding clause, wherein the spring arrangement comprises a plate spring coupled to the carrier and to the seal segment.
(555) The turbine engine of any preceding clause, wherein the plate spring extends within a reference plane perpendicular to the radial direction of the turbine engine.
(556) The turbine engine of any preceding clause, wherein the plate spring defines a rectangular shape in the reference plane.
(557) The turbine engine of any preceding clause, wherein the plate spring defines a circular shape in the reference plane.
(558) The turbine engine of any preceding clause, wherein the carrier comprises a forward wall and an aft wall, and wherein the plate spring is coupled to the forward wall and to the aft wall of the carrier.
(559) The turbine engine of any preceding clause, wherein the plate spring is coupled to the seal segment at a center of the plate spring.
(560) The turbine engine of any preceding clause, wherein the seal segment includes an attachment column extending outward along the radial direction of the turbine engine, wherein the plate spring is coupled to the attachment column.
(561) The turbine engine of any preceding clause, wherein the attachment column comprises a ledge with a post extending outwardly along the radial direction from the ledge, wherein the post extends through the plate spring and wherein the plate spring is pressed against the ledge.
(562) The turbine engine of any preceding clause, wherein the spring arrangement further includes a deflection limiter positioned outward of the plate spring along the radial direction of the turbine engine, inward of the plate spring along the radial direction of the turbine engine, or both.
(563) The turbine engine of any preceding clause, wherein the plate spring comprises a plurality of flex members extending from a perimeter to one or more attachment points defined with the seal segment.

(564) The turbine engine of any preceding clause, wherein the plurality of flex members includes a first flex member extending from the perimeter to a first attachment point defined with the seal segment and a second flex member extending from the perimeter to a second attachment point defined with the seal segment.
(565) The turbine engine of any preceding clause, wherein the first attachment point is spaced from the second attachment point along the axial direction of the turbine engine.
(566) The turbine engine of any preceding clause, wherein a first flex member of the plurality of flex members defines a variable geometry along a length of the first flex member.
(567) The turbine engine of any preceding clause, wherein the plate spring is coupled to the seal segment in a pre-strained condition.
(568) The turbine engine of any preceding clause, wherein the plate spring is moveable between a free condition and an engaged position, and wherein the plate spring moves towards a more planar geometry when moved from the free condition to the engaged position.
(569) The turbine engine of any preceding clause, wherein the seal plate is bolted to the carrier, press fit to the carrier, or both.
(570) The turbine engine of any preceding clause, wherein the spring arrangement comprises one or more elements formed of shape memory alloy material, a bimetallic material, or both.
(571) The turbine engine of any preceding clause, wherein the spring arrangement comprises a first spring extension formed of a shape memory alloy material.
(572) The turbine engine of any preceding clause, wherein the shape memory alloy material is a temperature-dependent shape memory alloy material.
(573) The turbine engine of any preceding clause, wherein the shape memory alloy material is a strain-dependent shape memory alloy material.
(574) The turbine engine of any preceding clause, wherein the spring arrangement further comprises a second spring extension, wherein the first and second spring extensions extend between the carrier and the seal segment in series.
(575) The turbine engine of any preceding clause, wherein the spring arrangement further comprises a second spring extension, the shape memory alloy material is a first shape memory alloy material, and wherein the second spring extension is formed of a second shape memory alloy material different than the first shape memory alloy material.
(576) The turbine engine of any preceding clause, wherein the spring arrangement is in airflow communication with a working gas flowpath of the turbine engine.
(577) The turbine engine of any preceding clause, wherein the spring arrangement comprises a first spring extension formed of a bimetallic material.
(578) The turbine engine of any preceding clause, wherein the bimetallic material comprises a layer of a shape memory alloy material and a metal material.
(579) The turbine engine of any preceding clause, wherein the spring arrangement includes a first cam and a second cam, wherein the first and second cams are rotatable about a common rotational axis, and wherein the spring arrangement includes a spring extension extending between the first and second cams.
(580) The turbine engine of any preceding clause, wherein the first and second cams are rotatably coupled to one of the seal segment or the carrier, and wherein the first and second cams extend to and engage with a surface of the other of the seal segment or the carrier.
(581) The turbine engine of any preceding clause, wherein the spring arrangement includes a cam extending between a first end and a second end, wherein the cam defines a rotational axis, and wherein the spring arrangement includes a first spring extension extending from the cam at the first end or between the first end and the rotational axis and a second spring extending from the cam at the second end or between the second end and the rotational axis.
(582) The turbine engine of any preceding clause, wherein the first cam is rotatably coupled to one of the seal segment or the carrier, and wherein the first cam extends to and engage with a surface of

the other of the seal segment or the carrier.
(583) The turbine engine of any preceding clause, wherein the spring arrangement includes a spring extension arranged in tension.
(584) The turbine engine of any preceding clause, wherein the spring arrangement forms a spherical joint with the seal segment for connecting the spring arrangement to the seal segment.
(585) The turbine engine of any preceding clause, wherein the spring arrangement includes a first body extension fixed along the radial direction relative to the carrier and a second body extension fixed along the radial direction relative to the seal segment, and wherein the first body extension is slidable relative to the second body extension.
(586) The turbine engine of any preceding clause, wherein the spring extension is a helical spring extension arranged coaxially with the first and second body extensions.
(587) A turbine engine, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment, the seal segment having a seal face configured to form a fluid bearing with the rotor; and a seal support assembly extending between the carrier and the seal segment and comprising a flexible extension and a driver extension, the driver extension coupled to and extending from the flexible extension, the driver extension formed of a first material, the first material being a material defining a different coefficient of thermal expansion than a material forming the flexible extension, a shape memory alloy material, a bimetallic material, or a combination thereof, wherein the driver extension is positioned to move the flexible extension during operation of the turbine engine.
(588) The turbine engine of any preceding clause, wherein the flexible extension is a first flexible extension, wherein the seal support assembly further comprises a second flexible extension, and wherein the driver extension extends between the first and second flexible extensions.
(589) The turbine engine of any preceding clause, wherein the first and second flexible extensions each include a diverging section and a converging section, and wherein the driver extension is coupled to the first flexible extension at a location between the diverging and converging sections of the first flexible extension and to the second flexible extension at a location between the diverging and converging sections of the second flexible extension.
(590) The turbine engine of any preceding clause, wherein the seal support assembly comprises an outer support section along a radial direction of the turbine engine and an inner support section along the radial direction, wherein the driver extension extends between a first connection point with the first flexible extension and a second connection point with the second flexible extension, wherein the seal support assembly defines an extension axis along the radial direction, wherein the first and second flexible extensions are coupled to the outer support section at one or more locations closer to the extension axis than the first and second connection points.
(591) The turbine engine of any preceding clause, wherein the outer support section is coupled to the carrier, and wherein the inner support section is coupled to the seal segment.
(592) The turbine engine of any preceding clause, wherein the first material defines a coefficient of thermal expansion less than a coefficient of thermal expansion of the material forming the first and second flexible extensions.
(593) The turbine engine of any preceding clause, wherein the seal support assembly comprises an outer support section along a radial direction of the turbine engine and an inner support section along the radial direction, wherein the driver extension extends between a first connection point with the first flexible extension and a second connection point with the second flexible extension, wherein the seal support assembly defines an extension axis along the radial direction, wherein the first and second flexible extensions are coupled to the outer support section at a location farther from the extension axis than the first and second connection points, respectively.
(594) The turbine engine of any preceding clause, wherein the first material defines a coefficient of thermal expansion greater than a coefficient of thermal expansion of the material forming the first and second flexible extensions.

(595) The turbine engine of any preceding clause, wherein the seal support assembly is coupled to the seal segment through a hanger attachment, and wherein the seal support assembly comprises a leaf spring support coupling the flexible extension and driver extension to the seal segment.
(596) The turbine engine of any preceding clause, wherein the first material comprises a bimetallic material.
(597) The turbine engine of any preceding clause, wherein the first material comprises a shape memory alloy material.
(598) The turbine engine of any preceding clause, wherein the first flexible extension includes a diverging section and a converging section, and wherein the seal support assembly further comprises a support member extending between the converging and diverging sections.
(599) The turbine engine of any preceding clause, wherein the support member is a bimetallic member.
(600) The turbine engine of any preceding clause, wherein the driver extension and the flexible extension are arranged in series.
(601) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.
(602) The turbine engine of any preceding clause, wherein the high-pressure side is located forward of the low-pressure side.
(603) The turbine engine of any preceding clause, wherein the turbine defines in part a working gas flowpath of the engine, and wherein the seal support assembly is in airflow communication with the working gas flowpath.
(604) The turbine engine of any preceding clause, further comprising: a compressor and a turbine arranged in serial flow order and together defining in part a working gas flowpath, wherein the rotor is a turbine rotor of the turbine and wherein the seal support assembly is in airflow communication with a portion of the working gas flowpath defined by the compressor.
(605) The turbine engine of any preceding clause, wherein the seal face is in airflow communication with the high-pressure side of the seal assembly, and wherein the high-pressure side of the seal assembly is in airflow communication with the portion of the working gas flowpath defined by the compressor.
(606) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine, wherein the turbine defines in part a working gas flowpath of the engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.
(607) The turbine engine of any preceding clause, wherein the seal support assembly defines a resistance along the radial direction of the turbine engine, wherein the turbine engine is operable at a high power operating mode and at a low power operating mode, wherein the turbine engine defines a first high-pressure at the high-pressure side of the seal assembly when the turbine engine is operated at the high power operating mode and a second high-pressure at the high-pressure side of the seal assembly when the turbine engine is operated at the low power operating mode, wherein the seal support assembly holds the seal segment at a radial distance away from the rotor when the turbine engine defines the second high-pressure, and wherein the seal support assembly moves the seal segment towards the rotor when the turbine engine defines the first high-pressure.
(608) A turbine engine defining a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment having a seal face forming a fluid bearing with the rotor; and a seal support assembly comprising a torsional spring extension extending from the carrier, from the seal segment, or both, to bias the seal segment along the radial direction.
(609) The turbine engine of any preceding clause, wherein the seal support assembly further comprises a cam coupled to a distal end of the torsional spring extension.
(610) The turbine engine of any preceding clause, wherein the torsional spring extension is coupled

at a base to the carrier.
(611) The turbine engine of any preceding clause, wherein the seal segment includes a support extension extending at least partially along an axial direction of the turbine engine, the support extension defining an inner support surface along the radial direction, and wherein the cam is positioned to engage the inner support surface.
(612) The turbine engine of any preceding clause, wherein the cam defines an outer engagement surface, and wherein the outer engagement surface has a pear shape.
(613) The turbine engine of any preceding clause, wherein the cam defines an outer engagement surface, and wherein the outer engagement surface has a non-pear shape.
(614) The turbine engine of any preceding clause, wherein the seal assembly defines a high-pressure side and a low-pressure side, wherein the torsional spring extension is a first torsional spring extension positioned at the high-pressure side, and wherein the seal support assembly further comprises a second torsional spring extension positioned at the low-pressure side.
(615) The turbine engine of any preceding clause, wherein the seal support assembly further comprises a first cam coupled to a distal end of the first torsional spring extension and a second cam coupled to a distal end of the second torsional spring extension, wherein the seal segment defines at least two inner support surfaces, and wherein the first and second cams are each positioned to engage a respective inner support surface of the seal segment.
(616) The turbine engine of any preceding clause, wherein the torsional spring extension is a first torsional spring extension, and wherein the seal support assembly further comprises a second torsional spring extension spaced from the first torsional spring extension along a circumferential direction.
(617) The turbine engine of any preceding clause, wherein the seal support assembly further comprises a first cam coupled to a distal end of the first torsional spring extension and a second cam coupled to a distal end of the second torsional spring extension, wherein the seal segment defines an inner support surface, and wherein the first and second cams are each positioned to engage the inner support surface of the seal segment.
(618) The turbine engine of any preceding clause, wherein the first and second cams are configured to rotate in a common circumferential direction in an aft looking forward view.
(619) The turbine engine of any preceding clause, wherein the first and second cams are configured to rotate in opposite circumferential directions in a forward looking aft view.
(620) The turbine engine of any preceding clause, wherein the seal support assembly further comprises a conjugate cam coupled to a distal end of the torsional spring extension, wherein the conjugate cam comprises a first cam member and a second cam member, and wherein the first and second cam members define different shapes, different orientations, or both.
(621) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.
(622) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, and wherein the high-pressure side is located forward of the low-pressure side.
(623) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the seal segment includes a lip and a body, wherein the lip extends from the body along an axial direction of the turbine engine on the high-pressure side, and wherein the lip includes an outer pressurization surface along the radial direction of the turbine engine.
(624) The turbine engine of any preceding clause, further comprising: a compressor, wherein the turbine and the compressor together define in part a working gas flowpath of the turbine engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.
(625) The turbine engine of any preceding clause, further comprising: a compressor and a turbine

arranged in serial flow order and together defining in part a working gas flowpath, wherein the rotor is a turbine rotor of the turbine and wherein the seal support assembly is in airflow communication with a portion of the working gas flowpath defined by the compressor.
(626) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the seal face is in airflow communication with the high-pressure side of the seal assembly, and wherein the high-pressure side of the seal assembly is in airflow communication with the portion of the working gas flowpath defined by the compressor.
(627) A turbine engine defining a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment having a seal face forming a fluid bearing with the rotor; and a seal support assembly comprising a spring extension and a cam coupled to an end of the spring extension, the seal support assembly extending between the carrier and the seal segment to bias the seal segment along the radial direction.
(628) A turbine engine defining a circumferential direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a plurality of seal segments supported at least in part by the carrier, the plurality of seal segments having a first seal segment and a second seal segment, the first and second seal segments each having a seal face forming a fluid bearing with the rotor; and a seal support assembly comprising a tangential spring extension extending between the first seal segment and the second seal segment for biasing the first seal segment away from the second seal segment in the circumferential direction
(629) The turbine engine of any preceding clause, wherein the first seal segment extends between a first circumferential end and a second circumferential end, wherein the second seal segment extends between a first circumferential end and a second circumferential end of the second seal segment, wherein the tangential spring extension is coupled to the first seal segment proximate the second circumferential end and to the second seal segment proximate the first circumferential end.
(630) The turbine engine of any preceding clause, wherein the first seal segment defines a tangential slot at a circumferential end adjacent to the second seal segment, wherein the second seal segment defines a tangential slot at a circumferential end adjacent to the first seal segment, wherein the tangential spring extension extends between a first end positioned in the tangential slot of the first seal segment and a second end positioned in the tangential slot of the second seal segment.
(631) The turbine engine of any preceding clause, wherein the tangential spring extension defines in part a first point of contact with the first seal segment and a second point of contact with the second seal segment, wherein the tangential spring extension defines a length along an axial direction of the turbine engine between 5% and 100% of an axial length of the seal face of the first seal segment.
(632) The turbine engine of any preceding clause, wherein the tangential spring extension is a bent plate spring.
(633) The turbine engine of any preceding clause, wherein the bent plate spring includes a middle section, a first segment extending between the middle section and a first end, and a second segment extending between the middle section and a second end.
(634) The turbine engine of any preceding clause, wherein the first segment, the second segment, or both defines one or more stiffness modifiers.
(635) The turbine engine of any preceding clause, wherein the tangential spring extension is engaged with the carrier to support the seal assembly.
(636) The turbine engine of any preceding clause, wherein the seal support assembly includes a pin fixedly or slidably coupled to the carrier, and wherein the tangential spring extension is engaged with the pin.
(637) The turbine engine of any preceding clause, wherein the seal support assembly includes a radial spring, and wherein the tangential spring extension is engaged with the pin through the radial

spring.
(638) The turbine engine of any preceding clause, wherein the radial spring is positioned inward of the carrier along a radial direction of the turbine engine.
(639) The turbine engine of any preceding clause, wherein the radial spring is positioned at least partially outward of the carrier along a radial direction of the turbine engine.
(640) The turbine engine of any preceding clause, wherein the tangential spring extension is engaged with the carrier through a radial spring operable to bias the tangential spring extension outward along a radial direction.
(641) The turbine engine of any preceding clause, wherein the first seal segment defines a first circumferential end adjacent to the second seal segment, wherein the wherein the second seal segment defines a second circumferential end adjacent to the first seal segment, and wherein the tangential spring extension is engaged with the first and second circumferential ends.
(642) The turbine engine of any preceding clause, wherein the seal support assembly includes a wedge body, and wherein the tangential spring extension is a first tangential spring extension positioned on a first side of the wedge body and engaged with the first seal segment and wherein the seal support assembly further includes a second spring extension positioned on a second side of the wedge body and engaged with the second seal segment.
(643) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.
(644) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, and wherein the high-pressure side is located forward of the low-pressure side.
(645) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the first seal segment includes a lip and a body, wherein the lip extends from the body along an axial direction of the turbine engine on the high-pressure side, and wherein the lip includes an outer pressurization surface along a radial direction of the turbine engine.
(646) The turbine engine of any preceding clause, further comprising: a compressor, wherein the turbine and the compressor together define in part a working gas flowpath of the engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.
(647) The turbine engine of any preceding clause, further comprising: a compressor and a turbine arranged in serial flow order and together defining in part a working gas flowpath, wherein the rotor is a turbine rotor of the turbine and wherein the seal support assembly is in airflow communication with a portion of the working gas flowpath defined by the compressor.
(648) A turbine engine defining a circumferential direction and a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a plurality of seal segments supported at least in part by the carrier, the plurality of seal segments having a first seal segment and a second seal segment, the first and second seal segments each having a seal face forming a fluid bearing with the rotor; and a seal support assembly comprising: a first engagement assembly extending between the first seal segment and the carrier, the first engagement assembly operable to bias the first seal segment along the radial direction; a second engagement assembly extending between the second seal segment and the carrier, the second engagement assembly operable to bias the second seal segment along the radial direction; and a tangential spring extension extending between the first engagement assembly and the second engagement assembly for biasing the first seal segment relative to the second seal segment in the circumferential direction.
(649) The turbine engine of any preceding clause, wherein the first engagement assembly includes a first radial extension, wherein the second engagement assembly includes a second radial extension, and wherein the tangential spring extension extends between the first and second radial

extensions.
(650) The turbine engine of any preceding clause, wherein the seal support assembly further includes a first ring member coupled to the first engagement assembly and a second ring member coupled to the second engagement assembly, and wherein the tangential spring extension extends between the first and second ring members.
(651) A turbine engine defining a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment having a seal face configured to form a fluid bearing with the rotor; and a seal support assembly including a magnet assembly having a magnet coupled to the carrier or the seal segment for biasing the seal segment along the radial direction.
(652) The turbine engine of any preceding clause, wherein the magnet is a first magnet coupled to the carrier, and wherein the magnet assembly further comprises a second magnet coupled to the seal segment, and wherein the second magnet is in a magnetic field of the first magnet.
(653) The turbine engine of any preceding clause, wherein the seal support assembly includes a first magnet base with the first magnet coupled to and positioned at least partially within the first magnet base, and wherein the seal support assembly includes a second magnet base with the second magnet coupled to and positioned at least partially within the second magnet base.
(654) The turbine engine of any preceding clause, wherein the first and second magnet bases are each formed of a non-ferromagnetic material.
(655) The turbine engine of any preceding clause, wherein the first magnet base is coupled to the carrier and wherein the second magnet base is coupled to the seal segment.
(656) The turbine engine of any preceding clause, wherein the first and second magnet bases together form a bumper to prevent the first magnet from contacting the second magnet.
(657) The turbine engine of any preceding clause, wherein the magnet assembly further comprises a non-ferromagnetic plate positioned between the first magnet and the second magnet.
(658) The turbine engine of any preceding clause, wherein the non-ferromagnetic plate is a first non-ferromagnetic plate coupled to the first magnet base over a surface of the first magnet facing the second magnet, and wherein the magnet assembly further comprises a second non-ferromagnetic plate coupled to the second magnet base over a surface of the second magnet facing the first magnet.
(659) The turbine engine of any preceding clause, wherein the seal support assembly further comprises a particle shield surrounding at least in part the magnet assembly.
(660) The turbine engine of any preceding clause, wherein the particle shield is a bellows assembly extendable along the radial direction and coupled to the carrier and the seal segment.
(661) The turbine engine of any preceding clause, wherein the first and second magnets form a magnetic attraction force.
(662) The turbine engine of any preceding clause, wherein the first and second magnets form a magnetic repelling force.
(663) The turbine engine of any preceding clause, wherein the first magnet comprises a first surface facing the second magnet, wherein the second magnet comprises a second surface facing the first magnet, and wherein the first surface has a non-planar geometry complementary to the second surface.
(664) The turbine engine of any preceding clause, wherein the magnet comprises a plurality of sections arranged linearly, and wherein each section defines a north pole facing in a unique direction relative to one or both adjacent sections.
(665) The turbine engine of any preceding clause, wherein the magnet is a permanent magnet.
(666) The turbine engine of any preceding clause, wherein the magnet defines a Curie temperature greater than 1200 degrees Celsius.
(667) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.

(668) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, and wherein the high-pressure side is located forward of the low-pressure side.
(669) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the seal segment includes a lip and a body, wherein the lip extends from the body along an axial direction of the turbine engine on the high-pressure side, and wherein the lip includes an outer pressurization surface along the radial direction of the turbine engine.
(670) The turbine engine of any preceding clause, further comprising: a compressor, wherein the turbine and the compressor together define in part a working gas flowpath of the turbine engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.
(671) A turbine engine defining a radial direction, the turbine engine comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment having a seal face configured to form a fluid bearing with the rotor; and a seal support assembly comprising a prestressed spring assembly extending from the seal segment for biasing the seal segment along the radial direction.
(672) The turbine engine of any preceding clause, wherein the prestressed spring assembly extends between the seal segment and the carrier.
(673) The turbine engine of any preceding clause, wherein the seal segment includes a support extension having a seal segment spring seat, wherein the prestressed spring assembly comprises a spring extension and a compression plate having a compression plate spring seat, wherein the spring extension is a prestressed spring extension extending from the seal segment spring seat to the compression plate spring seat.
(674) The turbine engine of any preceding clause, wherein the seal support assembly defines an assembly length from the seal segment spring seat to the compression plate spring seat, wherein the spring extension defines an unloaded length, and wherein the assembly length is less than or equal to 90% of the unloaded length of the spring extension.
(675) The turbine engine of any preceding clause, wherein the compression plate is fixed to the carrier.
(676) The turbine engine of any preceding clause, wherein the carrier comprises a bump stop to maintain the prestressed spring assembly in a prestressed condition.
(677) The turbine engine of any preceding clause, wherein the compression plate defines an assembly opening, wherein the carrier defines a threaded opening aligned with the assembly opening to facilitate assembly of the prestressed spring assembly.
(678) The turbine engine of any preceding clause, wherein the prestressed spring assembly comprises a spring extension, and wherein the spring extension is at least one of a linear helical spring, a progressive helical spring, a bellows assembly, a conical spring, an optimized super-elastic spring block, or an optimized metal foam structure.
(679) The turbine engine of any preceding clause, wherein the seal segment comprises a support extension, wherein the prestressed spring assembly comprises a support ring positioned inward of the support extension, and wherein the prestressed spring assembly includes a spring extension extending between the seal segment and the support ring.
(680) The turbine engine of any preceding clause, wherein the seal segment is a one of a plurality of seal segments extending along a circumferential direction, wherein each of the plurality of seal segments includes a support extension, wherein the prestressed spring assembly comprises a support ring positioned inward of the support extensions of the plurality of seal segments, and wherein the support ring extends 360 degrees in the circumferential direction.
(681) The turbine engine of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.

(682) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, and wherein the high-pressure side is located forward of the low-pressure side.
(683) The turbine engine of any preceding clause, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the seal segment includes a lip and a body, wherein the lip extends from the body along an axial direction of the turbine engine on the high-pressure side, and wherein the lip includes an outer pressurization surface along the radial direction of the turbine engine.
(684) The turbine engine of any preceding clause, further comprising: a compressor, wherein the turbine and the compressor together define in part a working gas flowpath of the turbine engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.
(685) A method for assembling a seal segment for a turbine engine, the method comprising: moving a compression plate of a seal support assembly over an attachment mechanism of a carrier, the attachment mechanism of the carrier extending along a radial direction of the turbine engine, the seal support assembly including a spring extension extending between the compression plate and the carrier; inserting a tightening rod through a tightening rod opening defined in the compression plate and into an engagement opening in the carrier; moving the tightening rod through the engagement opening in the carrier by contacting the compression plate with a ledge of the tightening rod to move the compression plate closer to the carrier; and affixing a mechanical fastener to the attachment mechanism to maintain a maximum distance between the compression plate and the carrier.
(686) The method of any preceding clause, further comprising: removing the tightening rod from the engagement opening and from the tightening rod opening after affixing the mechanical fastener to the attachment mechanism.
(687) The method of any preceding clause, wherein the mechanical fastener defines a threaded opening, and wherein affixing the mechanical fastener to the attachment mechanism comprises rotating the mechanical fastener about the attachment mechanism to engage the threaded opening with threads of the attachment mechanism.
(688) The method of any preceding clause, wherein moving the tightening rod through the engagement opening in the carrier comprises compressing the spring extension to an assembly length from an unloaded length, and wherein the assembly length is less than or equal to 90% of the unloaded length of the spring extension.
(689) The method of any preceding clause, wherein the compression plate is fixed to the carrier.
(690) The method of any preceding clause, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.
(691) A turbine engine defining a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a plurality of seal segments, the plurality of seal segments having a seal segment having a seal face forming a fluid bearing with the rotor; and a seal support assembly comprising a pneumatic engagement assembly operable to bias the seal segment along the radial direction during operation of the turbine engine.
(692) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly is in fluid communication with a working gas flowpath defined by the turbine engine.
(693) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly includes a high pressure air supply, wherein the high pressure air supply is in fluid communication with the working gas flowpath at a high pressure location.
(694) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly includes an extension member coupled to the seal segment, and wherein the extension member is moveable along the radial direction.

(695) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly defines a pressure chamber, and wherein the extension member is positioned at least in part within the pressure chamber.
(696) The turbine engine of any preceding clause, wherein the extension member includes a pressure head defining a pressure surface positioned in the pressure chamber.
(697) The turbine engine of any preceding clause, wherein the extension member further includes a rod extending from the pressure head, and wherein the pneumatic engagement assembly includes a bellows positioned around the rod within the pressure chamber.
(698) The turbine engine of any preceding clause, wherein the pressure head defines a head crosswise measure, wherein the pressure chamber defines a chamber crosswise measure, and wherein the head crosswise measure is within 5% of the chamber crosswise measure.
(699) The turbine engine of any preceding clause, wherein the extension member further includes a rod extending from the pressure head, and wherein the pneumatic engagement assembly includes a spring extension positioned within the pressure chamber to bias the pressure head outward along the radial direction.
(700) The turbine engine of any preceding clause, wherein the extension member includes a rod and an engagement guide positioned at an end of the rod, wherein the seal segment comprises a hanger defining an opening, and wherein the engagement guide is positioned within the opening.
(701) The turbine engine of any preceding clause, wherein the opening defines a height along the radial direction greater than a thickness of the engagement guide along the radial direction.
(702) The turbine engine of any preceding clause, wherein the engagement guide is a circular member, and wherein the opening is a circular opening.
(703) The turbine engine of any preceding clause, wherein the pressure head forms a seal with a wall of the pressure chamber.
(704) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly defines one or more openings to the pressure chamber to provide a low pressure airflow to a low-pressure side of the pressure head.
(705) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly defines a first pressure chamber and a second pressure chamber located downstream of the first pressure chamber, wherein the pneumatic engagement assembly includes a retraction member positioned within the first pressure chamber moveable between a first position and a second position along a length of the first pressure chamber, and wherein the first pressure chamber defines an air outlet in fluid communication with the second pressure chamber at a location between the first position and the second position.
(706) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly includes a diaphragm, a first engagement rod, a second engagement rod, a third engagement rod, and a radial spring extension, wherein the first engagement rod is coupled to the diaphragm and configured to move in a direction perpendicular to the radial direction, wherein the second and third engagement rods are pivotably coupled to the first engagement rod, wherein the second engagement rod is coupled to the carrier, wherein the third engagement rod is coupled to the seal segment, and wherein the radial spring extension extends between the carrier and the seal segment.
(707) The turbine engine of any preceding clause, wherein the seal segment defines an engagement surface having a slope relative to an axial direction, wherein the pneumatic engagement assembly defines a pressurized sliding chamber and includes a slider positioned at least partially within the pressurized sliding chamber, wherein the slider includes a radial extension engaged with the engagement surface of the seal segment.
(708) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly defines a pressurized sliding chamber and includes a slider positioned at least partially within the pressurized sliding chamber and a radial member extending to the seal segment, wherein the slider includes an engagement surface having a slope relative to an axial direction, wherein the

engagement surface is engaged with the radial member to move the seal segment along the radial direction.
(709) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly comprises an extension member moveable along the radial direction and a lever assembly, wherein the extension member is coupled to the seal segment through the lever assembly.
(710) The turbine engine of any preceding clause, wherein the seal segment is one of the plurality of seal segments of the seal assembly, wherein the pneumatic engagement assembly comprises a unison ring defining a plurality of angled slots, wherein the unison ring is slidably coupled to the plurality of seal segments through the plurality of angled slots, and wherein the pneumatic engagement assembly is operable to rotate the unison ring along a circumferential direction to move the plurality of seal segments along the radial direction.
(711) The turbine engine of any preceding clause, wherein the pneumatic engagement assembly comprises a piston arrangement coupled to the unison ring for rotating the unison ring along the circumferential direction based on a pressure differential within the piston arrangement.
(712) The turbine engine of any preceding clause, wherein the unison ring is a first unison ring, wherein the pneumatic engagement assembly further comprises a second unison ring slidably coupled to the plurality of seal segments through a plurality of angled slots of the second unison ring, and wherein the first and second unison rings are spaced along an axial direction of the turbine engine.
(713) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A turbine engine defining a radial direction, comprising: a rotor; a stator comprising a carrier; a seal assembly disposed between the rotor and the stator, the seal assembly comprising a seal segment having a seal face configured to form a fluid bearing with the rotor; and a seal support assembly including a magnet assembly having: a magnet, a magnet base, the magnet coupled to and positioned at least partially within the magnet base, and a plate coupling the magnet base to the carrier or the seal segment, wherein the magnet is configured for biasing the seal segment along the radial direction.
2. The turbine engine of claim 1, wherein the magnet is a first magnet coupled to the carrier, and wherein the magnet assembly further comprises a second magnet coupled to the seal segment, and wherein the second magnet is in a magnetic field of the first magnet.
3. The turbine engine of claim 2, wherein the magnet base is a first magnet base, the first magnet coupled to and positioned at least partially within the first magnet base, and wherein the seal support assembly includes a second magnet base with the second magnet coupled to and positioned at least partially within the second magnet base.
4. The turbine engine of claim 3, wherein the first and second magnet bases are each formed of a non-ferromagnetic material.
5. The turbine engine of claim 3, wherein the first magnet base is coupled to the carrier and wherein the second magnet base is coupled to the seal segment.
6. The turbine engine of claim 3, wherein the first magnet base includes a first bumper and the second magnet base includes a second bumper disposed inward of the first bumper in the radial

direction, wherein the first bumper and the second bumper are configured to contact to prevent the first magnet from contacting the second magnet in the radial direction.

7. The turbine engine of claim 3, wherein the magnet assembly further comprises a non-ferromagnetic plate positioned between the first magnet and the second magnet.

8. The turbine engine of claim 7, wherein the non-ferromagnetic plate is a first non-ferromagnetic plate coupled to the first magnet base over a surface of the first magnet facing the second magnet, and wherein the magnet assembly further comprises a second non-ferromagnetic plate coupled to the second magnet base over a surface of the second magnet facing the first magnet.

9. The turbine engine of claim 2, wherein the first and second magnets form a magnetic attraction force.

10. The turbine engine of claim 2, wherein the first and second magnets form a magnetic repelling force.

11. The turbine engine of claim 2, wherein the first magnet comprises a first surface facing the second magnet, wherein the second magnet comprises a second surface facing the first magnet, and wherein the first surface has a non-planar geometry complementary to the second surface.

12. The turbine engine of claim 1, wherein the seal support assembly further comprises a particle shield surrounding at least in part the magnet assembly.

13. The turbine engine of claim 12, wherein the particle shield is a bellows assembly extendable along the radial direction and coupled to the carrier and the seal segment.

14. The turbine engine of claim 1, wherein the magnet comprises a plurality of sections arranged linearly, and wherein each section defines a north pole facing in a unique direction relative to one or both adjacent sections.

15. The turbine engine of claim 1, wherein the magnet is a permanent magnet.

16. The turbine engine of claim 15, wherein the magnet defines a Curie temperature greater than 1200 degrees Celsius.

17. The turbine engine of claim 1, further comprising: a turbine, wherein the rotor is a turbine rotor of the turbine.

18. The turbine engine of claim 17, wherein the seal assembly includes a high-pressure side and a low-pressure side, and wherein the high-pressure side is located forward of the low-pressure side.

19. The turbine engine of claim 17, wherein the seal assembly includes a high-pressure side and a low-pressure side, wherein the seal segment includes a lip and a body, wherein the lip extends from the body along an axial direction of the turbine engine on the high-pressure side, and wherein the lip includes an outer pressurization surface along the radial direction of the turbine engine.

20. The turbine engine of claim 19, further comprising: a compressor, wherein the turbine and the compressor together define in part a working gas flowpath of the turbine engine, and wherein the outer pressurization surface is in airflow communication with the working gas flowpath.